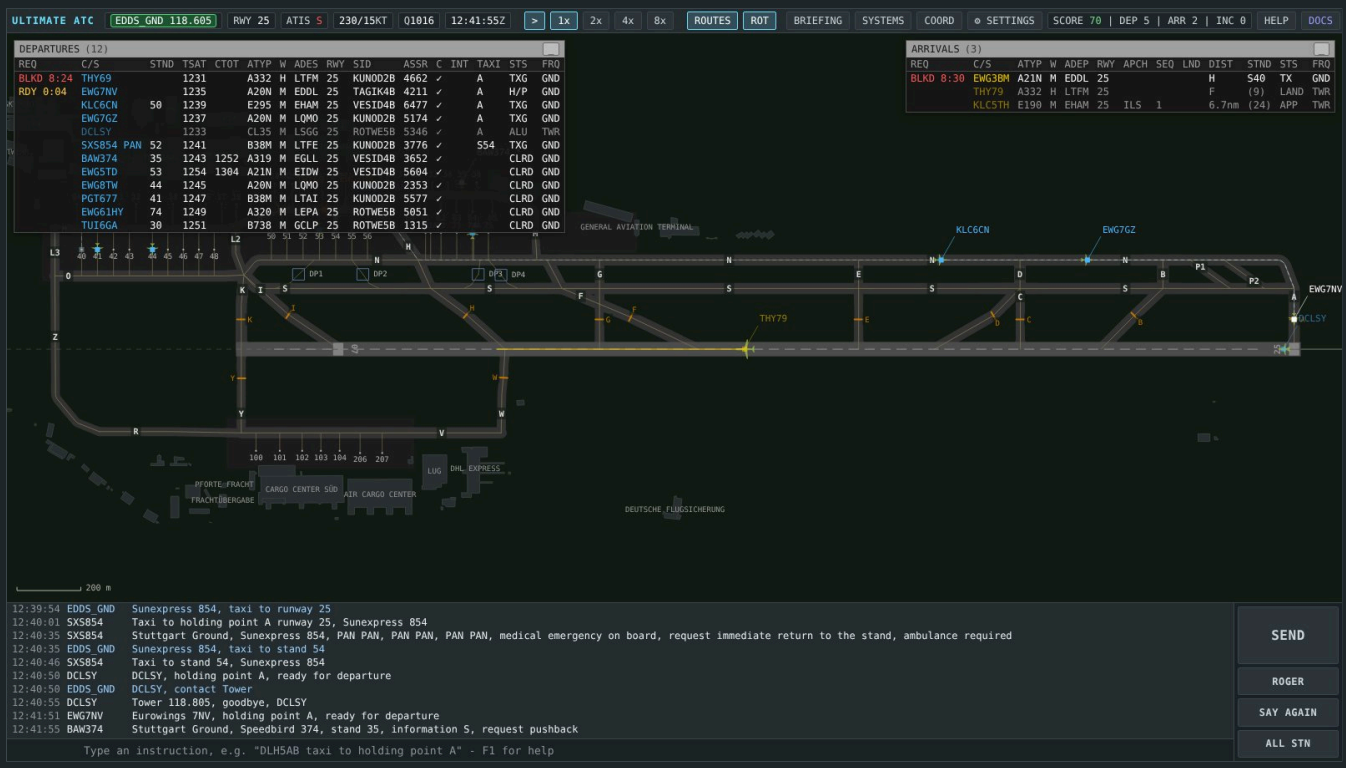
ULTIMATE ATC

CONTROLLER HANDBOOK

DEL | GND | TWR | VERSION 1.4.0 | 2026-10-25



A browser-based tower and ground control simulator

github.com/linoking09/ultimate-atc/

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For training and entertainment only - not for real-world navigation or air traffic control. Airport layouts from © OpenStreetMap contributors (ODbL) and OurAirports (public domain); facts digitised by hand from the AIP as a reference. Screenshots from the app. Developer documentation (architecture, airport data, contributing) is online in the repository.

1 GETTING STARTED

PLAY ONLINE

Every push to `main` builds the app and deploys it to GitHub Pages (see below). Once Pages is enabled for the repository, the simulator is available at:

```
https://linoking09.github.io/ultimate-atc/
```

To read the documentation on paper, print the handbook (PDF).

RUN LOCALLY

Requirements:

- Node.js **20 or newer** (CI uses Node 22)
- npm (comes with Node.js)
- A desktop browser. Chrome, Edge, Firefox and Safari are supported. Speech recognition (the MIC button) needs Chrome or Edge.

```
git clone https://github.com/linoking09/ultimate-atc.git
cd ultimate-atc
npm install
npm run dev
```

Open the URL Vite prints (normally `http://localhost:5173`). The page reloads automatically when you change the source code.

Production build

```
npm run build      # type check + bundle into dist/
npm run preview    # serve dist/ on http://localhost:4173
```

`dist/` is a static site: an `index.html`, a CSS file and two JavaScript files (the code and the generated airport data, split in `vite.config.ts`). You can host it on any static web server. The build uses relative asset paths (`base: './'` in `vite.config.ts`), so it also works from a sub-directory.

BROWSER NOTES

- **Text-to-speech** (TTS button) uses the browser's built-in voices. The available voices depend on the operating system. English voices are preferred when installed.
- **Speech recognition** (MIC button, or hold the key left of `1`, which is `^` on German keyboards and `` ` `` on US keyboards) uses the Web Speech API. Chrome sends the audio to Google's speech service for recognition. If you don't want that, don't use the microphone; everything can be typed.
- **Settings** (last airport, traffic density, special events, TTS on/off, route display) are stored in your browser's `localStorage`. If storage is blocked, for example in some private-browsing modes, the app still works with default settings.

2 USER GUIDE

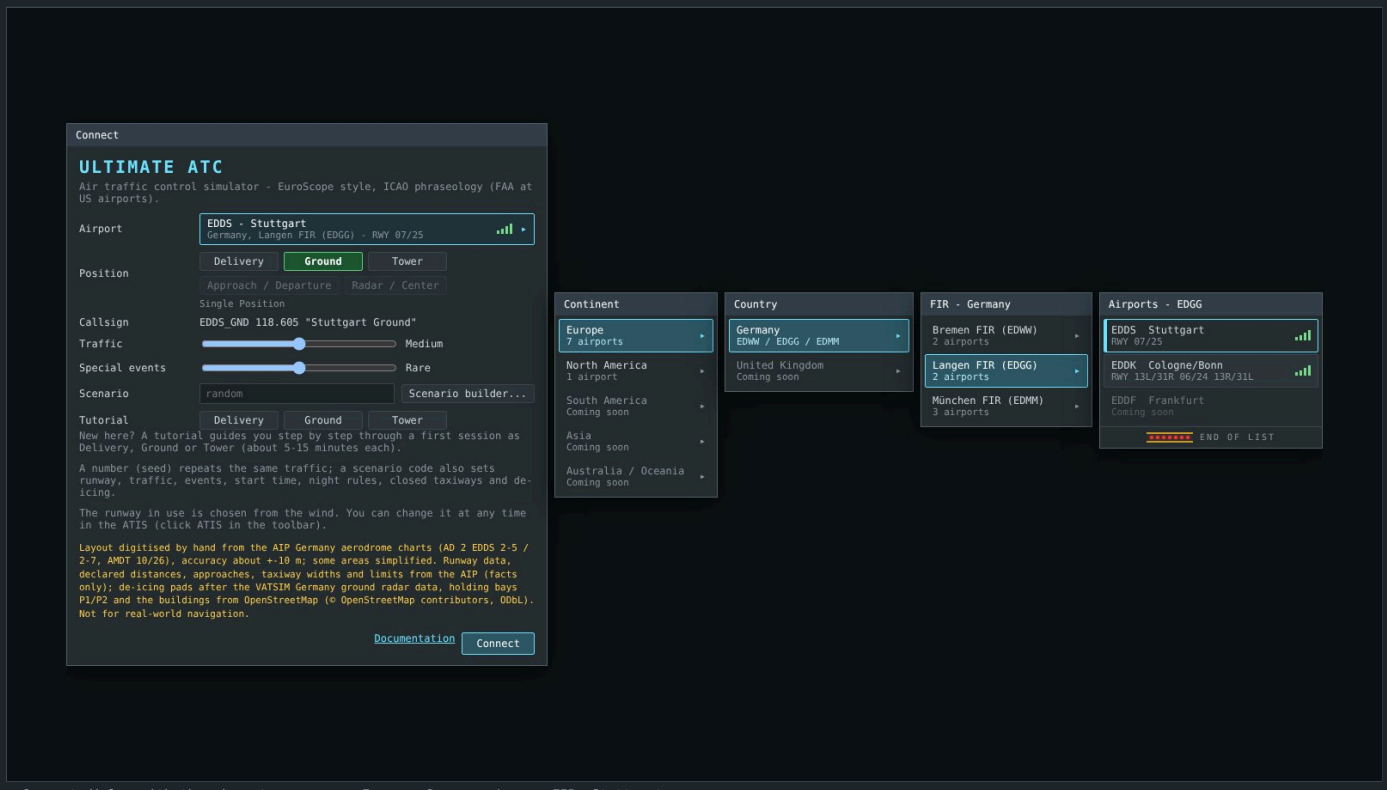
This guide explains the controller client screen by screen. It covers the **Delivery**, **Ground** (since v0.6) and **Tower** (since v0.7) positions at **EDDS** - each alone or combined (since v0.8 also with Tower).

1. CONNECTING

When the app starts, the **Connect** dialog opens. It works like the connect dialog in EuroScope or a VATSIM login:

FIELD	MEANING
Airport	Airport to control: Stuttgart (EDDS), Cologne/Bonn (EDDK), Nuremberg (EDDN), Leipzig/Halle (EDDP, two parallel runways), Bremen (EDDW), Memmingen (EDJA) or Sylt (EDXW). The field shows the airport with its country, FIR, runways and traffic bars; click it to open the **airport menu** to the right of the dialog (see below).
Position	Controller position: **Delivery**, **Ground** and **Tower** are available; Approach/Departure and Center are planned. Only the positions the airport has are offered (Nuremberg, Bremen and Memmingen: no Delivery, Ground gives the clearances; Sylt: Tower only). A selected position is coloured (Delivery dark blue, Ground green, Tower red). Click several to staff them together (**combined positions**, for example Delivery + Ground, Ground + Tower or all three, as one controller does at night): the label below the buttons changes from *Single Position* to a purple **Combined Position**, and the toolbar shows all your stations in their colours with `COMBINED`. Positions you don't staff are run by the simulator.
Callsign	Shows the resulting station callsigns, frequencies and radio names, for example `EDDS_DEL 121.915 "Stuttgart Delivery" + EDDS_GND 118.605 "Stuttgart Ground"`.
Traffic	Slider in five steps: light, light to medium, medium, medium to heavy, heavy (see simulation model).
Special events	Slider: how often medical emergencies (arrivals and departures) and rejected take-offs happen - *never*, *very rare*, *rare* (default), *occasional*, *often*. See simulation model.
Scenario	Optional. A **number** is a seed: the same seed with the same settings gives the same traffic. A **scenario code** (for example `EDDS-25-MB0-M10-K7Q2M`) also fixes the runway, traffic mix and scheduled events, and optionally the start time, night flying restrictions and closed taxiways. **Scenario builder...** creates one from presets or your own choices. Leave it empty for a random session. See Scenarios and seeds.
Tutorial	*Delivery*, *Ground*, *Tower*: starts a guided first session for that position right away (fixed traffic, a panel explains each step and shows the instruction). See Tutorials.

Airport menu. The menu opens in columns, each to the right of the previous one, so you can still change an earlier choice while the later columns are open: **Continent** (Europe, North America, South America, Asia, Australia / Oceania) → **Country** with its FIR (for example *United Kingdom - London FIR (EGTT)*) → for Germany the **FIR** (Bremen FIR EDWW, Langen FIR EDGG, München FIR EDMM) → **Airport**. Each airport shows its runways and **traffic bars**: four bars for the busiest airports (Leipzig/Halle about 29, Stuttgart and Cologne/Bonn about 26 movements per hour at medium traffic), one for the quietest (Sylt, about 7); hover for the number. Airports that are not available yet (Frankfurt, London Heathrow; Los Angeles, Aspen, Jackson Hole, Key West, St. Maarten; Buenos Aires Aeroparque, La Paz, Porto Alegre, Medellín, Quito; Nagoya Chubu, Penang, Colombo, Taipei Songshan, Phuket; Sydney, Melbourne, Perth, Brisbane, Gold Coast, Hobart, Auckland, Queenstown, Wellington) are listed as *Coming soon* and cannot be selected; continents, countries and FIRs without an airport say *Coming soon* too. The airport column ends with a lit stop bar - the row of red lights at a holding point - and *END OF LIST*. The menu always starts with the continents, whichever airport is selected: each column slides in with a short animation after a choice in the previous one (the dialog moves aside, so everything stays centred), and the selected airport is marked when you reach its column. Picking an airport closes the menu, so only the Connect dialog is left. Keyboard: arrow keys up / down within a column, right opens the next column, left goes back, Enter picks, Escape (or a click beside the menu) closes it without a change. The dialog and the menu are centred together; on a narrow screen the row scrolls sideways to the newest column. A shared scenario link selects its airport.

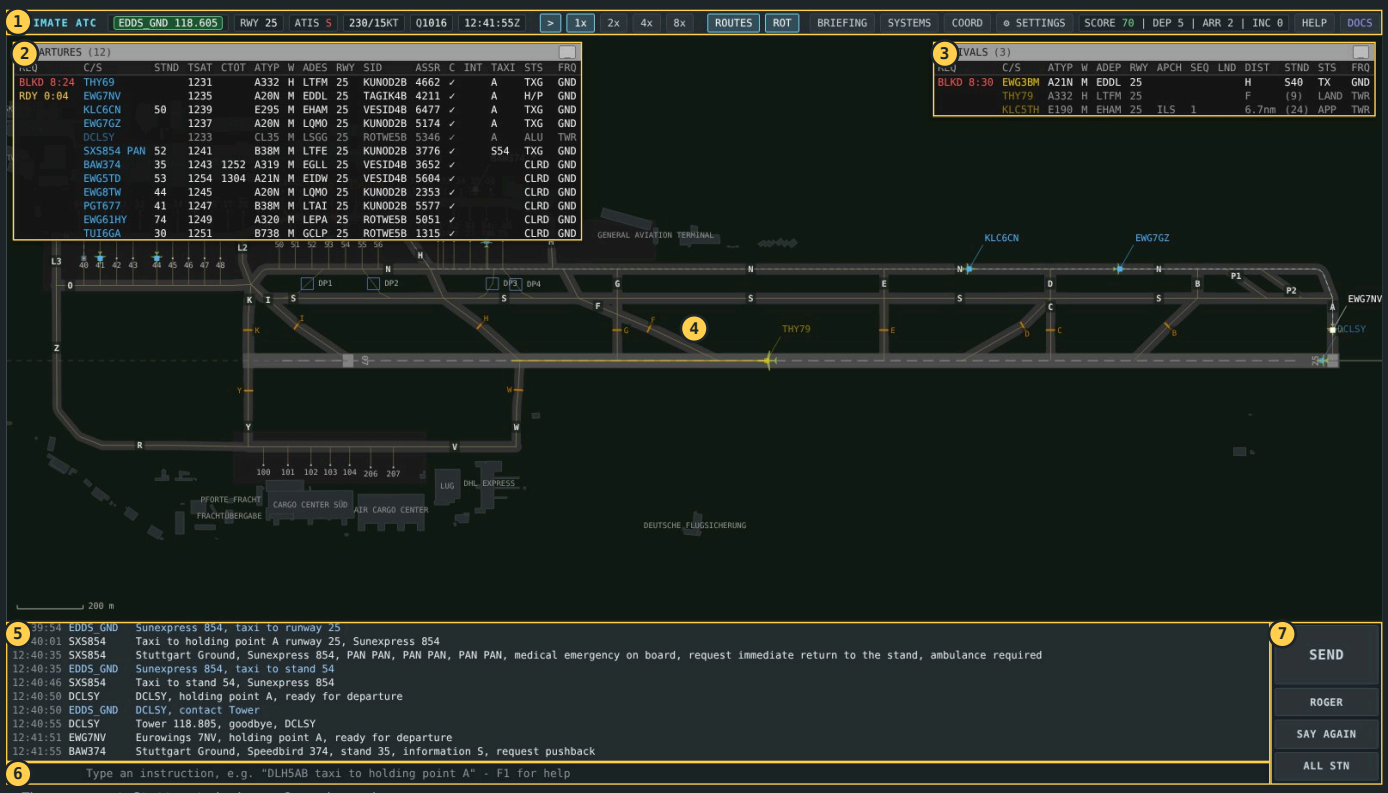


▸ Connect dialog with the airport menu open: Europe, Germany, Langen FIR, Stuttgart

There is no runway selection: the session starts with a random wind, and the **runway in use is the one with the most headwind** (runway 25 at EDDS unless the tailwind on it is above 3 kt; at Leipzig/Halle both parallel runways of that direction). The wind changes during the session with the weather (a new METAR every 30 minutes); when the tailwind on the runway in use grows above 5 kt, the METAR message suggests a runway change. You change the runway, wind, QNH and ATIS letter during the session in the ATIS editor.

The yellow notice shows how accurate the airport data is. Press **Connect** to start. The session begins immediately and the clock runs. Open the airport briefing (`BRIEFING`) to learn the local procedures.

2. SCREEN LAYOUT



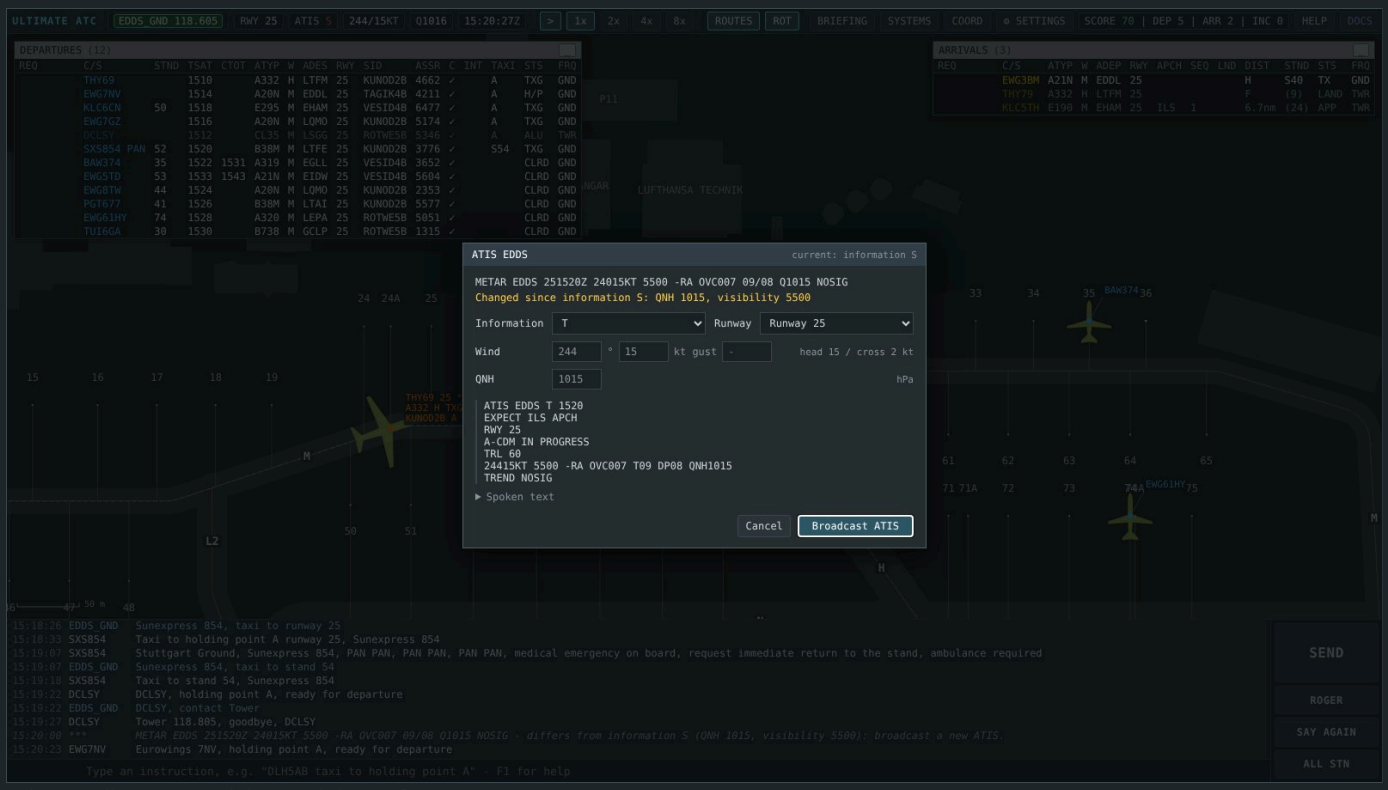
▸ The screen at Stuttgart during a Ground session

1. Toolbar: stations, runway, ATIS, wind, QNH, clock, speed, score, help
2. Departure list
3. Arrival list
4. Ground radar scope
5. Message window (radio log)
6. Command line with preview, MIC and SEND
7. Sidebar

Toolbar

ELEMENT	DESCRIPTION
`EDDS_GND 118.605`	Your station(s) and frequency, in the position colour (several plus `COMBINED` with combined positions). **Click a frequency to switch it off**: the simulator takes the position over (pilots waiting on it are answered by the AI) and it is shown dashed with `OFF`; click again to take it back. At least one frequency stays on.
`RWY 25`	Runway in use (with parallel runways both, `RWY 26R 26L`). Click to open the ATIS editor.
`ATIS E`	Current ATIS letter. Pilots report it on first contact. Red: a newer METAR differs from it (the tooltip says what) - broadcast a new ATIS. Click to open the ATIS editor.
`250/08KT`	Surface wind (magnetic), with gusts `070/18G32KT`. Shown in red if the tailwind on the runway in use is above 5 kt, and with `TS` while a thunderstorm is active. Click to edit.
`Q1013`	QNH in hPa (US airports: `A2992`, the altimeter setting in inches). Click to edit.
`14:32:10Z`	Simulation time in UTC. The session starts at the current real time, or at a scenario's start time. With night flying restrictions the local time follows (`2215L`).
`II` / `>`	Pause / resume. **Space** does the same when the command line is empty.
`1x 2x 4x 8x`	Simulation rate. Pilots and Tower run faster too; text-to-speech gets slightly faster.
`TTS`	Pilots read their transmissions aloud (text-to-speech). *Hidden in this version, see Voice.*
`ROUTES`	Shows the cleared taxi routes of all aircraft on your frequency. The selected aircraft's route is always shown.
`ROT`	Rotates the scope: the main runway **horizontal** like the aerodrome chart (default; Stuttgart 07 on the left, Sylt 14) or north-up.
`BRIEFING`	Opens the airport briefing.
`SYSTEMS`	Opens the systems window (also **F3**). Reads `SYSTEMS (n OFF)` in orange while systems are off.
`COORD`	**Coordination** by landline with the positions the simulator runs: *Langen Radar: no more arrivals* (Approach releases no new arrivals onto the final; those already on final continue) and *Stuttgart Ground: hold all departures* (no more pushbacks; aircraft already taxiing continue; only while the simulator runs Ground). Use it to make room, e.g. to clear a queue or a blocked runway. Each costs **2 points per minute** while it is on. A third switch (no cost) has Ground ask every departure for the **earliest intersection** it accepts: shown as `INT D` (or `INT FULL`) in the tag and in the `INT` column of the departure list; the button reads `COORD (ARR+DEP HOLD)` in orange. Click again to resume.
`SETTINGS`	Opens the settings menu (also **F2**).
`SCORE ...`	Score, departures handed off (`DEP`), arrivals parked (`ARR`) and incidents (`INC`). See Score.
`HELP`	Help window with the tabs *Airport briefing*, *Phraseology* and *Controls* (also **F1**; it opens on the tab you used last).
`DOCS`	Opens this documentation on GitHub.
`DISCONNECT`	Ends the session properly - reloading or closing the page asks first, because the debriefing is not saved. Opens the **debriefing** (the simulation pauses): *Continue session* goes back, *End session* returns to the Connect dialog. See Debriefing.

ATIS editor



▸ The ATIS editor: runway, wind, QNH and the ATIS letter

Click `RWY`, `ATIS`, the wind or `Q...` in the toolbar to edit the ATIS. The top shows the latest METAR (`METAR EDDS 010950Z 25008KT CAVOK 21/11 Q1018 NOSIG`, new at minute 20 and 50 of every hour) and, in yellow, what changed since the current information (`Changed since information N: visibility 6000`); the title shows the current letter.

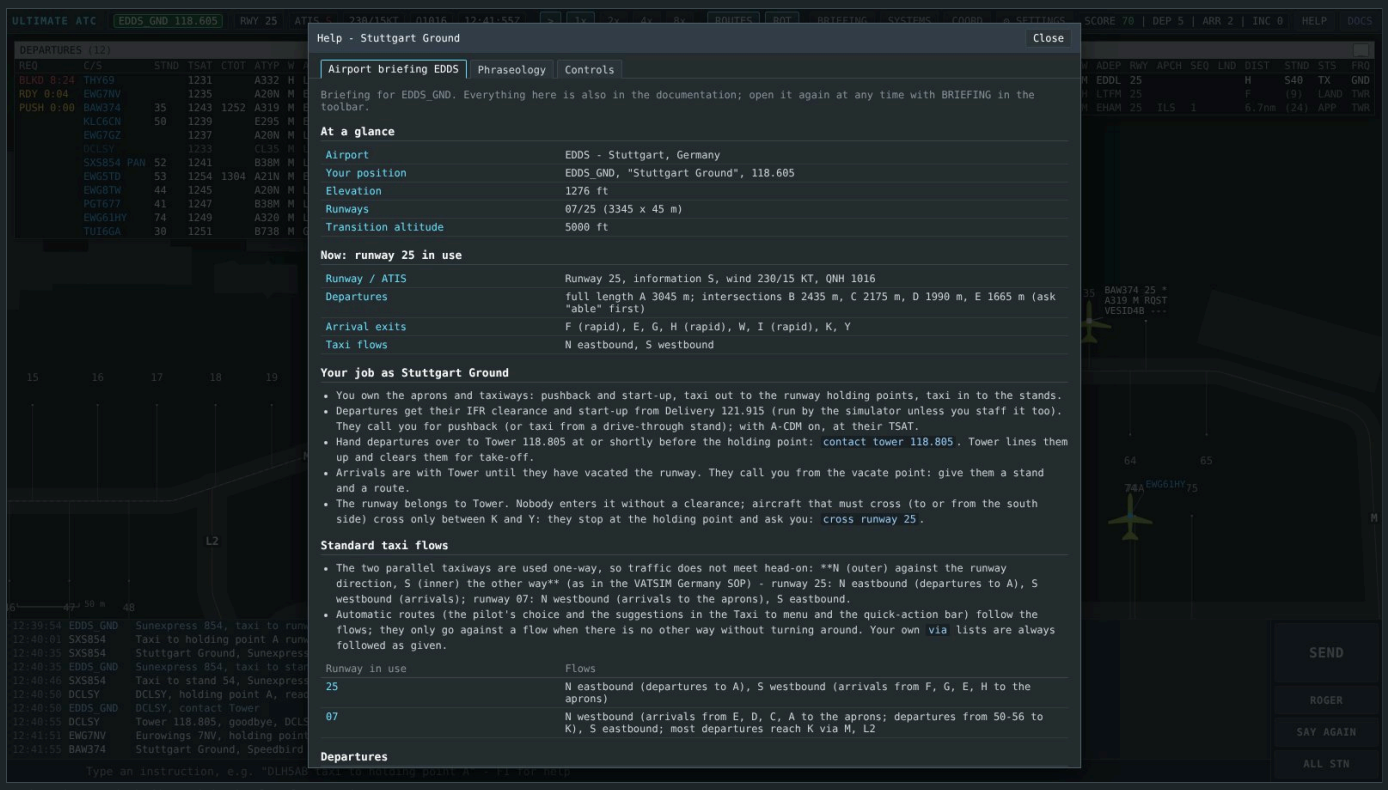
FIELD	MEANING
Information	The new ATIS letter (pre-set to the next letter)
Runway	Runway for departures and arrivals. At airports with parallel runways (Leipzig/Halle) first the parallel runways used together (`Runways 26R + 26L`), then each runway alone - see parallel runways
Wind	Direction, speed and gusts (`-` = none; broadcast as `gusting 32 knots`) in one row, with the head-/tailwind and crosswind for the selected runway (`head 9 / cross 4 kt`). Direction, speed and the tailwind turn red when the tailwind on the selected runway is above 5 kt
QNH	Pressure setting in hPa (pre-filled from the METAR; at US airports the dialog also shows the altimeter setting it is broadcast as, `hPa = A3030`)

Below the fields the dialog previews the new ATIS as D-ATIS text, with the METAR's visibility, clouds, temperature and trend and the notices (closed taxiways, de-icing, A-CDM, DCL out of service, windshear, the runway of each apron with parallel runways); **Spoken text** unfolds the spoken version. When you select another runway, a short yellow note says what happens to the traffic (see *Runway change* below). **Broadcast ATIS** publishes it. It appears in the message window, and crews that listen to the ATIS from now on quote the new letter. The ATIS does not change on its own.

Crews that listened before the broadcast still report the old letter (`..., information D, request pushback` with E current): the aircraft menu then offers `Information E is current` (`information E is current, QNH 1018`), or broadcast `all stations, information E is now current` with **ALL STN**. A crew that leaves your frequency with an old letter costs 3 points.

Runway change: departures that are not yet taxiing get the new runway and a matching SID. Arrivals further out than 3.5 NM are re-sequenced onto the new final by Approach; arrivals closer in still land on the old runway. Aircraft that are already taxiing keep their clearance, so re-route them to a holding point of the new runway. If an aircraft ends up at the wrong end, Tower sends it back to you.

Airport briefing



▸ The airport briefing with the local procedures

`BRIEFING` in the toolbar opens the help window on the **Airport briefing** tab. You can open it at any time; it always shows the current state:

- **At a glance**: airport, your position and frequency, elevation, runways, transition altitude.
- **Now**: runway in use, ATIS letter, wind and QNH, the full-length and intersection holding points, the arrival exits and the taxi flows of the runway in use; with a scenario also closed taxiways, the night flying restrictions (with the local time) and an active thunderstorm.
- **Local procedures** for your position: your job, standard taxi flows, departures (entries, push directions, typical routes), arrivals (exits, vacating, typical routes), stands, hot spots and pitfalls.
- **Frequencies** of all stations, and the data accuracy notice.

Every airport added in the future comes with its own briefing.

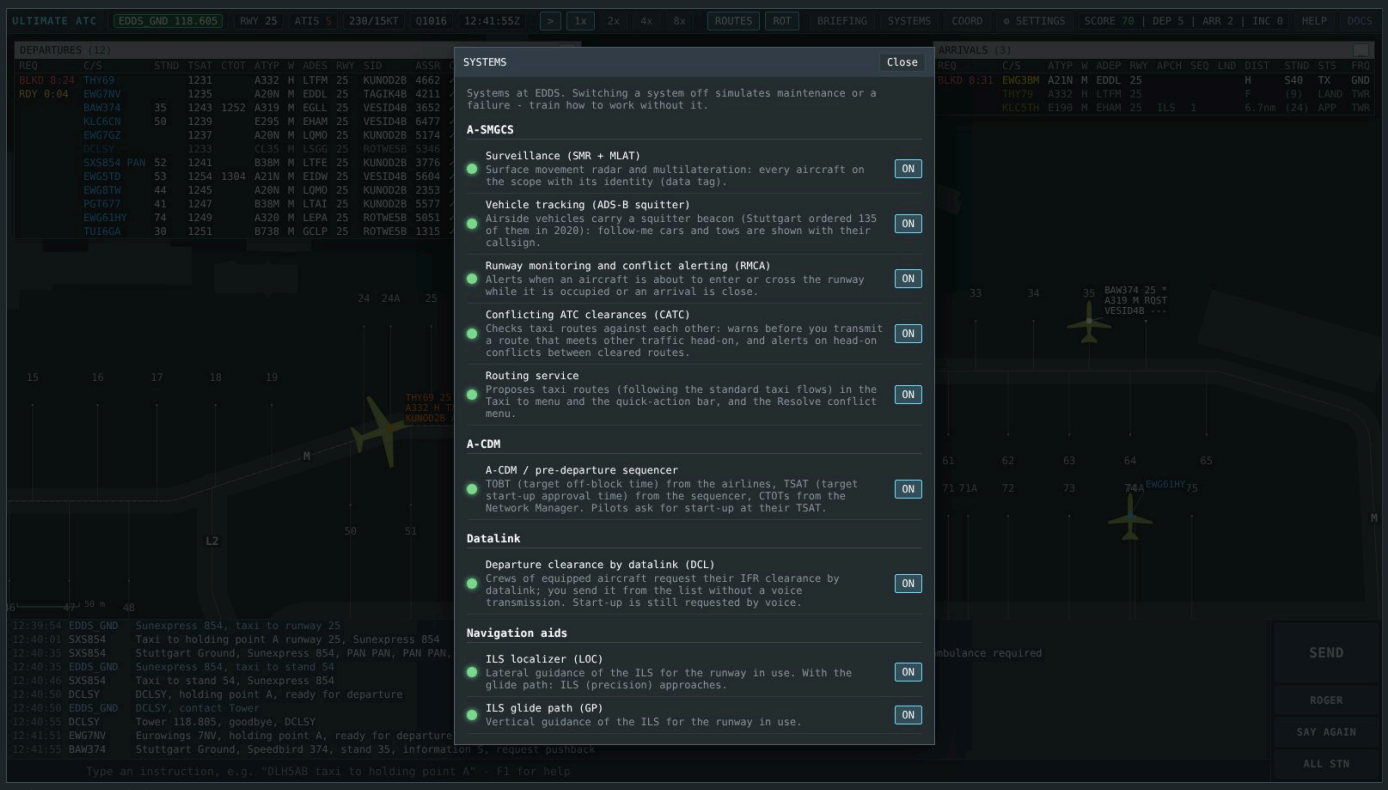
Settings

`SETTINGS` in the toolbar (or **F2**) opens the settings menu. Changes apply immediately, also in the middle of a session, and are stored in the browser.

SETTING	EFFECT
Device (phone - laptop slider)	**Mobile** or **PC** layout, see Mobile mode. Touch-only devices start in mobile mode.
Interface size	Size of the text and buttons (80-160 %)
Tag size	Size of the aircraft data tags on the scope (80-200 %)
Data tags	**Compact when zoomed out** (default): with the whole airport in view, tags show only the callsign - see tags; **Always full**
Scope orientation	Runway horizontal (like the chart) or north-up - the same as `ROT`
Scope style	**PHX night** (default), **PHX day** or **Classic** - see Scope styles
Show cleared routes	The same as `ROUTES`
Pilot voices (TTS)	The same as `TTS`
Voice volume / speed	Volume and speaking rate of the pilot voices
Send voice automatically	Transmit the recognised instruction when you release push-to-talk; off = check and press Enter
Recognition accent	English accent the speech recogniser expects (US, UK, Australia, India, Ireland)
Traffic density	Slider in five steps from light to heavy; applies to traffic generated from now on
Special events	Slider: never, very rare, rare, occasional, often

Reset to defaults restores everything except the airport and position.

Systems window



▸ The systems window

`SYSTEMS` (or **F3**) shows the status of the airport's systems - A-SMGCS (surveillance, runway monitoring RMCA, conflicting clearances CATC, routing service), A-CDM, datalink clearances (DCL) and the ILS (localizer, glide path) - with a lamp and an ON/OFF switch each. Switch a system off to train working without it (maintenance, failure). What each system does and what changes when it is off: Airport and ATC systems.

Mobile mode

Mobile mode is made for touch screens, above all the **iPad** (a phone screen is too small to control comfortably):

- **+ / - / home buttons** on the right of the scope zoom in, out and reset the view.
- **One finger** pans the scope (also when the drag starts on an aircraft), **two fingers** pan and zoom at the same time. The page itself never scrolls, bounces or zooms.
- **Tap once** on an aircraft to select it. **Tap it again** to open its menu; a second tap on a tag item opens that item's function (flight plan, taxi destinations, aircraft menu). A **long press** opens the menu directly. A single tap therefore never sends anything by accident.
- A **quick-action bar** for the selected aircraft: `PUSH` (push and start approved), `TAXI` (taxi destinations), `HOLD`, `RESOLVE` (head-on conflict), `CONT` (continue taxi), `TWR` (contact Tower) and `MENU`, depending on what the aircraft is doing. For tows: `TOW` (tow approved); for a crew that asked for a follow-me: `FLWM`. On Delivery frequency: `CLR` or `DCL` (IFR clearance), `RB OK` (readback correct), `START` (start-up approved), `GND` (contact Ground). **Phrase buttons are picked, not sent**: a tap highlights the button and writes the phrase into the command line, a second tap removes it; **SEND** transmits all picked phrases as one transmission, e.g. `RB OK` + `START` + `GND` gives `readback correct, start-up approved, contact ground 121.900`. Buttons that open a menu (`TAXI`, `CLR`, `RESOLVE`, `MENU`) work as before. Instructions that need not be said twice disappear once given: `RB OK` after the readback is confirmed, `CONT` (continue approach) and `T/O` on Tower once given, `HOLD` while the aircraft holds position; `CONT` (continue taxi) only shows while it waits for it (hold position, hold short, give way).

Menus on touch screens: a sub-menu opens with a tap (not when the finger merely passes over a row) next to its menu, or below the tapped row if there is no room beside it. A tap right after a menu opened is ignored, so one tap never selects an entry of the menu that just appeared.

- Larger buttons, menu entries, list rows, symbol hit areas and tags; sub-menus open with a tap. List rows react to the first tap (they are updated in place, never rebuilt under your finger).
- On narrow screens (portrait) the departure and arrival lists start collapsed so the scope has room.
- The command-line preview is hidden; use voice (`MIC`), the menus or a keyboard.

iPad with a hardware (Bluetooth) keyboard: just start typing, the first letter already goes into the command line. Tapping the command line does not push the page up any more: the layout shrinks to the area above the keyboard bar. The command line is not a form field, so Safari shows no AutoFill buttons (passwords, cards, contacts) and no form arrows for it. The small bar iPadOS shows at the bottom while a hardware keyboard is connected (language switch, shortcuts) belongs to the system and cannot be removed by a web page; iPadOS can hide the shortcut part under *Settings* > *General* > *Keyboard* > *Shortcuts*.

3. THE SCOPE

The scope is a ground radar showing the aerodrome chart. By default it is rotated like the official aerodrome chart, with the runway horizontal; **ROT** switches to north-up.

Scope styles

Scope style in the settings chooses the look:

- **PHX night** (default) and **PHX day**: like the EuroScope ground radar of VATSIM Germany, which is modelled on the **DFS PHOENIX** tower display used at German airports - a dark green-black background (day: light green), dark grey aprons, grey taxiways, a lighter grey runway, amber holding points, white taxiway designators; aircraft as **radar returns** (olive green) with a small **SSR symbol** (the transponder square) and five **history dots** behind moving targets; tags in boxes, **departures blue, arrivals yellow**, vehicles green, traffic on other frequencies dimmed; the PHX ground states (`SUG`, `PBG`, `TXG` ...). The lists get the PHX window colours (light grey title bars).
- **Classic**: the colours of Ultimate ATC before 1.2 (blue-grey background, black runway, white / grey silhouettes) with the EuroScope ground states (`ST-UP`, `PUSH`, `TAXI` ...).

ELEMENT	DRAWN AS (PHX; CLASSIC IN BRACKETS)
Runway	Grey strip (classic: black) with a dashed centre line, threshold bars and designators
Taxiways and taxilanes	Grey bands in their real width (where the airport data has it) with a thin yellow centre line; bends are rounded like the taxiway centre lines, junctions and runway entries have fillets (widened corners)
Taxiway designators	White (classic: yellow) letters in dark boxes (`N`, `S`, `R`, `D` ...). Long taxiways (N, S, M) repeat their designator about every 600 m when you zoom in
Closed taxiways	Red, hatched, designator in red (scenario option, see closed taxiways)
Permanently closed taxiway parts	Grey pavement with black crosses, as on the aerodrome chart; a taxiway closed everywhere has a grey designator (Memmingen S, the loops around the runway ends, Sylt G north of the glider site track)
Paved areas inside a runway strip	Drawn like taxiways (Sylt: between D, L and F at the 14 end, between D, M, P and A at the 06 end), but routes go through the holding points
De-icing pads	Outlined squares (PHX: blue) with the pad name (`DP1` ... `DP4`), with thin lead lines from N to S
Runway holding positions	Thick amber bar across the taxiway, labelled with the holding point (`A`, `K`, `W` ...)
Stands	Thin yellow lead-in line plus the stand number
Aprons	Dark grey areas (classic: slightly lighter than the background); also paved taxiway areas and helipads
Buildings	Dark blue-grey (classic: brown) areas: every building on the aerodrome (OpenStreetMap). Named ones are labelled when you zoom in - each name once, the largest buildings first; a name that would cover a stand number, a taxiway designator or another name is left out
Scale bar	Bottom left

Details appear and disappear depending on zoom: centre lines and stand numbers only show when zoomed in.

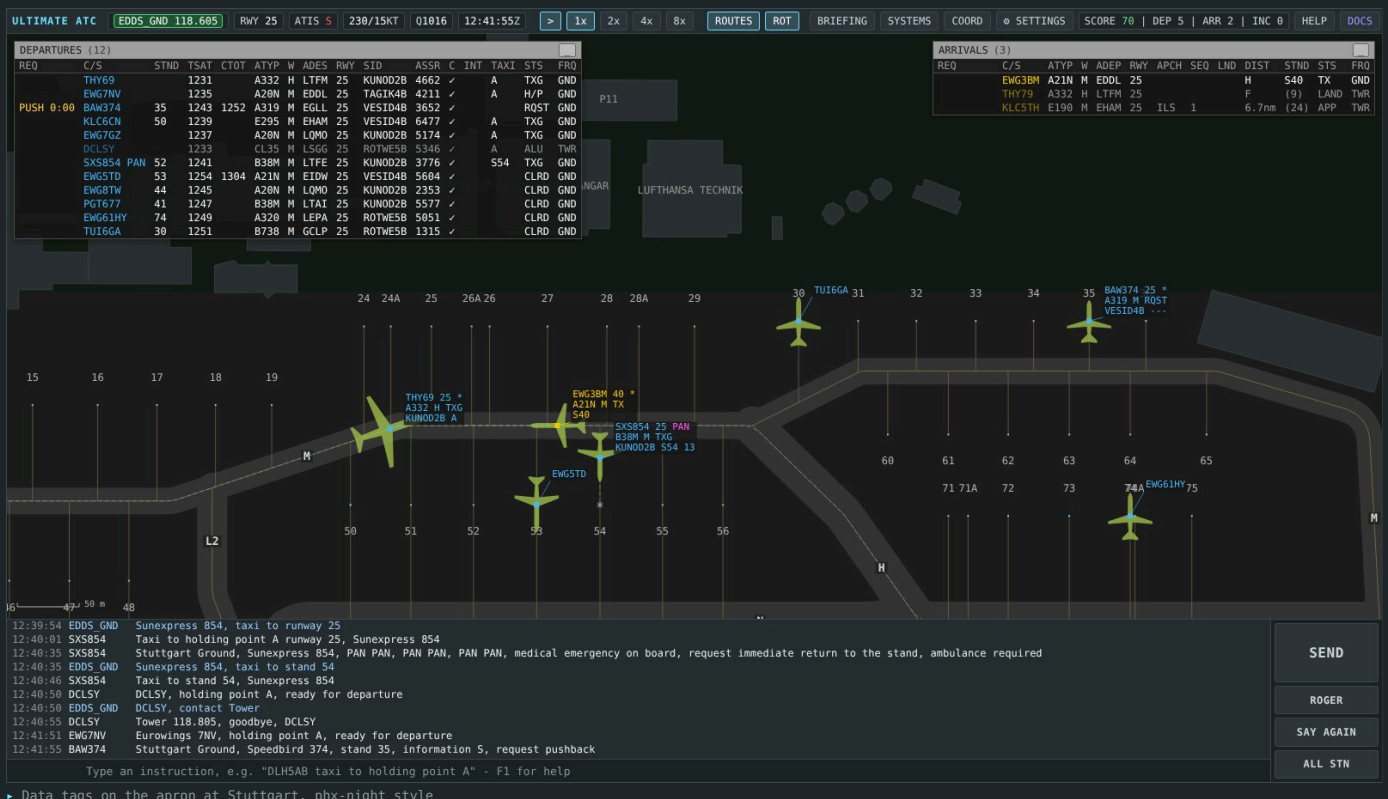
Navigation

- **Mouse wheel**: zoom around the mouse cursor. On touch screens, pinch with two fingers.
- **Drag** on empty space, with the left or right mouse button: pan.
- **Home**: reset the view to the whole airport.
- **Double-click a list row**, or use *Centre view* in the aircraft menu: centre on an aircraft.

Routes on the scope

- The **selected aircraft's** cleared route is a solid cyan line. With `ROUTES` on, the routes of all aircraft taxiing on your frequency are dashed lines.
- Small squares mark where the aircraft will stop: **red** = runway holding point without a crossing clearance, **yellow** = hold short of a taxiway, **cyan/grey** = destination.
- While you type a taxi instruction, the resulting route is previewed as a **green dashed line**.

4. AIRCRAFT TAGS AND COLOURS



▸ Data tags on the apron at Stuttgart, phx-night style

Every aircraft is drawn as a top-down shape at real size (with a minimum size, so it stays visible when zoomed out) - in the PHX styles as the radar return with the SSR symbol - and carries a data tag laid out like the ground labels of VATSIM Germany:

```
DLH5AB 25 PAN *     callsign + departure runway (arrivals: stand); "PAN" = emergency, "*" = waiting for your answer
A320 M TXG          aircraft type + wake category + ground state
ROTWE5B A 15 #2     SID (departures) + where it is cleared to (A = holding point, S14 = stand 14, HS F = hold short of F, >14 = suggested stand) + ground speed in knots + queue number ("number 2 for ...")
```

Like in EuroScope, **the tag items are clickable**. Hovered items are highlighted:

ITEM	LEFT CLICK
Callsign	Flight plan card: **radiotelephony callsign** (e.g. `SPEEDBIRD 947`), type, route, SID, runway, squawk, stand, frequency

ITEM	LEFT CLICK
Aircraft type	Same flight plan card
SID	On your Delivery frequency the clearance menu (IFR clearance with the SID, squawk, start-up), otherwise the flight plan card
Cleared-to (`A`, `S14`, `---`), arrival stand	Taxi menu: holding points for departures, stands for arrivals
Ground state (`TXG`, `RQST`, ...)	The full aircraft menu

Hovering over the callsign adds a line with the radiotelephony callsign below it, so you know how to address the aircraft. The selected aircraft's telephony is also shown in front of the command line (`[BAW947 SPEEDBIRD 947]`).

Airborne aircraft (arrivals on final, departures after take-off) show callsign, type and wake category, altitude in hundreds of feet and speed (`A034 140`), and the approach and runway (`ILS 25`) or the SID, plus a speed vector line.

Compact tags when zoomed out: below a zoom of 0.6 pixels per metre (about the whole airport in view, the typical Tower picture) every tag shows only the **callsign**, with `*` for a pending request and `PAN` for an emergency, like the minimal (untagged) label of EuroScope. The **selected** aircraft - tap it on a tablet - and the one under the mouse get the full tag; zoom in and all tags are full again. *Settings -> Data tags -> Always full* switches this off. To keep the apron readable, parked aircraft with no pending request only get a tag when you zoom in or select them. **Drag a tag** with the mouse to move it; *Reset tag position* in the aircraft menu puts it back.

Extended centrelines

From each runway threshold an extended centreline runs 15 NM outwards with a tick every NM and a longer tick every 5 NM (like the approach path / ILS localizer course lines on a EuroScope map). The runway in use is drawn solid and brighter, the other dashed. With the localizer switched off in the systems window, the line is marked `LOC U/S`; a runway without an ILS (Sylt 14) is marked with its approach, `RNP`. Zoom out to see the arrivals on final.

Ground status codes

The PHX styles use the ground states of the DFS PHOENIX display (as in the VATSIM Germany EuroScope profile), the classic style the EuroScope names:

PHX	CLASSIC	MEANING
`----`	`----`	Parked on the stand, no IFR clearance yet
`CLRD`	`CLRD`	Parked, IFR clearance received
`RQST`	`RQST`	Parked, with a request (clearance, start-up, pushback, taxi)
`PBG`	`PUSH`	Pushback in progress (pushback given)
`SUG`	`ST-UP`	Start-up given, or pushback complete and starting engines
`TXG`	`TAXI`	Taxiing out (departure)
`TX`	`TAXI`	Taxiing in (arrival, tow)
`DEICE`	`DEICE`	Being de-iced on a de-icing pad (scenario option de-icing)
`HOLD`	`HOLD`	Holding position on your instruction
`HS-R`	`HS-R`	Holding short of a runway, waiting for a crossing clearance
`H/P`	`H/P`	At the runway holding point (destination)
`ALU`	`L/U`	Lined up on the runway (Tower)
`DEP`	`DEPA`	Departing (take-off roll and climb)
`APP`	`APP`	On final approach
`LAND`	`LAND`	Landing roll
`VAC`	`VAC`	Vacating the runway
`VACD`	`VACD`	Vacated, waiting for taxi instructions
`ONB`	`PARK`	Arrived on the stand (on blocks)
`G/A`	`G/A`	Going around

Colours

COLOUR (PHX; CLASSIC)	MEANING
Blue tag (classic: white)	Departure on your frequency
Yellow tag (classic: white)	Arrival on your frequency
Green tag (classic: white)	Tow (an aircraft moved by a tug) on your frequency
Dimmed (classic: grey)	On another frequency (Tower, or a position run by the simulator)
Olive shape + small square	PHX: the radar return and the SSR (transponder) symbol; the dots behind a moving target are its last positions (every 2 s). Without A-SMGCS surveillance only the radar return is shown
Flashing red	A-SMGCS CATC: two aircraft rolling towards each other on cleared routes that meet head-on (not when one of them stands still; see Resolve conflict); *Acknowledge CATC alert* in the aircraft menu (`ACK` in the quick-action bar) stops the flashing when you know they will not meet
Flashing white (classic: yellow)	Waiting for your answer (a request is pending); `roger` / `standby` (or ROGER in the sidebar) stops it until the pilot calls again
Flashing orange	Waiting for more than a minute
Flashing magenta, `PAN`	Emergency (medical) - give priority
Red	Involved in a collision
Framed tag, light box (classic: cyan circle)	Selected aircraft
Yellow box `FOLLOW-ME 1`	Follow-me car (the label needs vehicle tracking in the systems window)

5. DEPARTURE AND ARRIVAL LISTS

The lists work like EuroScope's departure and arrival lists, with the columns in the order of the VATSIM Germany profile (the request and its waiting time first). Aircraft waiting for you are sorted to the top, the longest-waiting first. Drag a list by its title bar; the `_` button collapses it. **Cells with a function** (underlined when the mouse is over them) open it with a click, like the items of a data tag: `REQ` and `STS` the aircraft menu, `C/S` and `ATYP` the flight plan card, `SID`, `ASSR` and `C` the clearance menu (on your Delivery frequency; otherwise the flight plan card), `TAXI` / `STND` the taxi menu.

DEPARTURES

COLUMN	CONTENT
REQ	Pending request and how long the pilot has been waiting
C/S	Callsign (PHX: in the departure colour)
STND	Stand (empty once pushed back)
TSAT	Target start-up approval time from the A-CDM sequencer (empty with A-CDM off)
CTOT	Calculated take-off time (ATFM slot) if the flight has one
ATYP	ICAO aircraft type
W	Wake turbulence category (L, M, H)
ADES	Destination
RWY	Departure runway
SID	Standard instrument departure of the flight plan (towards the destination), or the one you cleared
ASSR	Assigned squawk
C	✓ once the crew has its IFR clearance (EuroScope clearance flag)
INT	Earliest intersection the crew accepts (when Ground asks, see `COORD`)
TAXI	Where the aircraft is cleared to taxi (holding point)
STS	Ground state (see above)
FRQ	Frequency the pilot is on: `DEL`, `GND` or `TWR`

ARRIVALS

COLUMN	CONTENT
REQ, C/S, ATYP, W	as above (C/S in the arrival colour)
ADEP	Origin airport
RWY	Landing runway
APCH	Approach procedure while on final: `ILS`, `LOC` (localizer only) or `RNP` (ILS switched off, or none published for the runway, e.g. Sylt 14) - see the systems window
SEQ	Position in the arrival sequence on final (1 = next to land)
LND	`CLR` once the arrival is cleared to land (Tower)
DIST	Distance to the threshold while on final; afterwards the exit used
STND	Cleared stand, or the **suggested stand in brackets** (from the stand allocation)
STS, FRQ	as above

Request codes in the `REQ` column:

CODE	THE PILOT ...
`PUSH`	requests pushback
`TAXI`	is ready for taxi (after pushback, or on a taxi-out stand)
`TXIN`	has vacated the runway and needs taxi instructions
`RDY`	is at the holding point, ready for departure (hand over to Tower!)
`XRWY`	is holding short of a runway on the route, requesting to cross
`BLKD`	is stuck in front of opposing traffic and needs instructions
`RTE?`	needs further instructions (route problem, long hold short)
`CLR`	requests its IFR clearance by voice (Delivery)
`DCL`	requests its IFR clearance by datalink - send it from the aircraft menu
`STUP`	is ready for start-up (with A-CDM: at its TSAT)
`FREQ`	has start-up and asks for the frequency for pushback (`contact ground`)
`FLWM`	has vacated and asks for a follow-me to the stand
`TOW`	a tug asks to tow an aircraft to another stand
`RDY`	(Tower) at the holding point, ready for departure
`LDG`	(Tower) on final, expects the landing clearance
`VACD`	(Tower) has vacated the runway (after landing or a crossing), expects `contact ground`
`RDR`	(Tower) airborne, asks for the frequency change to Radar

VEHICLES (only shown while vehicles are at work): tows - callsign of the tug (`TUG5`), the towed aircraft and its type, from and to stand, status, frequency, request - and follow-me cars: callsign (`FME1`), the aircraft it works for, status (`ASSG` assigned, `PROC` proceeding to the aircraft, `LEAD` leading, `DONE` waiting to return, `RTB` returning to base, `HOLD`) and request (`PROC` wants to proceed, `RTB` wants to return to base).

Click a row to select the aircraft, double-click to centre the scope on it, right-click for the aircraft menu.

6. MESSAGES AND THE COMMAND LINE

Sidebar

Right of the message window, from the top: a large **SEND** button, **ROGER** (acknowledges the call of the selected aircraft, or of the pilot who called last: no read-back, the aircraft stops flashing and the pilot waits 4 minutes before calling again - for example the departures queuing at the holding point), **SAY AGAIN** (the selected aircraft, or the pilot who called last, repeats its last transmission - `standby` is in the aircraft menu and on the command line) and **ALL STN** - broadcasts to all stations: `information B is now current, QNH 1013`, `runway 25 in use`, `standby` (puts every open request on hold) and `expect delays`. Nobody reads a broadcast back. On the desktop ALL STN opens a menu; in mobile mode the quick-action bar shows the broadcasts as phrase buttons (pick one or more, then SEND). Tab still selects the next aircraft with a pending request.

Message window

Shows all radio traffic on your frequency plus system messages:

COLOUR	LINE
Light blue	Your transmissions (`EDDS_GND`)
White	Pilot transmissions
Grey italic	System information (for example ATIS changes)
Red bold	Incidents (collisions, runway incursions, go-arounds) and warnings (A-SMGCS alerts RMCA and CATC, readback errors not caught, missed CTOTs)
Yellow	Hints for you, not transmitted (for example "DLH5AB is not on your frequency")

Click a line to select the aircraft it belongs to.

There is **one frequency**: only one station talks at a time. Pilots wait until the frequency is free, and read-backs come before new calls. Your own transmissions go out immediately, so avoid "stepping on" a pilot by answering too fast in a busy situation.

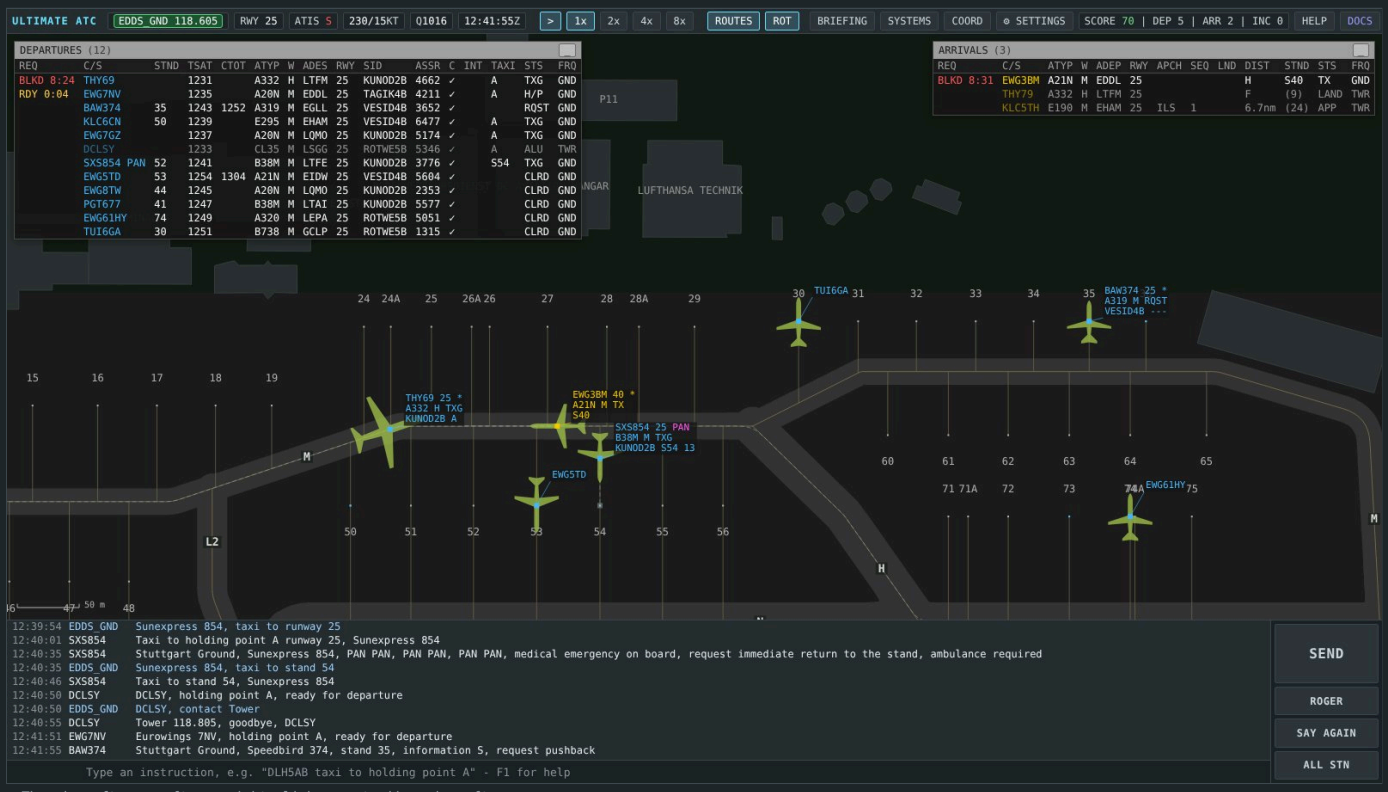
Command line

Type an instruction and press **Enter** (or click **SEND**). The full grammar is in the phraseology reference; at US airports (San Diego) the **FAA forms** apply as well - `runway 27, taxi via B` , `push back approved, tail east` , `altimeter 2992` - and the pilots answer in US wording (FAA phraseology). In short:

```
[callsign] instruction[, instruction ...]
DLH5AB pushback approved facing east
Lufthansa 5AB, taxi to holding point A via L2, S, hold short of taxiway H
taxi to stand 14 via N R          <- goes to the selected aircraft
```

- The **target** on the left (`[DLH5AB]`) is the selected aircraft. It receives instructions without a callsign.
- The **preview** on the right shows how your text was understood:
  - **green**: understood, and the route (if any) is valid. The route is drawn on the scope.
  - **orange**: understood and valid, but there is a catch: the A-SMGCS (CATC) warns that the route meets other traffic head-on (`-- CATC: head-on with EWG7TK on N`), or the stand is taken (`-- stand 14 taken by DLH5AB` ; the crew will accept it and only notice when it gets there). You can still transmit it.
  - **red**: understood, but it won't work, for example `unable to follow route via S` , or the aircraft is not on your frequency.
  - **grey** `? ...` : words that were not understood.
- When you transmit, the message window shows your instruction in clean phraseology. For example, `dlh5ab taxi a via l2 s` becomes `Lufthansa 5AB, taxi to holding point A via L2, S` , unless parts of it were not understood; then your raw text is shown.
- **Up / Down** browse the history of your last 50 transmissions. **Esc** clears the line, or deselects if the line is already empty.
- The **×** button right of the text clears the line (it appears as soon as there is text; in mobile mode the keyboard stays open).
- Typing anywhere on the page focuses the command line.

7. AIRCRAFT MENU



▸ The aircraft menu after a right-click on a taxiing aircraft

Right-click an aircraft symbol, its tag, or its list row to open the menu. Items are disabled when they don't make sense in the current state.

ITEM	SENDS	AVAILABLE WHEN
Pushback approved -> facing	`pushback approved [facing east/west/...]`	Departure parked on a pushback stand
Push and start approved -> ...	`push and start approved [facing ...]`	same
Start-up approved	`start-up approved`	Parked
Taxi to -> holding point	`taxi to holding point A via ...` (shortest route, following the standard flows); with de-icing the pads come first (`De-icing DP2` , occupied pads greyed out) while the departure still needs de-icing	Departure after pushback, or on a taxi-out stand

ITEM	SENDS	AVAILABLE WHEN
Taxi to stand -> stand	`taxi to stand 14 via ...` (assigned stand first, then free stands nearby)	Arrival on the ground, or a departure returning to a stand
Stop pushback / Cancel pushback	`stop pushback` / `cancel pushback`	During pushback
Able intersection? -> intersection	`advise able for departure from intersection D`	Departure on the ground; the hint shows the answer once given
Hold position	`hold position`	Taxiing
Continue taxi	`continue taxi`	Taxiing
Hold short of -> taxiway	`hold short of taxiway D`	Taxiways the remaining route joins or crosses
Cross runway 25	`cross runway 25`	A runway holding point is ahead on the route
Give way to -> traffic	`give way to EWG7TK`	Other ground traffic (not parked) within 800 m
Resolve conflict with ... -> option	a re-route (`taxi to holding point A via ...`, another entry or a free stand nearby), a turn-around (`turn around, taxi to ...`) for one of the two, or `cancel pushback`	Two aircraft block each other (see below)
Follow the follow-me	`follow the follow-me` (to the allocated stand if the aircraft has no route yet)	Aircraft on the ground, not parked
Tow approved to stand 45 / Tow approved to stand -> stand	`tow approved` / `tow approved to stand 52`	A tug on a stand (tow request)
Contact Tower 118.805	`contact tower 118.805`	Departure taxiing or at the holding point
Number ... for pushback/taxi/departure	`number 2 for pushback`	A request is pending
Standby	`standby`	A request is pending
Say again	`say again`	always
Centre view	-	always
Reset tag position	-	always

Hovering over a destination in the *Taxi to* sub-menus draws the proposed route on the scope. The hint on the right shows the `via` list that will be transmitted, and `! head-on EWG7TK` when the A-SMGCS (CATC) sees a conflict. With the routing service off, the menu offers the destinations without a route. Menu actions are transmitted exactly like typed text, so they show up in the message window and get a read-back.

Resolve conflict: when two aircraft face each other on a taxiway, the menu (and `RESOLVE` in the mobile quick-action bar) offers the ways out a real Ground controller has: one aircraft turns off via another taxiway while the other waits (each option is checked to keep clear of the other aircraft's route) - if its own destination cannot be reached that way, to another entry of its runway (a departure, if it accepts the take-off run) or to a free stand nearby (an arrival) - or, last, one aircraft turns around (`turn around, taxi to ...`): an airliner needs a **tug** for it. Airliners cannot make a U-turn on a taxiway, so the tug takes 5 to 10 minutes (the pilot tells you the expected time). Hover an option to see its route.

Follow-me cars (right-click the car or its row in the VEHICLES list) have their own menu: *Proceed to DCEEO*, *Return to base*, *Hold position*, *Continue*, *Standby* - see the vehicle phraseology. In mobile mode the quick-action bar shows `PROCEED`, `BASE`, `HOLD`, `CONT`. **Tab** also selects vehicles with a pending request.

On Tower frequency (you staff Tower) the menu offers the Tower clearances: *Taxi to* (another holding point, for example the intersection B, while the departure is still taxiing), *Taxi to stand* (a vacated arrival you keep on your frequency), *Able intersection?* (`advise able for departure from intersection B`), *Line up and wait runway 25*, *Behind ... line up and wait* (behind a landing aircraft), *Cleared for take-off* (with the wind; the hint shows - updated live while the menu is open - whether the runway is occupied, how long the departure spacing still needs, or when the next arrival comes), *Cleared for immediate take-off*, *Backtrack runway 24, line up and wait* / *Backtrack runway 24, cleared for take-off* (from a holding point away from the runway end, e.g. Memmingen N; hint `requested` when the crew asked for it), *Speed* (180 kt until 6 NM, 170 kt until 5 NM, 160 / 150 kt until 4 NM, reduce to final approach speed; without an instruction arrivals fly 160 kt until 4 NM), *Cancel take-off* / *Stop immediately*, *Cleared to land*, *Continue approach*, *Continue, expect late landing clearance*, *Go around*, *Vacate via*, *Cross runway 25*, *Contact Ground* and *Contact Langen Radar* (for every aircraft in the air: departures and go-arounds). In mobile mode the quick-action bar shows `LUP`, `B/T`, `T/O`, `LAND`, `CONT`, `G/A`, `CROSS`, `GND`, `RDR`, `STOP`, `HOLD`, `TAXI` and `ROGER` while a call is open.

Request hand-off: for an aircraft that is still with a position the simulator runs (shown as *On GND frequency*), the menu (and `REQ H/O` in the quick-action bar) offers *Request hand-off to TWR* (or GND) when it would come to you next: a departure taxiing on Ground (to you as Tower), an arrival on final still with Approach, a vacated arrival still with Tower (to you as Ground), a cleared departure still with Delivery. The position hands it over after a few seconds of coordination (the item reads *Hand-off requested* meanwhile) and the crew calls you.

On Delivery frequency the menu offers instead:

ITEM	SENDS	AVAILABLE WHEN
Send DCL (datalink) -> SID	Datalink clearance (no voice), squawk shown in the hint	The crew requested its clearance by DCL
IFR clearance -> SID	`cleared to Frankfurt, ETASA4B departure, flight planned route, climb via SID to altitude 5000 feet, squawk 2101[, CTOT 1435]` . The SID of the flight plan's first fix is listed first; the squawk is the next free code	Departure on Delivery frequency (*Amend IFR clearance* once cleared)
Readback correct	`readback correct`	Cleared
Squawk -> ...	`negative, squawk 2101` (correct a wrong readback) or a new code	Cleared
Start-up approved	`start-up approved` (the hint shows the TSAT)	Cleared, still on the stand
CTOT 1435	`CTOT 1435`	The flight has a CTOT
Contact Ground 118.605	`contact ground 118.605`	Cleared

8. KEYBOARD AND MOUSE REFERENCE

INPUT	ACTION
Enter	Transmit the command line
Esc	Clear the command line / deselect / close menus
Tab	Select the next aircraft with a pending request, longest-waiting first
Up / Down	Command history
Space (command line empty)	Pause / resume
Home	Reset the scope view

INPUT	ACTION
F1	Help
F2	Settings
F3	Systems window
Hold the key left of `1` (`^` / `` ` ``), **Right Ctrl** or **Insert**	Push-to-talk (speech recognition)
Mouse wheel / pinch	Zoom
Drag (left or right button)	Pan; drag a tag to move it
Left click	Select aircraft (or deselect when clicking empty space); on a tag item: that item's function
Right click	Aircraft menu
Double click (list)	Centre on aircraft
Mobile mode: tap / tap again / long press	Select / menu (or tag item) / menu

9. VOICE

> **Switched off in this version.** Pilot voices (TTS) and speech recognition (MIC) are not reliable enough on all devices yet and are disabled for 1.0 (the `TTS` and `MIC` buttons and the voice settings are hidden). The description below applies when they come back.

Pilots speaking (TTS)

Press **TTS** in the toolbar. Every pilot transmission is read aloud with the browser's speech synthesis. Each callsign always gets the same voice, pitch and speaking rate, so you can tell pilots apart. Designators are spelled phonetically: "Lufthansa five Alpha Bravo, taxi to holding point Golf one".

On the **iPad** (and iPhone) TTS is off until you switch it on (in mobile mode the toolbar scrolls sideways - `TTS` is in it, or use the settings). Safari only lets a page speak after a tap: switching `TTS` on speaks a short confirmation (`Stuttgart Ground, pilot voices on`), and the first tap anywhere unlocks the voices too. **Settings -> Pilot voices -> Test** speaks a radio check. Only voices installed on the device are used (Apple's effect voices are left out). Speech follows the media volume and is muted while the iPad is in silent mode (Control Centre bell icon, or the side switch on older iPads) - if the test phrase stays silent, check those first.

You speaking (speech recognition)

Hold a push-to-talk key and speak your instruction: the key left of `1` (`^` on German, `` ` `` on US keyboards), **Right Ctrl** or **Insert**. On a touch screen **hold MIC** while you speak (release ends the transmission, like a push-to-talk key), or tap MIC once to start and tap again to stop. Safari (iPad) ends its recognition at every short pause; the simulator restarts it at once and keeps everything you said until you release or tap MIC again (at most 60 s). For example: *"Lufthansa five alpha bravo, taxi to holding point golf one via november golf"*. The recognised text appears in the command line while you speak and is transmitted when you release the key.

The goal is to work **by voice only**. Several things help with recognition errors:

- **Alternatives**: the recogniser returns up to five alternatives per phrase. The simulator parses all of them and transmits the one that makes the most sense: known callsign, understood instructions, valid route. If it picks an alternative, the message window shows what it heard (`Heard "..." - using the alternative "..."`).
- **Forgiving callsigns**: telephony designators are matched even when slightly misrecognised ("Lufthanza", "Euro wings", "speed bird"). If the flight number is off by one character, or left out and only one aircraft of that airline is on frequency, the right aircraft is still found.
- **Typical misrecognitions** are corrected: "push back" -> pushback, "gulf" -> golf, "run way" -> runway, "holding position" -> holding point, "twenty five" -> 25, "point" -> decimal in frequencies, "to"/"for" -> 2/4 after "runway", "stand" and "number". See phraseology.md.
- **No callsign heard and nothing selected**: the instruction goes to the pilot who called last and is still waiting for an answer.
- **Pilots stop talking** while your push-to-talk key is pressed (or MIC is on), so the microphone doesn't hear them; what they said meanwhile is spoken afterwards (the last three messages).
- **Sent or cleared is gone**: after SEND, Enter, `×` or Escape the text you spoke does not come back into the command line, even while the microphone is still on. SEND also switches the microphone off.
- Works in Chrome, Edge and Safari (Web Speech API, also on the iPad). The MIC button is disabled in browsers without it.
- If recognition still gets a word wrong, edit the text in the command line and press Enter.

10. YOUR JOB AS GROUND

Responsibilities in this simulator:

- **Departures** come to you with their IFR clearance and, with A-CDM, at their TSAT. If you don't staff Delivery, the simulator runs it: the crews already have their clearance and ask you for start-up with the pushback. Departures on drive-through stands (EDDS 40-48, 50-56) don't need a pushback; they call `request taxi` and leave forwards.
  1. Approve **pushback** (and start-up), choosing a facing direction that fits the traffic.
  2. When the pilot is **ready for taxi**, give a **taxi route to a runway holding point**. For runway 25 that is normally `A` (full length); `B`, `C` and `D` are intersection departures. For runway 07 it is `K` (or `Y` from the south), `I`, `H` and `W` are intersections.
  3. **Hand the aircraft over to Tower** (`contact tower 118.805`) when it is at or close to the holding point. Tower then lines it up and clears it for take-off on its own.
- **Arrivals** are landed by Tower. After vacating, they switch to you and call `vacated runway 25 via E`. **Taxi them to a stand.** The suggested stand is shown in brackets; you can give any free stand that is big enough.
- **Runway crossings**: some routes cross the runway, for example from the south apron or for arrivals that vacated to the south. Aircraft stop at the runway holding point and request crossing. Clear them with `cross runway 25` **only when the runway is free**: no arrival on short final, nobody lining up or rolling. Otherwise you cause a runway incursion and possibly a go-around.
- **Keep traffic flowing**: avoid head-on encounters on the same taxiway (the A-SMGCS warns you; see *Resolve conflict* in the aircraft menu), give way instructions at intersections, and don't push an aircraft onto a taxilane where another one is taxiing. Conditional clearances help here: `behind the A320 passing from left to right, push and start approved facing east`.
- **Busy hours**: give queue positions (`number 2 for pushback`) or expected delays (`expect pushback in 5 minutes`). The pilots then wait without reminding you.
- **Special events**: a crew may declare **PAN PAN (medical emergency)**. Arrivals call after vacating; departures call while taxiing and want to return to a stand. Give them a direct route to a free stand. If they are on the stand within 6 minutes of the call, you get +15 points. After a **rejected take-off**, the aircraft vacates the runway and calls you. It wants to go back to a stand or to taxi for another departure.

11. YOUR JOB AS DELIVERY

Clearance Delivery (EDDS: `Stuttgart Delivery` 121.915) gives departures their IFR clearance, a squawk and the start-up, then hands them to Ground. If you don't staff Ground, the simulator runs it (pushback, taxi, hand-off to Tower).

1. **Clearance request**: about 10 minutes before off-block (TOBT) the crew calls: `Stuttgart Delivery, Eurowings 51WM, A319, stand 45, information R, request clearance to Berlin`. Crews of datalink-equipped aircraft (about 4 in 10, airline flights only) request it silently: the list shows `DCL` in the REQ

column.

1. **IFR clearance**: clearance limit (the destination), SID, initial climb, squawk: `cleared to Berlin, OKIBA4B departure, flight planned route, climb via SID to altitude 5000 feet, squawk 2101`. Pick the SID of the **runway in use** that leads to the **first fix of the flight plan** (the aircraft menu lists it first; flight plans go via the exit fix towards the destination, see EDDS SIDs). If something is missing or wrong, the crew queries it (`confirm OKIBA4H departure, information R says runway 25 in use`). For a DCL request use *Send DCL* in the aircraft menu: no voice, no readback, the crew answers WILCO.
2. **Readback**: listen to it. About 4 % of crews read back a wrong squawk (two digits swapped). Correct it with `negative, squawk 2101`; otherwise say `readback correct`. A wrong readback that you let pass costs 10 points, a caught one gives 5.
3. **CTOT**: flights with an ATFM slot show it in the CTOT column. Include it in the clearance (`..., CTOT 1435`) or tell the crew separately (`CTOT 1435`). The aircraft must take off between CTOT -5 and +10 minutes; Tower holds it at the holding point until it fits. A missed slot gives a new CTOT 20 to 40 minutes later (8 to 15 minutes if the flight is ready at the holding point) and -10 points.
4. **Start-up**: with A-CDM the crew calls at its TSAT (`ready for start-up, TSAT 1452`): `start-up approved`. Without A-CDM it calls when ready. Start-up needs the clearance first.
5. **Hand-off**: the crew asks for the frequency for pushback: `contact ground 118.605`.

With **combined positions** (Delivery + Ground, Ground + Tower, or all three) you do all the jobs; the aircraft still change frequency as on split positions: Delivery -> Ground (`contact ground`), Ground -> Tower at the holding point (`contact tower`), Tower -> Ground after vacating (`contact ground`; the crew then calls you on Ground for the taxi to the stand). Runway crossings you may clear on either frequency - you are Tower too, so no coordination is needed.

12. YOUR JOB AS TOWER

Stuttgart Tower (118.805) owns runway 07/25. When you staff Tower, Ground and Delivery are run by the simulator, Approach (Langen Radar) brings the arrivals onto final.

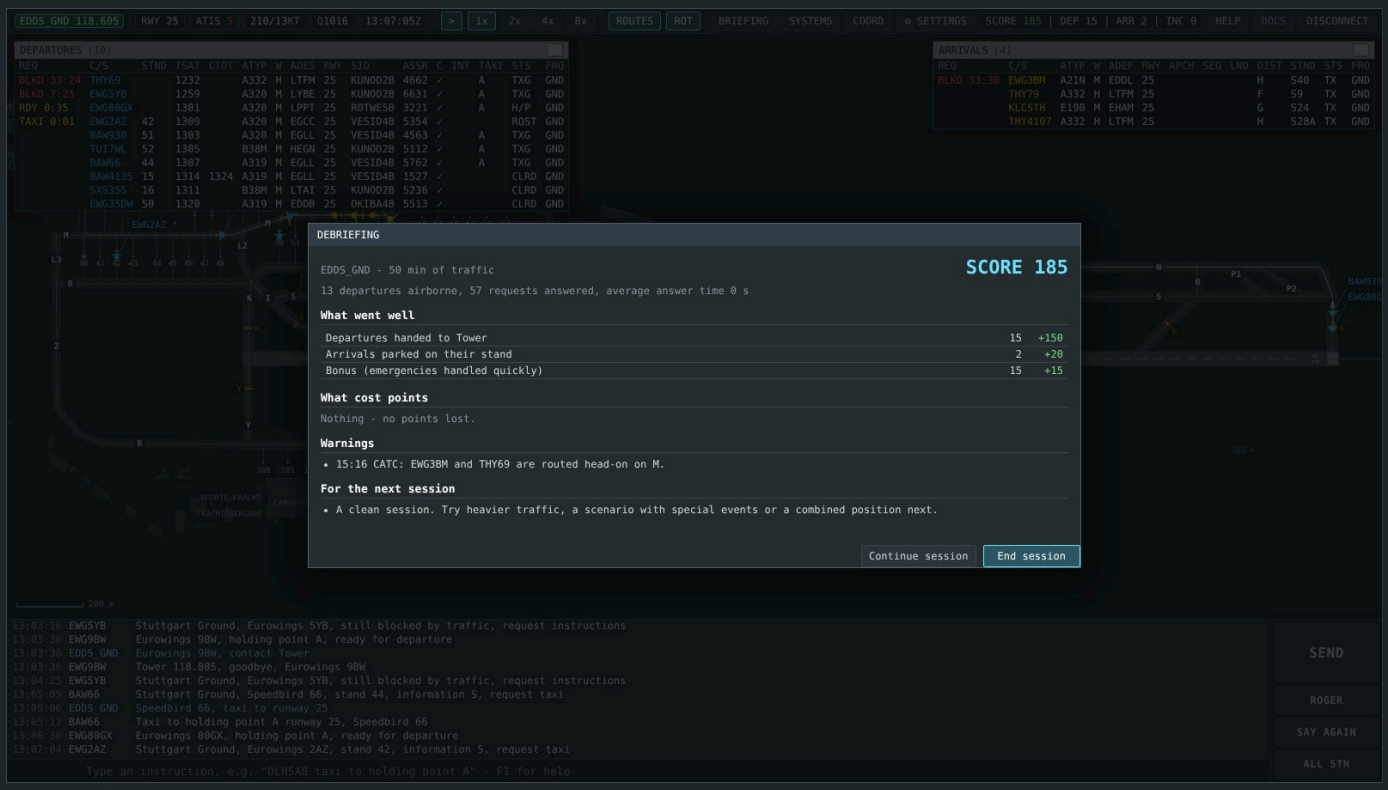
1. **Departures** reach the holding point with the AI Ground and call: `Stuttgart Tower, Lufthansa 5AB, holding point A, ready for departure`. Give `line up and wait runway 25` (it lines up and waits) or straight away `wind 250 degrees 8 knots, runway 25, cleared for take-off`. Line up behind landing traffic with `behind landing EWG7TK, line up and wait behind`. A crew with a take-off clearance does not roll while someone is on the runway in front of it.
2. **Spacing**: 2 minutes behind a heavy (3 from an intersection), 2 minutes behind a departure on the same SID fix, 1 minute on diverging SIDs - counted from the previous departure being airborne. Taking off earlier counts as a separation loss (-10). The take-off menu shows how long you still have to wait. CTOT flights may only go from CTOT -5 minutes: the take-off menu warns you, but a crew you clear earlier takes off and it counts as a slot violation (-10).
3. **After take-off** the crew asks for a frequency change: `contact radar 125.050` (+10) - at Stuttgart the southern departures (ROTWE, SUL, KUNOD, ABTAL) go to 119.200; the take-off menu offers the right frequency. If it leaves the control zone (4000 ft above the airport) still on your frequency, it costs 5 points.
4. **Arrivals** call on final: `Stuttgart Tower, Eurowings 7TK, ILS approach runway 25` (or LOC / RNP if the ILS is switched off). Clear them in time: `wind 250 degrees 8 knots, runway 25, cleared to land`. Without a landing clearance the crew goes around at 0.5 NM (an incident, -15); `continue approach` tells it the clearance comes later; `go around` sends it around (-5). `vacate via E` asks for a particular exit.
5. **After landing** the crew reports `runway 25 vacated via E`: `contact ground 118.605` (+10). Until then it waits at the vacate point - and Tower cannot use an exit whose vacate point is occupied. You can also hand it over earlier, right after touchdown (*Contact Ground* in the menu, `GND` in the quick-action bar): it then calls Ground itself once it has vacated.
6. **Runway crossings**: Ground hands aircraft that must cross the runway over at the runway holding point (`holding short runway 25 at W, request crossing`). `cross runway 25` only when nobody is landing or taking off; after crossing the crew reports the runway vacated, then `contact ground`.

The A-SMGCS runway monitoring (RMCA) warns when you give a take-off or landing clearance while the runway is occupied.

13. SCORE

EVENT	POINTS
Departure handed over to Tower	+10
Arrival parked on a stand	+10
IFR clearance delivered (departure handed from Delivery to Ground)	+10
Tow brought to its stand	+5
Tower: departure handed to Radar after take-off	+10
Tower: arrival handed to Ground after vacating	+10
Tower: departure left the control zone on your frequency	-5
Tower: take-off with too little departure spacing	-10
Tower: go-around you instructed	-5
Per minute Approach holds all arrivals / Ground holds all departures for you (`COORD`)	-2
Wrong readback caught and corrected	+5
Wrong readback not caught	-10
CTOT missed or violated (no take-off inside -5 / +10 minutes)	-10
Night rules: jet ready in time that missed the curfew	-10
Medical emergency on a stand within 6 minutes of the call	+15
"Say again" (pilot did not understand)	-2
Crew left your frequency with an old ATIS letter	-3
Slow answer: per 15 s of waiting beyond the first 30 s	-1
Go-around caused by an occupied runway (or, as Tower, a missing landing clearance)	-15
Runway incursion	-50
Collision	-100

Debriefing



▸ The debriefing after a session

`DISCONNECT` opens the debriefing of your session before you leave:

- your positions, the time worked, the score, departures airborne, requests answered, the **average answer time** and the five longest waits,
- **What went well**: everything that earned points (count and points), largest first,
- **What cost points**: everything that lost points, largest loss first,
- the **incidents** (collisions, incursions, go-arounds) and the last system **warnings** (RMCA/CATC alerts, missed CTOTs, separation, missed hand-offs) with the time since the session start,
- **For the next session**: advice for what cost the most (for example answering sooner with `standby` or queue numbers, waiting for the departure spacing, clearing arrivals to land in time).

Continue session goes back to work (the simulation resumes), *End session* returns to the Connect dialog.

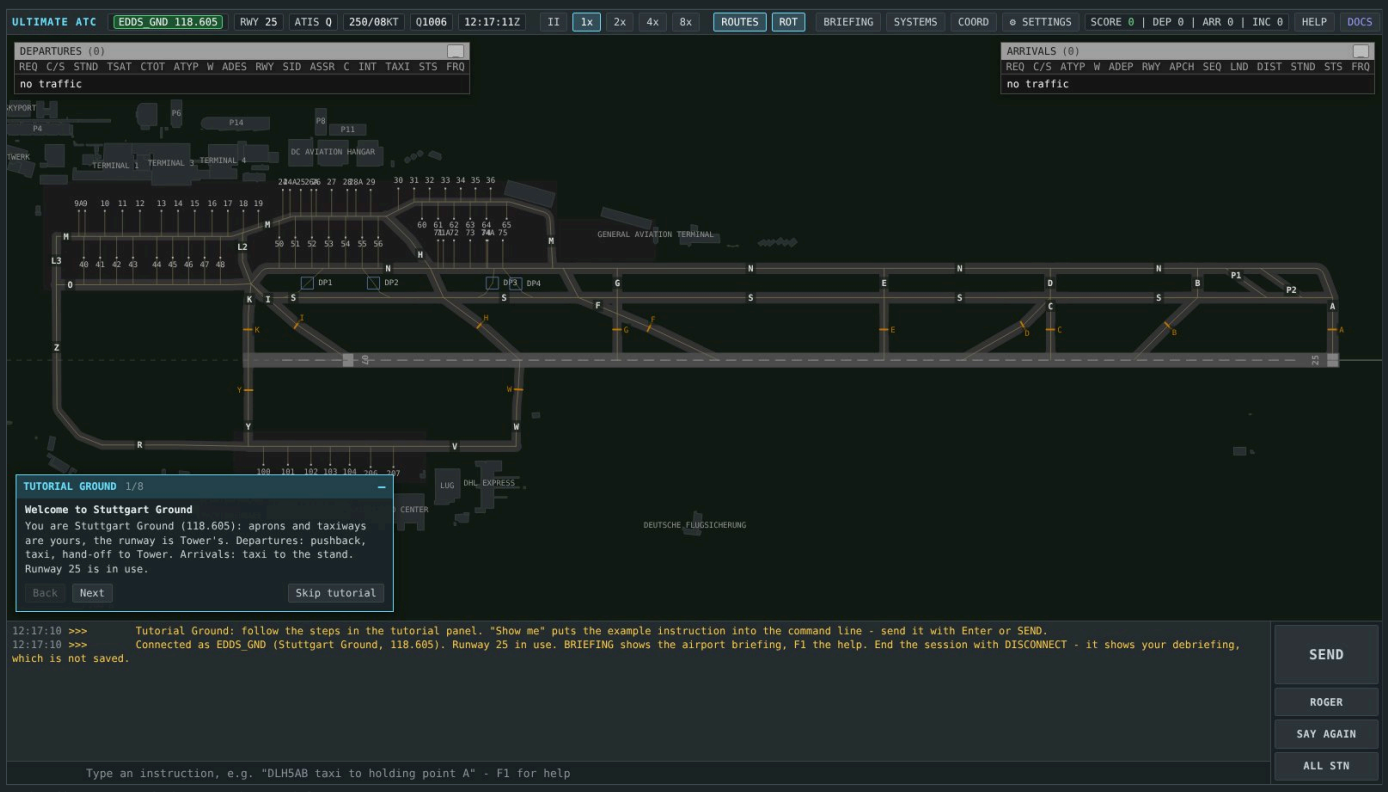
14. TIPS

- Use **Tab** to work through requests in order. Then you only need to type the instruction, without the callsign.
- Hover over destinations in the **Taxi to** menu to compare routes before you send one. The suggestions (also those of the quick-action bar in mobile mode) follow the airport's **standard taxi flows** (at EDDS: N eastbound, S westbound), see the airport page and the airport briefing.
- Plan **pushback direction** with the departure runway in mind. For runway 25 the holding point A is at the east end, so departures from the terminal usually push facing **east** and taxi via L2 or H onto S (the standard flow for 25). If you don't specify, the pilot picks the direction with the shortest taxi.
- Use `standby` if you can't answer right away. The pilot then waits two minutes before calling again, instead of reminding you every minute.
- If two aircraft meet **head-on** on a taxiway, neither can pass. Crews who see the other aircraft in time stop short of the junction between them and ask (`opposite traffic on taxiway N ... holding short of G`): *Resolve conflict* then offers a turn-off at that junction. Prevent it by holding one at an intersection; the orange CATC warning in the preview tells you before you transmit. If it happens anyway, use *Resolve conflict*: turn one off via a junction, or - last resort - a tug (5 to 10 minutes).
- Run at **2x or 4x** during quiet phases and pause (**Space**) when it gets busy.

3 TUTORIALS

A tutorial is a guided first session for one position: a fixed, calm traffic situation, explained step by step. Start one in the **Connect** dialog with the buttons next to **Tutorial** (*Delivery*, *Ground*, *Tower*).

HOW A TUTORIAL WORKS



▸ The first step of the Ground tutorial

- The session is always the same: EDDS, **runway 25**, wind **250/08**, no random traffic, no special events, no random readback errors. The positions you don't staff are run by the simulator, as in a normal session.
- The **tutorial panel** (bottom left of the scope; in mobile mode above the quick-action bar) shows the current step: what to do and why, and the **example instruction**.
  - **Show me** selects the aircraft of the step, centres the scope on it and puts the example into the command line. Send it with **Enter** or **SEND** - or use the aircraft menu or the quick-action bar instead.
  - Steps with a task continue on their own as soon as it is done (for example when the pushback has started). **Next** continues after an explanation, or skips the task of the current step.
  - **Back** shows the previous step again (marked *review* in the title bar). A step you go back to is only shown: its traffic is not started again and it does not continue by itself - **Next** brings you forward again, up to the step you were working on.
  - **Skip tutorial** ends the tutorial at once and starts live traffic (see below).
  - Tap the panel's title bar to fold it away and see the whole airport.
  - When the tutorial is finished or skipped, the panel disappears.
- Waiting for an aircraft? **2x** / **4x** in the toolbar speed the simulation up.
- Score, debriefing (DISCONNECT) and all menus work as usual.

DELIVERY

Position: Stuttgart Delivery 121.915 (Ground and Tower by the simulator). About 5 minutes.

STEP	YOU LEARN
Welcome	What Delivery does
A clearance request	Selecting an aircraft (DLH5AB, stand 14)
IFR clearance	`cleared to Athens, KUNOD2B departure, flight planned route, climb via SID to altitude 5000 feet, squawk 2101` - destination, SID by the first fix of the route (towards the destination), initial climb, free squawk
Check the readback	Listen to the squawk; `readback correct`
Start-up	`start-up approved` at the TSAT (A-CDM)
Hand-off to Ground	`contact ground 118.605`
Catch a readback error	EWG7TK reads back a wrong squawk (always, in this step)
Correct it	`negative, squawk 2102`
Start-up and Ground	The routine on your own
Datalink clearance	CFG53 requests its clearance by DCL: aircraft menu *Send DCL (datalink)*
Well done	CTOTs and what else to practise

GROUND

Position: Stuttgart Ground 118.605 (Delivery and Tower by the simulator). About 7 minutes.

STEP	YOU LEARN
Welcome	What Ground does, the scope and the lists
A departure calls	Selecting an aircraft (DLH5AB, stand 14)
Pushback	`pushback approved, facing east`
Taxi to the holding point	`taxi to holding point A via M, H, N`, the live route preview, the standard flows
An arrival is landing	EWG7TK lands and calls you after vacating
Taxi in	`taxi to stand 32` - automatic route without "via"
Hand-off to Tower	`contact tower 118.805` at the holding point
Well done	Hold short, give way, conditional clearances, standby, follow-me, CATC

TOWER

Position: Stuttgart Tower 118.805 (Delivery and Ground by the simulator). About 12 minutes (the departure needs a while to taxi out; use 2x / 4x).

STEP	YOU LEARN
Welcome	What Tower does
An arrival on final	Selecting an aircraft (EWG7TK); airborne tags
Landing clearance	`wind 250 degrees 8 knots, runway 25, cleared to land`
Vacated: to Ground	`contact ground 118.605`
Line up	`line up and wait runway 25` (DLH5AB, handed over by Ground)
Take-off	`wind 250 degrees 8 knots, runway 25, cleared for take-off`; departure spacing
Hand-off to Radar	`contact radar 125.050` (the frequency of the departure's SID sector)
Well done	Conditional line-up, immediate take-off, go around, vacate via, speed control, crossings, COORD

AFTER THE TUTORIAL

The last step offers **Live traffic**: from then on new departures and arrivals are generated as in a normal session (light density), so you can practise right away; **Close** ends the tutorial without new traffic. Either way the panel disappears. End the session with **DISCONNECT** for the debriefing. Next: a scenario such as *First steps*.

4 PHRASEOLOGY REFERENCE

Ultimate ATC understands controller instructions written or spoken in **ICAO phraseology** (ICAO Doc 4444 / Doc 9432 style, as used on VATSIM) - and at US airports (San Diego) the **FAA** forms as well, see FAA phraseology. This page lists everything the parser understands, what the pilots read back, and what the pilots say on their own.

The parser lives in `src/core/phraseology/parser.ts`. Its behaviour is covered by `tests/parser.test.ts`.

GENERAL RULES

- **Case and punctuation don't matter.** `DLH5AB Taxi To Holding Point A, via L2, S.` and `dlh5ab taxi to holding point a via l2 s` are the same.
- **Filler words are ignored:** `roger`, `please`, `thanks`, `good day`, `bye`, `correction`, `the`, `and` ...
- **Words that are not understood** are listed in the preview (`ignored: ...`). If nothing at all is understood, the pilot replies `Say again, <callsign>?` (-2 points).
- **Designators** can be typed (`A`, `L2`, `25`, `118.805`) or spelled in words (`alpha`, `lima two`, `two five`, `one one eight decimal eight zero five`). Taxilanes with a dot (Leipzig `Z4.1`) are typed `z4.1` or said `zulu 4 point 1`.

CALLSIGNS

The callsign can be written in any of these ways:

FORM	EXAMPLE
ICAO callsign	`DLH5AB`, `THY1734`, `DCMGB`
Radiotelephony designator + flight number	`Lufthansa 5AB`, `Turkish 1734`, `Wizz Air 12AB`
Spelled out	`Lufthansa five alpha bravo`
Flight number only, if unique	`5AB`, `1734`
At the end of the transmission	`pushback approved, DLH5AB`
Omitted: the selected aircraft is used	`taxi to stand 14 via N, R`

Radiotelephony designators of the simulated operators:

ICAO	TELEPHONY	ICAO	TELEPHONY	ICAO	TELEPHONY
AFR	Airfrans	DLH	Lufthansa	RYR	Ryanair
AUA	Austrian	EWG	Eurowings	SWR	Swiss
BAW	Speedbird	ITY	Itarrow	SXS	Sunexpress
CFG	Condor	KLC	City	THY	Turkish
CLH	Hansaline	KLM	KLM	TUI	Tuijet
PGT	Sunturk	VLG	Vueling	WZZ	Wizz Air

US operators (San Diego):

ICAO	TELEPHONY	ICAO	TELEPHONY	ICAO	TELEPHONY
AAL	American	HAL	Hawaiian	SKW	Skywest
ACA	Air Canada	JAL	Japanair	SWA	Southwest
ASA	Alaska	JBU	Jetblue	UAL	United
DAL	Delta	MXY	Moxy	UPS	UPS
FDX	Fedex	NKS	Spirit Wings	WJA	Westjet
FFT	Frontier Flight	QXE	Horizon	SCX	Sun Country

General aviation aircraft use their registration as callsign, for example `DCMGB`, spoken "Delta Charlie Mike Golf Bravo"; in the US an N-number such as `N436GR`, spoken "November four three six Golf Romeo".

INSTRUCTIONS

Notation: `[optional]`, `a | b` = alternatives, `X` = taxiway, `HP` = holding point, `RWY` = runway designator, `STAND` = stand number.

Pushback and start-up

YOU SAY	EFFECT
`pushback approved` / `push back approved` / `push approved`	Pushback. The pilot chooses the facing that gives the shortest taxi to the departure runway.
`pushback approved facing east` (`face west`, `nose north`)	Pushback so the nose points in that direction afterwards
`pushback approved tail east`	Same with the tail direction (tail east = facing west)
`push and start approved [facing ...]`	Pushback with engine start during the push. The aircraft is ready for taxi sooner.
	While a thunderstorm is overhead, ground handling has stopped: `unable, ground handling has stopped, lightning alert` (also for taxi from a drive-through stand).
`start-up approved` / `startup approved`	Engine start on the stand. Combine with a later pushback; it shortens the start-up after the push.
`cancel pushback` / `pushback cancelled`	Before the tug moves: the pushback is cancelled, the aircraft stays on the stand. Already moving: the tug **tows the aircraft back onto the stand**. Either way it calls again in 1-2.5 minutes.
`stop pushback` / `stop the push`	The push stops immediately (read-back `stopping pushback`)
`continue pushback` / `continue push`	Resumes a stopped pushback

Notes:

- Facing is matched to the nearest compass direction of the taxilane: at EDDS terminal stands (taxilane M), `east` means towards L2 / H and `west` towards L3.
- A pilot who hasn't called yet and is still boarding replies `negative, we are still boarding, we will call you when ready`.
- Stands without pushback (none at EDDS yet) reply `no pushback required ...`.

Taxi

```
taxi [to] DESTINATION [via|along X [,] X ...] [hold short of ...] [cross runway RWY]
taxi via X X ... to DESTINATION
taxi via|along X [,] X ..., hold short of TAXIWAY|runway RWY        (incomplete taxi instruction)
```

DESTINATION	EXAMPLE
Holding point	`taxi to holding point A via L2, S`
Holding point with runway	`taxi to holding point D runway 25 via H, S`
Runway (the pilot picks the full-length holding point on the shortest route)	`taxi to runway 25 via N`
Runway at an intersection	`taxi to runway 25 at D`
Stand	`taxi to stand 14 via N, R` (also `gate 14`, `parking position 14`; lettered stands `stand G1` / `stand golf 1`)
De-icing pad (scenario option de-icing)	`taxi to de-icing pad DP2 via N` (also `de-icing position 2`, `deicing pad delta papa 2`; `taxi to de-icing via N` = the free pad nearest to the aircraft)

The `via` list must name the taxiways in the order they are used. The router finds the shortest path that follows exactly these taxiways. Taxiways that controllers usually leave out may be omitted:

- the apron taxilane the aircraft starts on (for example `R` after pushback at the terminal),
- the taxiway the aircraft is currently on,
- the taxilane or taxiway the destination lies on (for example `M` for a terminal stand, `A` for holding point A).

Coloured guide lines for parallel taxiing are taxiways of their own: at Cologne/Bonn `M BLUE` / `M ORANGE` and `N BLUE` / `N ORANGE` next to the yellow centre lines M and N. Say the colour after the letter: `taxi to runway 13L via november blue, mike blue` or `via N BLUE, M BLUE`; `hold short of mike orange` works the same way.

So after a pushback from stand 14 facing east, `taxi to holding point A via L2, S` is accepted and gives the route M -> L2 -> S -> A. Without any `via`, the pilot takes the **shortest route**, avoiding runway crossings where possible. This is convenient, but not proper phraseology.

Left-out taxiways: crews are not pedantic. If the via list leaves out a connecting taxiway (`taxi to holding point A via S` from stand 14, where L2 links M and S), the crew takes the obvious connection - at most two added taxiways and 600 m between the ones you named; after the last one it follows one taxiway to the destination - and reads back the whole route: `Taxi to holding point A via M, L2, S, Lufthansa 5AB`. Check the read-back.

If the route is impossible, the pilot replies `unable to follow route via S, say again route`. An unknown taxiway gives `confirm taxiway Q, we can't find it`. A taxiway closed by the scenario (shown red and hatched, in the ATIS) gives `unable, taxiway S closed` - also for a holding point or `vacate via` on it; see closed taxiways. The live preview in the command line shows these problems **before** you transmit.

Incomplete taxi instructions (clearance limit): `taxi via S, hold short of E` or `taxi along R, D, N, hold short of runway 25`. There is no destination; the aircraft taxis along the via taxiways and stops at the hold-short point, which is the clearance limit: at the intermediate holding position before the junction with E where the charts have one, otherwise with the nose 40 m from the centre line of E; or at the runway holding point. The read-back is `Taxi via S, hold short of taxiway E, ...`. Then give the rest of the route with a normal taxi instruction. `continue taxi` at a clearance limit gets `confirm where to taxi`.

No turning around on the spot: an airliner (wingspan above 25 m) cannot make a 180 degree turn on a taxiway. If the route you give would need one, for example `taxi to holding point A via N` for an aircraft facing west on N when A is to the east, the pilot replies `unable, we are facing west and cannot turn around here, say again route`. Give a route that continues in the direction the aircraft is facing, or loops around. Smaller aircraft (business jets, CRJ900) can turn around. Exceptions:

- at a runway holding point (there is room to turn), and
- when the aircraft has been **stuck** for over 30 seconds (blocked by other traffic): a route that needs a turn is accepted, the pilot reads it back with `we need a tug to turn around, expect about 7 minutes`, and starts taxiing 5-10 minutes later. Use this to solve a deadlock when there is no junction left to turn off (the *Resolve conflict* menu offers it).

Turning around on purpose: `turn around, taxi to stand 208` (or `taxi to stand 208, turn around`) - the route starts the other way round. An airliner needs a tug for it (`... we need a tug to turn around, expect about 7 minutes`), a smaller aircraft turns on the spot. *Resolve conflict* uses it when an aircraft can only get out of a conflict by turning around; without it the crew would take a route forwards that leads back into the conflict.

Intersection departures: ask first, as in real operations:

YOU SAY	PILOT REPLIES
`advise able for departure from intersection D` / `are you able intersection D` / `confirm able to depart from D`	`Affirm, able intersection D, ...` or `Negative, we require full length, ...`

The crew compares the take-off run available from that intersection (the published value where the airport data has one, e.g. Stuttgart 25 from C 2175 m) with what their aircraft needs today (a figure per type, e.g. A320 about 2100 m, A330 about 2800 m, Citation about 1200 m, varied per flight by -10 % / +15 %). An aircraft that is not able replies `unable intersection D, we require full length` when you send it to that holding point anyway. Asking for a holding point that is not an entry to the runway in use gets `confirm intersection X, it is not an entry to runway 25`. The aircraft menu has *Able intersection?* for the intersections of the runway in use.

Stand assignments: the crew knows from its stand charts whether its aircraft fits (`unable, stand 22 is too small for us`), but not whether the stand is free - it accepts the instruction and only finds out when it sees the stand. About 150 m before it, it stops and calls: `Stuttgart Ground, Eurowings 7TK, stand 14 is occupied, request another stand` (or `stand 72 is blocked, not enough wingtip clearance to the A332 on stand 71A, request another stand`). An aircraft just pushing back from the stand (or taxiing onto the one next door) is waited for without a call. You can see it before: the preview warns in orange (`-- stand 14 taken by DLH5AB`).

During pushback or start-up a taxi instruction is accepted and read back. The aircraft starts taxiing once its engines are running.

Hold short, hold, continue

YOU SAY	EFFECT
`hold short of taxiway D` / `hold short D`	Stops before the route joins or crosses taxiway D: at the intermediate holding position of the charts (the dashed line before the junction) where there is one, otherwise with the nose 40 m from the centre line of D - clear of traffic on D
`hold short of runway 25`	Cancels a crossing clearance; the aircraft stops at the runway holding point
`hold position` / `hold` / `stop`	Stops immediately
`continue taxi` / `continue`	Cancels *hold position*, *give way* and the current or next *hold short of taxiway*
`expedite taxi`	Taxis about 5 kt faster

Hold short can be part of the taxi instruction (`taxi to holding point A via L2, S, hold short of taxiway H`) or sent on its own to an aircraft that is already taxiing. `continue taxi` does **not** clear an aircraft across a runway. The pilot then asks `confirm cleared to cross runway 07/25`.

Runway crossing

YOU SAY	EFFECT
`cross runway 25`	Clears the aircraft to cross runway 07/25 at the holding point it is waiting at, or at the next one on its route
`taxi ... , cross runway 25`	Crossing clearance as part of the taxi instruction

Every runway holding position on a route is a **mandatory stop** unless a crossing clearance was given. Aircraft never enter the runway on their own. Read the simulation model to see when a crossing counts as an incursion.

Conditional clearances

```
behind TRAFFIC [description], INSTRUCTION
when clear of TRAFFIC [description], INSTRUCTION
after TRAFFIC [has passed], INSTRUCTION
```

`TRAFFIC` is a callsign in any form (`behind DLH5AB`, `behind Lufthansa 5AB`) or `the` + an aircraft type: `the A320`, `the A321`, `the 737`, `the Boeing`, `the Airbus`, `the Embraer`, `the Dash`, `the ATR`, `the jet`, `the heavy`. The description after the traffic ("passing from left to right", "on M", "coming out of L2") is read back but not interpreted.

```
DLH5AB, behind the A320 passing from left to right, push and start approved, facing east
EWG7TK, when clear of the Boeing, taxi to holding point A via L2, S
```

- The pilot has to **identify the traffic**: a callsign must be on the ground; a type must match a moving (not parked) aircraft within 1500 m, and the nearest one is taken. If not, the reply is `Negative contact with the A320, say again` and nothing is executed.
- Pushbacks, taxi instructions, crossings and *continue* then wait until the traffic **has passed**: it came closest and is now at least 40 m further away again, clear of the wingtips. They also start if the traffic stops more than 300 m away, or after at most 4 minutes.
- The read-back starts with the condition: `Behind the A320 passing from left to right, push and start approved, facing east, Lufthansa 5AB`.

Give way / follow

YOU SAY	EFFECT
`give way to EWG7TK [from the left]`	The aircraft stops until the named traffic has passed (see conditional clearances), then continues on its own
`follow EWG7TK`	Currently handled like *give way*: wait until the traffic has passed, then continue

Words after the callsign ("from the left", "passing left to right") are ignored. The callsign of the traffic can be in any form listed under Callsigns.

Follow-me and tows

YOU SAY	EFFECT
`follow the follow-me` / `follow follow-me 1`	A follow-me car drives from its base to the aircraft and leads it along its route (ICAO: FOLLOW (description of vehicle)). Without a route of its own (just vacated) the aircraft is led to its allocated stand.
`follow the follow-me to stand 14 [via N, L2]` / `taxi to stand 14 via N, follow the follow-me`	The same with a destination
`Tug 5, tow approved [to stand 45] [via M, N]`	Approves a tow (ICAO PANS-ATM: TOW APPROVED VIA (routing)). Without a destination the tug tows to the stand it asked for. A tug on a stand that needs a pushback pushes the aircraft off first.
`Tug 5, hold position` / `continue` / `hold short of ...` / `cross runway 25` / `give way to ...`	As for aircraft

Tugs are addressed by their callsign: `Tug 5` (TUG5). They refuse `pushback approved` (`we are a tow, request tow approval`) and hand-offs (`we stay on your frequency until the tow is complete`).

Vehicles

Vehicles on the manoeuvring area are controlled like aircraft, with one difference in wording: vehicles get **"proceed"**, aircraft get "taxi" (vehicles are never "cleared"). Callsigns: `Follow-me 1` / `Follow-me 2` (FME1, FME2) and `Tug 1` ... `Tug 9` (TUG1 ...).

YOU SAY	EFFECT	READ-BACK
`Follow-me 1, proceed [to DCEEO] [via N, F]`	The follow-me drives to the aircraft it is assigned to (along the taxiways you give, otherwise the shortest way)	`Proceeding to DCEEO via N, F, Follow-me 1`
`Follow-me 1, proceed to base [via ...]` / `return to base` / `return to the fire station`	Drives back to its base (job finished, or cancelled)	`Proceeding to base, Follow-me 1` / `Returning to base, Follow-me 1`
`Follow-me 1, report position` / `say position`	Where the car is and what it does	`On taxiway N, leading DCEEO, Follow-me 1` / `At the base, Follow-me 1`
`Follow-me 1, hold position`	Stops (while leading: the aircraft behind it stops too)	`Holding position, Follow-me 1`
`Follow-me 1, continue`	Drives on	`Continuing, Follow-me 1`
`Follow-me 1, standby`	Acknowledges the request; the driver waits a little longer before asking again	-
`Tug 5, proceed [to stand 45] [via M, N]`	Same as `tow approved`	`Tow approved to stand 45 via M, N, Tug 5`

Calls from vehicles:

SITUATION	EXAMPLE
Follow-me assigned to an aircraft (after `follow the follow-me`)	`Stuttgart Ground, Follow-me 1, request proceed to DCEEO at taxiway F`
Follow-me has brought the aircraft to its stand	`Stuttgart Ground, Follow-me 1, DCEEO is at the stand, request return to base`
Tow request	`Stuttgart Ground, Tug 5, request tow Eurowings A320 from stand 14 to stand 45`

A follow-me only drives onto the taxiways after your `proceed`; the aircraft waits until the car is in front of it. Unanswered vehicles call again like pilots (every 60-90 s, up to five times). When the AI runs Ground (you staff Delivery only), vehicles do not call: their requests are approved automatically.

Tower

YOU SAY	EFFECT
`line up and wait [runway 25]`	Departure at (or taxiing to) the holding point lines up and waits. It is not a take-off clearance.

YOU SAY	EFFECT
`behind landing EWG7TK, line up and wait behind` / `behind the landing A320, line up and wait behind`	Conditional line-up: the departure enters the runway once the landing aircraft has passed.
`[wind 250 degrees 8 knots [gusting 25 knots],] [runway 25,] cleared for take-off`	Take-off clearance. From the holding point the aircraft lines up and rolls without stopping. The wind is information only. In a thunderstorm overhead the crew refuses (`unable, thunderstorm overhead with windshear, we will wait until it has passed`); a jet in the night curfew refuses take-off and line-up (`unable, night curfew, jet departures are not permitted until 0600 local`).
`hold position, cancel take-off [, I say again, cancel take-off]` / `stop immediately`	Cancels a take-off clearance; during the roll the crew stops below 80 kt (`unable, we are taking off` above).
`[wind ...,] [runway 25,] cleared to land`	Landing clearance (required, otherwise the crew goes around at 0.5 NM)
`continue approach` (also just `continue` to an aircraft on final)	The landing clearance comes later
`continue approach, expect late landing clearance` (also `expect late landing clearance` alone)	The landing clearance will come late, for example because a departure is still rolling or an arrival is vacating. Common practice at European airports; it is **not** a landing clearance - the crew reads back `continue approach, expect late landing clearance` and still needs `cleared to land`
`maintain 160 knots until 4 miles` / `reduce speed to 150 knots` / `speed 170 knots`	Speed control on final (default until 4 NM, then the crew slows to its approach speed). Without it, arrivals fly 170-190 kt on the intercept and 160 kt until 4 NM anyway. Below the approach speed: `unable, our minimum speed is 135 knots`; above 210 kt or inside the given distance: unable; too fast to slow down by 2.5 NM: `unable 180 knots until 4 miles, we could not be stabilised` (180 kt until 6 NM works)
`reduce to final approach speed`	The crew slows to its final approach speed at once (to open a gap to the next arrival)
`resume normal speed`	Cancels a speed instruction: back to the normal speeds on final (160 kt until 4 NM)
`cleared for immediate take-off` / `cleared for take-off, no delay`	Take-off into a tight gap: brisk line-up and roll without waiting
`go around [, I say again, go around]`	The arrival goes around
`vacate via E`	The exit to take after landing (used if it can still be reached; an exit marked for backtracking behind the aircraft is reached by turning around on the runway)
`backtrack runway 24, line up and wait` / `backtrack runway 24` / `enter runway 24 and backtrack` / `backtrack runway 24, cleared for take-off`	**Backtrack**: the departure enters the runway at an intersection, taxis along it against the take-off direction to the runway end, turns around on the runway and lines up for a full-length departure. The runway is occupied all the time. From a holding point at the runway end: `unable to backtrack from W, we are at the runway end`. Where the airport uses an entry with a backtrack (Memmingen N for 24) the crew asks for it (`holding point N, ready for departure, request backtrack`) and refuses a plain line-up if the runway ahead is too short (`unable, we require a backtrack for departure, 1009 metres available from N`)
`backtrack, vacate via N`	Arrival: after landing turn around on the runway and backtrack to N (the exit is behind it)
`contact radar [125.050]` / `contact Langen Radar 119.200` / `contact departure`	Departure to Radar after take-off (`confirm contact Radar, we are not airborne yet` on the ground). Every radar frequency of the airport is accepted (Stuttgart: 125.050, and 119.200 for the southern SIDs); without a frequency the crew reads back the one of its SID
`contact ground [118.605]`	Arrival to Ground - from touchdown on (it calls Ground once vacated) - or a crossing aircraft once it has vacated
`cross runway 25`	Crossing for an aircraft Ground handed over at the runway holding point

A wrong runway is queried: `confirm runway 07, we are departing runway 25`. A departure with a CTOT that is not due yet still takes off when cleared (as a real crew would; slot compliance is the controller's job): the take-off counts as a slot violation. The take-off menu warns `CTOT: not before 1430!`.

Clearance delivery

YOU SAY	EFFECT
`cleared to Frankfurt, ETASA4B departure, flight planned route, climb via SID to altitude 5000 feet, squawk 2312`	IFR clearance (German form; the short form `cleared to Frankfurt via ETASA4B departure, climb 5000 feet, squawk 2312` works too). Clearance limit: destination name (`Frankfurt`, `Paris Charles de Gaulle`) or ICAO code (`EDDF`). SID: designator, also spelled (`echo tango alpha sierra alpha four bravo`). Optional: `runway 25`, `CTOT 1435`. Altitude: `5000 feet`, `five thousand feet`, `flight level 70`, also after `maintain`. Also `..., ETASA4B departure, ...` without `via`.
`readback correct` / `read back correct`	Confirms a readback
`squawk 2312` / `negative, squawk 2312`	Assigns (or corrects) a squawk code (octal digits 0-7)
`CTOT 1435` / `slot 1435`	Tells the crew its calculated take-off time
`start-up approved`	Start-up (needs the clearance)
`contact ground 118.605`	Hand-off to Ground (needs the clearance)

`cleared to cross runway 25` is a crossing clearance, not an IFR clearance. A datalink clearance (DCL) is sent from the aircraft menu, not by voice.

Frequency change

YOU SAY	EFFECT
`contact tower 118.805` / `contact Stuttgart Tower 118.805`	Hand-off to Tower. The pilot reads back and changes frequency.
`contact tower`	Same; the pilot fills in the frequency
`monitor tower 118.805`	Treated like *contact*
`frequency change approved`	Departures: switch to Tower

A wrong frequency gets `confirm frequency 118.700 for Tower`. Departures that are still on the stand or pushing back reply `confirm contact Tower, we are not yet taxiing`. Station words understood: `tower`, `ground`, `apron`, `delivery`/`clearance`, `approach`/`radar`/`director`, `departure`, `center`/`centre`.

Other

YOU SAY	EFFECT
`standby`	Acknowledges a request; the pilot doesn't call again for 2 minutes
`roger`	Acknowledges a call without an instruction (no read-back); the pilot doesn't call again for 4 minutes and stops flashing on the scope
`number 2 [for pushback|start-up|taxi|departure]`	Queue position in busy periods. The pilot reads it back (`Number 2 for pushback`) and waits 60 s + 45 s per position before reminding you. The number is shown in the tag (`#2`) and in the list.
`expect pushback|start-up|taxi|departure in 5 minutes`	Expected delay. The pilot waits that long (plus 20 s) before reminding you.
`say again`	The pilot repeats their last transmission

YOU SAY	EFFECT
`information K is current[, QNH 1018]`	To a crew that reported an old ATIS letter: they read back `Information K, QNH 1018`. A letter that is not broadcast gets `confirm information J, the ATIS is broadcasting information K`. `QNH 1018` alone is read back too.
`report position` / `say position` (aircraft on the ground)	The pilot reports where it is: `Taxiing on taxiway N`, `At holding point A`, `On stand 14`
`all stations, ...`	Broadcast to everybody on your frequencies, no read-back: `all stations, information B is now current, QNH 1013`, `all stations, runway 25 in use`, `all stations, expect delays`. `all stations, standby` puts every open request on hold for 2 minutes. Sent as `All stations, Stuttgart Ground, ...`
`line up ...`, `cleared for take-off`, `cleared to land` (to an aircraft on Ground frequency)	Not Ground's job. The pilot asks `confirm, we are on Ground frequency, contact Tower?`

RADIO DISCIPLINE

After an instruction the frequency belongs to the station you addressed: **nobody else calls until it has read back** (at most 6 s after your transmission has ended; then waiting calls may go ahead). If you address a second station before the first one has answered, the frequency is kept for the second one, and the first answers afterwards.

COMBINING INSTRUCTIONS

Several instructions can be sent in one transmission, in any order:

```
DLH5AB push and start approved facing east
EWG7TK taxi to holding point A via L2, S, hold short of taxiway H
THY1734 hold position, give way to DLH5AB
CFG123 cross runway 25, taxi to stand 206 via Y, V
```

The pilot reads back all instructions in one transmission. If one of them can't be executed, that part is answered with the reason (for example `unable ...`), and the others are still executed.

PILOT READ-BACKS AND REPLIES

Read-backs repeat the safety-relevant parts and end with the callsign:

INSTRUCTION	READ-BACK
`pushback approved facing east`	`Pushback approved, facing east, Lufthansa 5AB`
`push and start approved`	`Push and start approved, Lufthansa 5AB`
`taxi to runway 25 via L2, S`	`Taxi to holding point A runway 25 via L2, S, Lufthansa 5AB` (the chosen holding point is named)
`taxi to stand 14 via N, L2, hold short of taxiway H`	`Taxi to stand 14 via N, L2, hold short of taxiway H, Eurowings 7TK`
`cross runway 25`	`Cross runway 25, Turkish 1734`
`hold position`	`Holding position, Lufthansa 5AB`
`continue taxi`	`Continue taxi, Lufthansa 5AB`
`give way to EWG7TK`	`Give way to Eurowings 7TK, Lufthansa 5AB`
`contact tower`	`Tower 118.805, goodbye, Lufthansa 5AB`
`line up and wait runway 25`	`Line up and wait runway 25, Lufthansa 5AB`
`wind 250 degrees 8 knots, runway 25, cleared for take-off`	`Cleared for take-off runway 25, Lufthansa 5AB`
`runway 25, cleared to land`	`Cleared to land runway 25, Eurowings 7TK`
`go around`	`Going around, Eurowings 7TK`
`maintain 160 knots until 4 miles`	`160 knots until 4 miles, Eurowings 7TK`
`cleared for immediate take-off`	`Cleared for immediate take-off runway 25, Lufthansa 5AB`
`hold position, cancel take-off`	`Holding position, take-off cancelled, Lufthansa 5AB`
`contact radar 125.050`	`Radar 125.050, goodbye, Lufthansa 5AB`
`tow approved` (to a tug)	`Tow approved to stand 45, Tug 5`
`follow the follow-me` (just vacated)	`Follow the follow-me to stand 14, DCEEO`
`cleared to Frankfurt, ETASA4B departure, flight planned route, climb via SID to altitude 5000 feet, squawk 2312`	`Cleared to Frankfurt, ETASA4B departure, flight planned route, climb via SID to altitude 5000 feet, squawk 2312[, CTOT 1435], Lufthansa 5AB` - now and then with a **wrong squawk**: correct it
IFR clearance with a missing or wrong part	`Confirm ETASA2H departure, information R says runway 25 in use, ...` / `Confirm clearance limit, our destination is Frankfurt` / `Request squawk` / `Confirm initial climb`
`squawk 2312`	`Squawk 2312, Lufthansa 5AB`
`readback correct`	*(no read-back)*
`start-up approved` without a clearance	`Negative, we have no clearance yet, Lufthansa 5AB`
`taxi to runway 25` before the de-icing	`Negative, we require de-icing first, Lufthansa 5AB`
`taxi to de-icing pad DP1` with DP1 occupied	`De-icing pad DP1 is occupied, request another pad, Lufthansa 5AB`
`taxi to de-icing` with every pad taken	`All de-icing pads are occupied, request instructions, Lufthansa 5AB`
a taxi instruction while being de-iced	`Negative, de-icing in progress, about 4 more minutes, Lufthansa 5AB`
a route that needs a tug	`Taxi to ..., we need a tug to turn around, expect about 7 minutes, Lufthansa 5AB`
`expedite taxi`	`Expediting, Lufthansa 5AB`
`standby`	*(no read-back)*
`information K is current, QNH 1018`	`Information K, QNH 1018, Lufthansa 5AB`
`number 2 for pushback`	`Number 2 for pushback, Condor 11`
`cancel pushback`	`Pushback cancelled, Lufthansa 5AB`
`taxi via S, hold short of E`	`Taxi via S, hold short of taxiway E, Lufthansa 5AB`
`advise able for departure from intersection D`	`Affirm, able intersection D, Lufthansa 5AB` / `Negative, we require full length, ...`
`behind DLH5AB, taxi to ...`	`Behind Lufthansa 5AB, taxi to ..., Eurowings 7TK`

PILOT CALLS

Pilots on your frequency call on their own:

SITUATION	EXAMPLE
Clearance request (Delivery, first contact, 10 min before off-block)	`Stuttgart Delivery, Lufthansa 5AB, A320, stand 14, information E, request clearance to Frankfurt` (DCL crews: no call, `DCL` in the list)
Start-up request (Delivery; with A-CDM at the TSAT)	`Stuttgart Delivery, Lufthansa 5AB, stand 14, ready for start-up, TSAT 1452`
After start-up approval (Delivery)	`Stuttgart Delivery, Lufthansa 5AB, start-up approved, request frequency for pushback`
Departure ready (first contact)	`Stuttgart Ground, Lufthansa 5AB, stand 14, information E, request pushback` (sometimes `request push and start`). The letter is the one the crew listened to 2-8 minutes earlier - after a new ATIS it can be the old one
Pushback and start-up complete	`Lufthansa 5AB, ready for taxi` (with de-icing: `ready for taxi, request de-icing`)
(De-icing) De-icing on the pad finished	`Stuttgart Ground, Lufthansa 5AB, de-icing complete on DP2, ready for taxi`
Arrival has vacated the runway (first contact)	`Stuttgart Ground, Eurowings 7TK, vacated runway 25 via E` (crews unfamiliar with the airport add `, request follow-me to the stand`)
(Tower) Departure handed over on the way to the holding point	`Stuttgart Tower, Lufthansa 5AB, approaching holding point A, ready for departure`
(Tower) Departure at the holding point (if not called before)	`Stuttgart Tower, Lufthansa 5AB, holding point A, ready for departure`
(Tower) Arrival on final (first contact)	`Stuttgart Tower, Eurowings 7TK, ILS approach runway 25` (LOC / RNP when the ILS is off or the runway has none, e.g. Sylt 14); reminder `2 miles final runway 25`
(Tower) No landing clearance at 0.5 NM	`Eurowings 7TK, going around, no landing clearance received`
(Tower) Arrival has vacated	`Eurowings 7TK, runway 25 vacated via E`
(Tower) Crossing handed over by Ground	`Stuttgart Tower, Condor 11, holding short runway 25 at W, request crossing`; afterwards `Condor 11, runway 25 vacated`
(Tower) Departure airborne, still on Tower	`Lufthansa 5AB, passing 2800 feet, request frequency change`
Tug ready to tow an aircraft to a remote stand	`Stuttgart Ground, Tug 5, request tow Eurowings A320 from stand 14 to stand 45` (ICAO: REQUEST TOW (company) (type) FROM (location) TO (location))
Departure at the holding point and still with you	`Lufthansa 5AB, holding point A, ready for departure`
Runway holding point on the route, no crossing clearance	`Turkish 1734, holding short runway 25 at W`
Opposite traffic seen in time (stops short of a junction)	`Lufthansa 5AB, opposite traffic on taxiway N, Eurowings 7TK, holding short of G, request instructions`
Head-on with other traffic for over a minute	`Lufthansa 5AB, we have opposite traffic ahead, EWG7TK, request instructions`
Stopped for over a minute behind an aircraft that waits for instructions (for example at a vacate point)	`Lufthansa 5AB, we are blocked by Eurowings 7TK, waiting on the taxiway ahead, request instructions`
Departure on Tower frequency stuck short of the holding point and far from it	`Stuttgart Ground, Lufthansa 5AB, Tower sent us back to you, we are short of the holding point, request taxi`
Held short of a taxiway for over two minutes	`Lufthansa 5AB, holding short of D, request to continue`
Tower refused the departure (too little runway left)	`Stuttgart Ground, Lufthansa 5AB, tower sent us back, not enough runway at K for departure 25, request taxi`
Medical emergency, arrival (after vacating)	`Stuttgart Ground, Eurowings 7TK, PAN PAN, medical emergency on board, vacated runway 25 via E, request expedited taxi to the stand, ambulance requested`
Medical emergency, departure (while taxiing)	`Stuttgart Ground, Lufthansa 5AB, PAN PAN, PAN PAN, PAN PAN, medical emergency on board, request immediate return to the stand, ambulance required`
Night curfew missed (jet still taxiing at 2300 local)	`Stuttgart Ground, Eurowings 1AB, we have missed the night curfew, request taxi back to stand 12`
Thunderstorm, go-around on final	`Lufthansa 5AB, going around, windshear` (or `..., tailwind out of limits`)
Rejected take-off (after vacating)	`Stuttgart Ground, Lufthansa 5AB, we rejected take-off due to a technical problem, vacated runway 25 via D, request taxi back to the stand` (or `..., problem solved, ..., request taxi for another departure`)

If you don't answer, pilots **call again** after 60-90 seconds, up to five times. Reminders are shorter: `Stuttgart Ground, Lufthansa 5AB, stand 14, request pushback`. `standby` stops the reminders for two minutes.

SPOKEN INPUT

Speech recognition (see the user guide) produces plain words, which are normalised before parsing:

SPOKEN	BECOMES
`zero` ... `nine`, `niner`, `tree`, `fife`, `won`, `ate`	`0` ... `9`
`ten` ... `ninety`, `twenty five`	`10` ... `90`, `25`
`to` / `too`, `for` after `runway`, `stand`, `gate`, `number`, `in`	`2`, `4`
`alpha` (`alfa`) ... `zulu`, `x-ray`	`a` ... `z`
`two five`	`25` (after `runway`)
`lima two`	`L2`
`one one eight decimal eight zero five`, `... point ...`	`118.805`
`lufthansa five alpha bravo`	`DLH5AB`

Frequent misrecognitions are corrected before parsing:

HEARD	BECOMES
`push back`, `pushed back`, `push bag`	`pushback`
`start up`, `stand by`	`startup`, `standby`
`run way`, `holding points`, `hold in point`, `holding position`	`runway`, `holding point`
`gulf`, `eco`, `charley`, `mic`, `fox trot`, `x ray`	`golf`, `echo`, `charlie`, `mike`, `foxtrot`, `xray`
`approve`, `improved`, `phasing`	`approved`, `facing`
`euro wings`, `speed bird`, `sun express`, `hansa line`, `luft hansa`, `air france`, `tui jet`, `ryan air`, `wiz air`	the telephony designator

Fuzzy callsigns: a telephony word within 1-2 letters of the correct spelling is accepted ("lufthanza"). If the flight number doesn't match exactly, the aircraft of that operator whose flight number is at most one character off is used. If only one aircraft of that operator is on the frequency, it

is used even without a flight number.

Alternatives: the client parses up to five recognition alternatives and transmits the one that scores best: recognised callsign, number of understood instructions, valid taxi route, no unknown words.

FAA PHRASEOLOGY (US AIRPORTS)

At US airports (`phraseology: 'faa'` in the airport data, so far San Diego KSAN) you work with the FAA forms (FAA JO 7110.65). The ICAO forms above are still understood, so mixing does no harm.

YOU SAY	MEANING
`runway 27, taxi via B`	Taxi to runway 27 (the runway in front is the clearance limit)
`runway 9 at B8, taxi via B`	Taxi to the intersection B8 of runway 9
`push back approved, tail east`	Pushback facing west (`tail` = the opposite of `facing`)
`cleared to KSFO airport, PADRZ2 departure, then as filed, climb via SID except maintain 5000, expect FL350 10 minutes after departure, departure frequency 125.15, squawk 1234`	IFR clearance; `then as filed`, `expect ...` and the departure frequency are accepted (the crew notes them)
`runway 27 at B1, line up and wait`	Line up
`wind 250 at 8, runway 27, cleared for takeoff`	Take-off clearance (wind in the FAA form `250 at 8`)
`cross runway 27 at C6`	Crossing where the crew holds
`turn left at B5`	Vacate via B5 (a landing aircraft)
`contact ground point niner`	Hand-off with the abbreviated frequency
`contact SoCal Departure` / `contact departure`	Hand-off to Departure (Approach)
`altimeter 2992` / `altimeter 29.92`, `information K is current, altimeter 2992`	Altimeter setting in inches of mercury

Runways are written and said without a leading zero ("runway 9"); typing `runway 09` works too.

What the pilots say differently (the same calls as elsewhere, in US wording):

ICAO WORDING	FAA WORDING
`Taxi to holding point B9 runway 09`	`Taxi to runway 9 at B9`
`holding point B1, ready for departure`	`holding short at B1, ready`
`runway 27 vacated via B5`	`clear of runway 27 at B5`
`flight planned route, climb via SID to altitude 5000 feet`	`then as filed, climb via SID except maintain 5000`
`request pushback` / `request push and start`	`ready to push`
`QNH 1018`, `CTOT 1234`	`altimeter 3006`, `EDCT 1234`
`Radar 125.150, goodbye`	`Departure 125.150, goodbye` (frequencies spoken with "point")

No start-up approval: after the clearance, the crew calls Delivery when ready - `stand 36, ready to push, request frequency` - and you hand it to Ground (`contact ground`), where it calls `ready to push`. Datalink clearances are called **PDC** (pre-departure clearance).

ATIS and weather: the METAR has the US format (`METAR KSAN 251150Z 11007KT 10SM SCT030 14/10 A2983`: visibility in statute miles, `CLR` for no clouds, altimeter in inches, no trend); the ATIS starts with the weather, gives the altimeter in inches, "ILS approach runway 9 in use, departing runway 9" and ends with "Advise on initial contact you have information K". The toolbar shows `A2983` instead of the QNH.

NOT SUPPORTED (YET)

- Taxi via a runway (backtrack), `line up runway 25 at D` (intersection line-up by Tower), traffic information and wake turbulence cautions.
- `follow` with real follow-the-leader behaviour.
- Expected-start-up phrases with a time (`expect start-up at 1452`); use `expect start-up in 5 minutes`.
- Clearance amendments in parts (`climb amended ...`) - give the full clearance again.
- Non-English phraseology (German "Rollkontrolle" phrases).
- FAA: progressive taxi, `monitor` hand-offs without a call, and US runway designators in the flight strips and menus (they keep "09"; the toolbar, the ATIS and the pilots say "9").

5 SCENARIOS AND SEEDS

Every session is driven by a **seeded random number generator**. This page explains what the seed does, and how **scenario codes** let you choose what you want to train and repeat it as often as you like.

WHAT THE SEED DOES

All random decisions in the simulation come from one pseudo-random generator (mulberry32), started with the **seed**:

- the initial wind, QNH, ATIS letter and therefore the runway in use,
- which airlines, aircraft types, callsigns, stands, destinations and SIDs appear,
- when departures are ready and when arrivals show up,
- pilot reaction times, reminders, random special events.

The same seed with the same settings gives the same session, as long as your instructions are the same. When you give different instructions (or give them at different times), the traffic soon develops differently, because the pilots react to you.

Leaving the field empty in the Connect dialog starts a random session. A plain number (for example `4711`) is used as the seed with the density and special-events settings from the dialog. A seed alone cannot choose *what* happens. Scenario codes can.

SCENARIO CODES

A scenario code packs everything a training session needs into one string:

```
EDDS-25-HD1-M10R20W30-K7Q2M
EDDS-25-MB0-S20-T2215.N.CE.CS-K7Q2M      (with the optional options part)
```

PART	EXAMPLE	MEANING
Airport	`EDDS`	Airport ICAO code
Runway	`25`	Runway in use at the start. `AUTO` = from the random wind. Parallel runways used together are joined with a dot: `26R.26L` (Leipzig/Halle)
Traffic	`HD1` or `4213`	**Letter form**: density (`L` light, `M` medium, `H` heavy), traffic mix (`B` balanced, `D` departure rush, `A` arrival rush), flags (`0` none, `1` heavies 35 %, `2` random special events "rare", `3` both). **Digit form** for the other slider steps: density 1-5, mix 1-5 (1 departure rush ... 3 balanced ... 5 arrival rush), heavies 0-4, random events 0-4 (see the tables below). The letter form is used whenever it fits, so older codes stay the same
Events	`M10R20W30`	Scheduled events: a letter and the minute after the start. `M` medical emergency (arrival), `D` medical emergency (departure), `R` rejected take-off, `W` wind shift, `S` thunderstorm. `X` = none
Options	`T2215.N.CE`	*Optional* (codes without options have five parts and are still valid). Tokens separated by `.`: `T2215` start at 22:15 local airport time, `N` night flying restrictions, `C` + taxiway (`CE`, `CL2`) taxiway closed - one token per closed taxiway, `I` de-icing (winter operations)
Seed	`K7Q2M`	The random seed, in base 36 (digits and letters)

Codes are not case-sensitive. Type or paste one into the **Scenario** field of the Connect dialog; the dialog shows what it means (or why it is invalid). With a code, the density and special-events sliders of the dialog show the code's values and are locked.

The four traffic **sliders** have five steps each (digit in the code in brackets):

STEP	DENSITY (DEP / ARR PER HOUR)	TRAFFIC MIX (DEP X / ARR X)	HEAVIES	RANDOM EVENTS (CHANCE X)
1st	Light 8 / 7 (`1`, `L`)	Departure rush 1.6 / 0.5 (`1`, `D`)	Normal: the operators' fleets (`0`)	Never 0 (`0`)
2nd	Light to medium 11 / 10 (`2`)	More departures 1.3 / 0.75 (`2`)	10 % wide-bodies (`1`)	Very rare 0.5 (`1`)
3rd	Medium 14 / 12 (`3`, `M`)	Balanced 1 / 1 (`3`, `B`)	20 % (`2`)	Rare 1 - the standard rates (`2`)
4th	Medium to heavy 18 / 15 (`4`)	More arrivals 0.75 / 1.3 (`4`)	35 % (`3`)	Occasional 2 (`3`)
5th	Heavy 22 / 18 (`5`, `H`)	Arrival rush 0.5 / 1.6 (`5`, `A`)	50 % (`4`)	Often 4 (`4`)

Heavies: with a share above *normal*, that share of the new flights is a wide-body (A330, B767, B787), picked from operators that fly them (at an airport without such operators, like Sylt, the traffic stays as it is). Wide-bodies cannot turn around on taxiways, need the full runway length and only fit a few stands - with a large share some wait for a stand; while no stand for a wide-body is free, the new flight is a normal one. **Random events** multiply the chances of the random special events (see simulation model); scheduled events happen at any step.

THE SCENARIO BUILDER

Click **Scenario builder...** next to the Scenario field in the Connect dialog:

1. Pick a **preset**, or set the runway and the sliders for density, traffic mix, heavies and random events yourself.
2. Under **Conditions**, set a **start time** (local airport time; empty = now), the **night flying restrictions**, **de-icing** and **closed taxiways**. A closure that would cut stands off from a runway is shown in red and cannot be used.
3. Add **scheduled events** with the minute when they should happen.
4. **New seed** keeps the scenario but gives different traffic.
5. The **scenario code** updates as you go. **Copy code** / **Copy link** copy it, **Use this scenario** puts it into the Connect dialog.

PRESETS

PRESET	SETTINGS	TRAINS
First steps	light, balanced, no events	The basics without pressure
Departure push	heavy, departure push	Sequencing pushbacks, queue numbers, conditional clearances
Arrival rush	heavy, arrival rush	Quick taxi-in instructions, keeping exits and vacate points free
Heavy metal	medium, more heavies	Wide-bodies: no turning around, full length, stand sizes
Emergencies	medium; medical arrival at 6 min, rejected take-off at 15, medical departure at 25	Priorities, returning aircraft to stands
Runway change	medium, runway 25, wind shift at 15 min	Changing the runway with traffic on the move
Night shift	medium, start 22:15 local, night flying restrictions	The last wave before the curfew: jets out by 2300, landings until 2330 (delayed ones until 2400)
Construction works	medium, runway 25, taxiways E and S closed	Routing around closed taxiways

PRESET	SETTINGS	TRAINS
Winter morning	medium, departure rush, start 06:30 local, de-icing	Every departure via a de-icing pad: pad allocation, two taxi clearances per departure
Thunderstorm	medium, thunderstorm at 12 min	Gusts, a tailwind shift, windshear go-arounds, a departure stop
Full shift	heavy, more heavies, random events; medical arrival at 12, wind shift at 35	Everything at once

Each preset gets a new random seed; press **New seed** for another variant.

SCHEDULED EVENTS

Scheduled events happen even when random special events are off:

EVENT	WHAT HAPPENS AT THE GIVEN MINUTE
Medical emergency, arrival	The **next arrival** that appears on final declares a medical emergency (PAN PAN), see special events.
Medical emergency, departure	The departure on your frequency that is ready earliest and has not started to taxi gets a medical emergency **40 s after it starts taxiing**. If there is none, the simulator tries again every 30 s.
Rejected take-off	The **next take-off** is rejected at 60-110 kt.
Thunderstorm	A storm cell crosses the airport for 20 minutes - see Thunderstorm. If one is still active, the next one waits.
Wind shift	The surface wind turns to the opposite direction (9-15 kt). A *Supervisor* message tells you the new wind and the tailwind; the toolbar wind turns red. The ATIS does not change by itself: open the ATIS editor (it is pre-filled with the new wind), broadcast a new ATIS with the other runway and re-route the traffic.

START TIME AND NIGHT FLYING RESTRICTIONS

Start time (`T2215`): the session starts at this local airport time (Stuttgart: Europe/Berlin, summer or winter time of today's date). Without it, the session starts at the current time. The toolbar clock stays in UTC; with night rules the local time is shown next to it (`2215L`).

Night flying restrictions (`N`) apply the airport's rules. Stuttgart (source: Flughafen Stuttgart):

RULE	LOCAL TIME
No jet departures	23:00 - 06:00
No landings	23:30 - 06:00
Delayed landings (scheduled before 23:30)	until 24:00

How the simulator applies them (turboprops are not affected by the departure curfew; exemptions such as mail flights, emergencies or permits are not modelled):

- **Traffic**: no jet is generated that could not be airborne before 23:00 (ready time + 20 minutes taxi estimate). From 04:45 local on, departures are planned for the morning wave: ready at 05:50-06:00. Arrivals that would land between 23:30 and 24:00 appear only as *delayed flights* (about 35 % of the normal rate; Approach announces them with their scheduled time); after 24:00 none until 06:00.
- **Messages** at the session start, 15 minutes before the curfew, at 23:00, 23:30, 24:00 and 06:00.
- **Jets still on the ground at 23:00** cannot leave: a jet still on its stand is postponed to the morning (new TOBT about 05:50-06:00 local); one that is off-block or taxiing asks to taxi back to a stand (`we have missed the night curfew, request taxi back to stand 14`) - when Ground or Tower is run by the simulator, it sends it back itself. Crews refuse a take-off or line-up clearance in the curfew (`unable, night curfew, jet departures are not permitted until 0600 local`).
- **Score**: a jet that was ready at least 15 minutes before the curfew and still missed it costs 10 points (*Departures that missed the night curfew*).

CLOSED TAXIWAYS

Closed taxiways (`CE`, `CS` ...) are closed for the whole session, e.g. for resurfacing:

- The scope draws them red and hatched, with the designator in red. The ATIS (`taxiways E and S closed`), the briefing (*Now* table) and a NOTAM message at the start say so.
- The router never uses them, and their runway entries (holding points) and exits are closed too. Automatic routes (AI Ground, *auto route*, conflict resolution) go around them; arrivals vacate elsewhere.
- A crew refuses an instruction over a closed taxiway: `unable, taxiway S closed` (also for a holding point or `vacate via` on it).
- Only taxiways can be closed (not apron taxilanes or the runway). The builder checks the closure: every stand must still reach a full-length entry of each runway and be reachable from an exit of each runway (Stuttgart: closing N, O, V or A is therefore not possible).

THUNDERSTORM

A storm cell crosses the airport (event `S`), in three phases:

PHASE	DURATION	WHAT HAPPENS
Warning	5 minutes	*MET* warns of the storm. The wind starts to gust (speed +4 kt, gusts 10-16 kt above it). Approach spaces the arrivals at least 8 NM apart and sends half as many.
Overhead	15 minutes	Gust front: the wind turns to blow along the runway in use from behind (±20°), 14-20 kt, gusts 12-18 kt above that - a strong tailwind. **Lightning stops the ground handling**: no pushbacks (crews don't call, `unable, ground handling has stopped, lightning alert`), no tows. **No take-offs**: crews refuse a take-off clearance (`unable, thunderstorm overhead with windshear, we will wait until it has passed`), the AI Tower lines nobody up; aircraft already lined up and cleared still go. **On final**, at 1 NM, crews go around with more than 10 kt tailwind, and in windshear about 4 in 10 times.
Passed	-	The wind settles from the new direction at 6-11 kt; the other runway is usually better now. Departures and ground handling resume.

What you do: broadcast a new ATIS (the editor is pre-filled with the gusts), change the runway to stop the tailwind go-arounds, keep the departures waiting at the holding points in order, and get the queue moving when the storm has passed. Weather go-arounds **do not cost points**; the toolbar shows the wind in red with `TS` while the storm is active.

DE-ICING

Winter operations (option `I`): frost on the wings, every departure has to be de-iced on one of the **de-icing pads** before take-off. Stuttgart has four, **DP1-DP4**, between taxiways N and S - DP1 / DP2 between I and H, DP3 / DP4 between H and F. Every pad is a drive-through bay **entered from N and left onto S** along its lead line: DP1 and DP3 lead west, DP2 and DP4 east, so DP2 / DP4 are entered eastbound on N (e.g. from the circle via L2) and DP1 westbound (from H or M). After the pad: runway 25 departures from DP2 / DP4 continue on S towards A, runway 07 departures from DP1 / DP3 on S towards K. See EDDS.

- The crew asks for it: `ready for taxi, request de-icing` (the ATIS says `de-icing in progress on the de-icing pads`, the briefing has a *De-icing* row).

- **Taxi it to a pad**: `taxi to de-icing pad DP2 via L2, N` (also `de-icing position 2`, `delta papa 2`; without a pad the crew takes the free pad nearest to it). The *Taxi to* menu lists the pads first, occupied ones greyed out. A pad is taken from the taxi clearance until the aircraft has left it; crews refuse an occupied pad (`de-icing pad DP2 is occupied, request another pad`).
- On the pad the status shows `DEICE` for **6-10 minutes** (3 more for aircraft over 40 m wingspan). A taxi instruction meanwhile: `negative, de-icing in progress, about 4 more minutes`.
- Then: `de-icing complete on DP2, ready for taxi` - taxi it to the runway as usual.
- **Before the de-icing** crews refuse the runway or a holding point: `negative, we require de-icing first`.
- Pad sizes: **DP1 up to 65 m** wingspan (code E), **DP2 up to 68.5 m** (code E, also the B747-8), **DP3 and DP4 up to 36 m** (code C) - wide-bodies use DP1 or DP2. An aircraft too large for every pad would be de-iced on its stand (none at Stuttgart).
- Not modelled: in reality S is closed next to the pads in use (I-H for DP1 / DP2, H-F for DP3 / DP4).
- The AI Ground does the same: it keeps a departure on its stand while all pads are busy, taxis it to a free pad, and on to the runway afterwards.
- Not modelled: holdover times (a de-iced aircraft stays de-iced), de-icing fluid types, snow on the taxiways, braking action.

SHARING A SCENARIO

Copy link gives a link such as `https://linoking09.github.io/ultimate-atc/?scenario=EDDS-25-MB0-W15-K7Q2M`. Opening it pre-fills the Connect dialog with the code, so others can train exactly the same session - for example to compare scores.

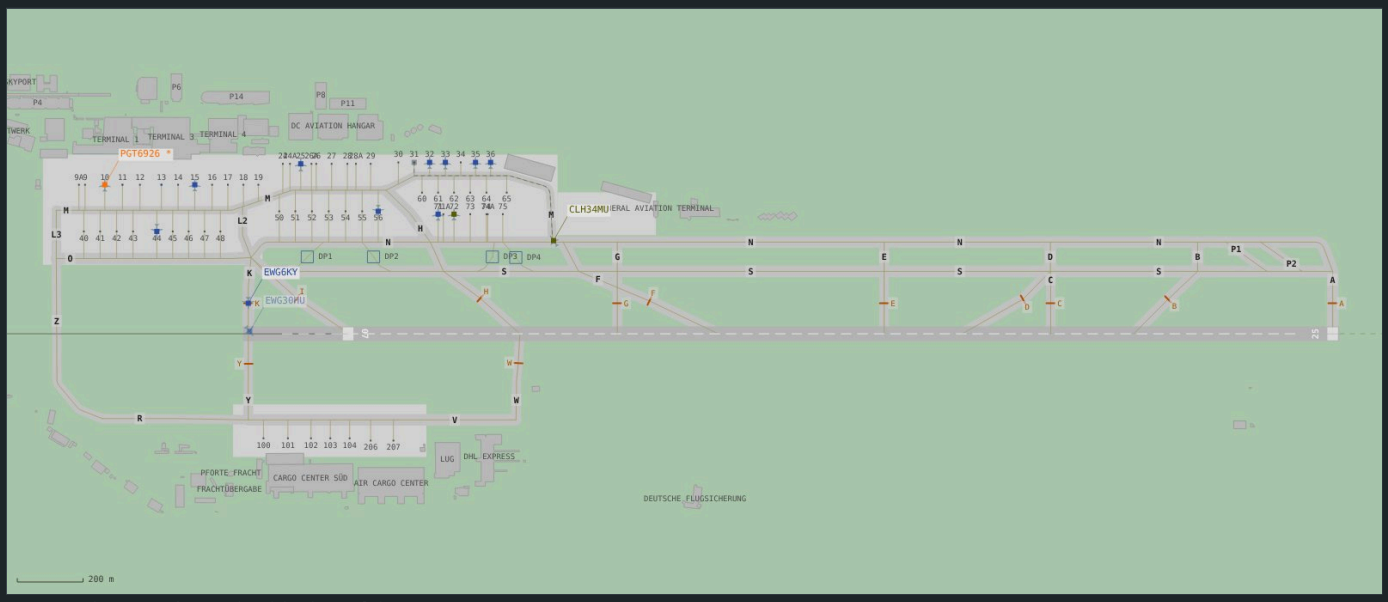
6 AIRPORTS

Every airport is built after the airport recipe: OpenStreetMap layout, facts from the national AIP (DFS; in the US the FAA Chart Supplement), procedures after VATSIM, checked against the charts.

AIRPORT	FIR	RUNWAYS IN USE	POSITIONS	WHAT MAKES IT SPECIAL
EDDS Stuttgart	Langen	07/25	Delivery, Ground, Tower	The home airport: tutorials, every scenario option, de-icing pads, holding bays
EDDK Cologne/Bonn	Langen	13L/31R (06/24 and 13R/31L drawn and crossed)	Delivery, Ground, Tower	Cargo hub (UPS, FedEx, DHL), parallel guide lines, crossing of 06/24
EDDN Nuremberg	München	10/28	Ground, Tower	Single runway, GA and cargo aprons
EDDP Leipzig/Halle	München	08L/26R and 08R/26L **at the same time**	Delivery, Ground, Tower	DHL hub, dependent parallel approaches, runway per apron
EDDW Bremen	Bremen	08/26	Ground (with clearances), Tower	Night rules, Beluga take-off runways drawn
EDJA Memmingen	München	06/24	Ground, Tower	Departures and arrivals **backtrack** on the runway
EDXW Sylt	Bremen	14/32 (06/24 crossed)	Tower only	Small island airport, crossing runways
KSAN San Diego	Los Angeles (KZLA)	9/27	Clearance Delivery, Ground, Tower	**FAA phraseology**, US weather and ATIS, busy single runway with displaced thresholds

Planned airports are listed as *Coming soon* in the airport menu; see the roadmap.

7 EDDS - STUTTGART



▸ EDDS at the start of a session (phx-day style)

ITEM	VALUE
ICAO / IATA	EDDS / STR
Location	Stuttgart / Leinfelden-Echterdingen, Germany
Reference point	48°41'23.56"N 009°13'19.07"E
Elevation	1276 ft
Magnetic variation	3.9° E (World Magnetic Model, 2026; AIP AD 2.2: 3.0° E, 2026.01)
Transition altitude	5000 ft
Runway	**07/25**, 3345 m x 45 m, true heading 074° / 254°; landing threshold 07 displaced by 300 m (984 ft); threshold elevations 1267 ft (07) / 1181 ft (25). Take-off run available 07: 3345 m, **25: 3045 m** (the 300 m behind the displaced threshold 07 do not count for take-offs 25)
Data file	`src/data/airports/edds.ts`

The simulator shows a condensed version of this page as the **airport briefing** (`BRIEFING` in the toolbar, data in `briefing` in `edds.ts`). Keep both consistent.

DATA SOURCES AND ACCURACY

DATA	SOURCE
Runway end coordinates	OurAirports (public domain); cross-checked against the AIP thresholds (AD 2.12): within 4 m
Declared distances, approaches, taxiway widths, de-icing pad limits, code letter limits, low visibility procedures, night restrictions	DFS AIP Germany AD 2 EDDS (AIRAC 2026OCT01): text pages AD 2.8, 2.12, 2.13, 2.20, 2.22 and the approach and SID charts, used as a reference only (facts read by hand)
ARP, elevations, frequencies, taxiway designators, holding points, stand numbers, layout	AIP Germany, **AD 2 EDDS 2-5 Aerodrome Chart** and **AD 2 EDDS 2-7 Aerodrome Ground Movement Chart** (AMDT 10/26)
Night flying restrictions, time zone	Flughafen Stuttgart (published night flying restrictions); applied only in scenarios with night rules, see scenarios
Stand sizes, wide-body positions, pushback / taxi-out stands	AD 2 EDDS 2-7 (AMDT 10/26, code letter markings), the older **Aircraft Parking/Docking Chart** (AMDT 09/13), the stand data of the VATSIM Germany ground radar (EuroScope package `edgg-package`, 2026) - see Stands
De-icing pads DP1-DP4 (position, lead lines, wingspan limits), take-off run available (TORA) from every entry, the current Apron South	VATSIM Germany EuroScope package (`edgg-package`, 2026): its ground radar maps are generated from the AIP data (AIXM). Used as a reference only - positions were read off and converted into the simulator's frame; no files or map content are included
Holding bays P1 / P2	OpenStreetMap (the two unnamed links between N and S at the 25 end)
Buildings	OpenStreetMap - © OpenStreetMap contributors, ODbL 1.0: every building on the aerodrome (terminals 0-4, hangars, cargo, the tower ...), generated with `scripts/osm-import.mjs EDDS --buildings-only` into `edds.buildings.ts` (do not edit by hand)
SIDs	Designators from a public chart index (2026) and the VATSIM Germany vSID configuration; destination restrictions and the radar sectors from the VATSIM Germany Delivery SOP; exit fix positions from public waypoint lists
Frequencies	VATSIM Germany station data (`datahub`, matching the AIP values used before); ATIS from the VATSIM Germany vATIS profile
Magnetic variation	NOAA World Magnetic Model (October 2026)

The geometry was **digitised by hand**. Positions were measured on the charts, using the 3345 m runway as scale and the 300 m displaced threshold as a check, and converted into the simulator's runway-aligned frame. The accuracy is roughly ±10 m. Only facts are used (topology, designators, approximate positions); no chart graphics are reproduced or distributed.

Checked on 2026-10-21 against OpenStreetMap and the VATSIM Germany ground radar data: all runway holding positions lie within 10 m of the OSM positions, the stands within 20 m of the ground radar stand positions; the published take-off run from every entry matches the simulator's geometry within 100 m (the published values are used).

Simplified or not modelled:

- the run-up areas, the general aviation apron (GAT exits 1-3), the US Army airfield, helipads and helicopter routes,
- on Apron South (layout since AIRAC 2512) the small stands 300-303 and the new 105-107 between 206 and 207 (the simulator has 100-104, 206 and 207),
- while a de-icing pad is in use, the real operation closes S next to it (I-H for DP1 / DP2, H-F for DP3 / DP4); the simulator does not,
- the intersection geometry inside the "circle" (the hot spot west of N/S, chart note 1), reduced to one junction node,
- the SIDs: all exit fixes with their 2026 designators (B = runway 25, H = runway 07), but no routes or altitude restrictions; flight plans pick the SID by the direction of the destination (see SIDs). The local SIDs STG / TEDGO and the RNP K-SIDs (runway 07) are not modelled,
- the initial climb in IFR clearances is the published SID altitude (5000 ft, `initialClimbFt`, AIP SID charts; also the transition altitude); the SID routes and their speed and level restrictions are not flown,
- the squawk codes (a simulator range, not the real code allocation).

Never use this simulator for real-world navigation.

Reference data (research 2026-10-11, checked again 2026-10-21)

Facts collected from public sources to bring the simulation closer to reality. Confidence: **certain** = official body or the airport operator; **likely** = VATSIM Germany SOP or another secondary source (modelled on the real operation, but not the real procedures); **estimate** = derived. Values marked open were not found. The official source for most of them is the AIP Germany AD 2 EDDS (sections 2.13, 2.20, 2.22), which still has to be checked.

TOPIC	VALUE	CONFIDENCE / SOURCE	IN THE SIMULATOR
Control zone	class D, GND - 3500 ft AMSL; extent roughly 15 NM east-west, 10 NM north-south along the runway axis	likely (VATGER Tower SOP); extent estimate	departures leave the scope at 10 NM
Approach / Director	Langen Radar 125.050 and 119.200, Stuttgart Arrival 119.850 (0500-2100 UTC)	certain (AIP AD 2.18)	Langen Radar 125.050; southern departures (ROTWE, SUL, KUNOD, ABTAL) 119.200 - the VATSIM Germany sector split (the AIP SID charts say "contact Langen Radar 125.050" for every SID)
Approach -> Tower hand-off	no fixed value; typically when established on the ILS, about 8-10 NM	estimate	8-10 NM
Departure hand-off	VATSIM: auto hand-off (crews call Langen Radar when airborne); north SIDs to sector Stuttgart, south SIDs (ROTWE, SUL, KUNOD, ABTAL) to sector Reutlingen 119.200	likely (VATGER)	you hand departures to Radar (`contact radar`); the crew reads back the frequency of its SID's sector
SIDs	exit fixes ABTAL, DKB, ETASA, GEBNO, KRH, KUNOD, OKIBA, ROTWE, STG, SUL, TAGIK, TEDGO, VESID; restrictions: ETASA only Frankfurt area, KRH only EDDR / EDRZ / EDSB / ETAR / ETIP, SUL jets only EDTL / EDNY / LSZH / LSZR, DKB only via N869 or EDDN / EDTY, GEBNO via Z76 max FL180, TAGIK via ABUMO / ASKIK max FL240, OKIBA min FL200	likely (VATGER Delivery SOP, chart index)	all but STG / TEDGO; flight plans by destination direction and these restrictions
Frequencies	ATIS 126.130 (call sign STUTTGART ATIS), Delivery 121.915, Ground 118.605 (one Ground frequency, no Apron), Tower 118.805 and 119.055	certain (AIP AD 2.18)	as listed
Runway	3345 m, threshold 07 displaced 300 m; 07: TORA / TODA / ASDA 3345, LDA 3045 m; 25: TORA / TODA 3045, ASDA 3345, LDA 3345 m	certain (AIP AD 2.12 / 2.13)	modelled
Approaches	ILS CAT II / III both ways (RVR 75 m); 25 also LOC, VOR and RNP; 07 also RNP (LNAV) - the LOC and VOR approaches 07 only for aircraft up to 5.7 t	certain (AIP AD 2.19, approach charts)	07 `['ILS', 'RNP']`: with the ILS 07 out of service airliners fly RNP; 25 `['ILS', 'LOC', 'RNP']`
Intersection departures	crews are asked for every intersection; smaller aircraft usually C (25) or G (07)	likely (VATGER)	`advise able for departure from intersection ...`; take-off run per type and flight; E (25) and G (07) are entries since 1.2
Take-off run available	25: A 3045, B 2435, C 2175, D 1990, E 1665 m; 07: K / Y 3345, I 3090, H 2515, W 2520, G 2210, F 2020, E 1400 m	certain (AIP AD 2.13)	the published values, every one an entry
Code letter limits, widths	G, E and C max. code letter D; widths A, B, D, F, H, I, K, N, O, S, V 30 m, C, E, G, Y 23 m, W, Z 22.5 m	certain (AIP AD 2.8, aerodrome chart)	modelled (G, E, C 52 m)
Planned exits	heavy B (07) / H (25), medium jet D (07) / F (25), medium prop and light jet E (07) / F (25)	certain (AIP AD 2.20)	exits chosen by speed and type
Taxi flows	N against the runway direction, S the other way: runway 25 N eastbound / S westbound, runway 07 N westbound / S eastbound	likely (VATGER Ground SOP)	modelled (07 changed on 2026-10-11)
Holding bays / de-icing	bypass P1/P2 (max B739/A321); de-icing only on DP1-DP4, all entered from N and left via S (DP1 / DP3 westbound, DP2 / DP4 eastbound); DP1 code E (65 m), DP2 code E plus B747-8 / C-5 (68.5 m, AIP AD 2.20), DP3 / DP4 code C; S closed next to a pad in use	DP1-DP4 certain (A-CDM briefing), rest likely (VATGER SOP and ground radar map)	P1/P2 at the OSM positions; DP1-DP4 at the map positions with these limits since 1.2 (S stays open); de-icing as the scenario option `I`
Runway crossings	only between K and Y	likely (VATGER)	automatic routes cross only between K and Y (elsewhere only if K / Y is closed)
Taxiway Z	west of Y, reserved for military aircraft	likely (VATGER)	automatic routes avoid it
Taxi-out stands	40-56 taxi out; 61-64 only with the opposite stand free; jet bridges only at 9-16	likely (VATGER)	40-48, 50-56 taxi out
Stand sizes	9 up to 60.5 m (9A 60.5-65 m), 11 code D, 24A / 26A / 28A code D-E, 48 30 m, 51 / 53 / 55 / 56 45 m, 52 / 54 code D, 65 29 m, 71A / 74A code E-F (A380 diversions), Apron South 100-104 code D, 206 / 207 code E-F	likely (VATGER ground radar stand data)	modelled since 1.2
Simultaneous pushbacks	at least two stands between them	likely (VATGER)	AI Ground: nothing taxiing within 250 m
Initial climb / transition	climb via SID to 5000 ft; transition altitude 5000 ft, transition level in the ATIS	likely (VATGER)	5000 ft; TL from the QNH: FL60 from 1013 hPa, FL70 below (the lowest flight level at least 1000 ft above the transition altitude)
Capacity	max. 48 movements/h (40 departures without arrivals; 40 arrivals with max. 8 departures), 14 in LVP	likely (VATGER)	traffic densities below that
Traffic 2025	95 765 movements, 9.6 million passengers	certain (aerotelegraph)	-
Runway direction	2025: 55 % of departures to the west; long-term average 60/40 west/east	certain (RP Stuttgart noise report)	session wind 60 % westerly
Night restrictions	no jet departures 23:00-06:00, no landings 23:30-06:00 (delayed landings until 24:00); the same for propeller aircraft not meeting the noise standards	certain (AIP AD 2.20 2.1, Flughafen Stuttgart)	adopted as the scenario option *night flying restrictions* (`nightRestrictions`, time zone Europe/Berlin); exemptions not modelled
A-CDM	TOBT from 90 min before EOBT, TSAT published 40 min before TOBT; start-up within TSAT ±5 min; push / taxi at the latest 5 min after start-up approval; TOBT changeable 3 times after the TSAT	certain (STR A-CDM flight crew briefing)	start-up at TSAT ±5 min
IFR clearance (Germany)	`cleared to Frankfurt, ETASA4B departure, flight planned route, climb via SID to altitude 5000 feet, squawk 1000`	likely (VATGER Delivery SOP)	used since 2026-10-11 (the older short form is still understood)
Conditional clearance	`SAS941, behind DC9 on short final, line up behind` (ICAO Doc 4444 12.2.7)	certain	understood; menu: `behind landing X, line up and wait behind`

TOPIC	VALUE	CONFIDENCE / SOURCE	IN THE SIMULATOR
"Expect late landing clearance"	not a DFS / ICAO standard phrase; German practice is to not withhold the landing clearance when the runway will be clear at the threshold	open / likely	kept as an extra phrase
LVP	preparation at RVR < 1000 m and / or ceiling <= 300 ft; in force at RVR <= 600 m and / or ceiling < 200 ft (or taxiing in RVR < 550 m); CAT IIIa / b and low-visibility take-offs both ways; stop bars at the CAT II / III holding positions	certain (AIP AD 2.22)	not modelled yet

Sources: VATSIM Germany EDDS SOPs (Tower, Ground, Delivery, Arrival), STR A-CDM Flight Crew Briefing, SkyVector EDDS, BAV EDDS briefing, RP Stuttgart noise report 2025, Flughafen Stuttgart noise reports, aerotelegraph, SKYbrary (LVP), ICAO Doc 4444.

STATIONS

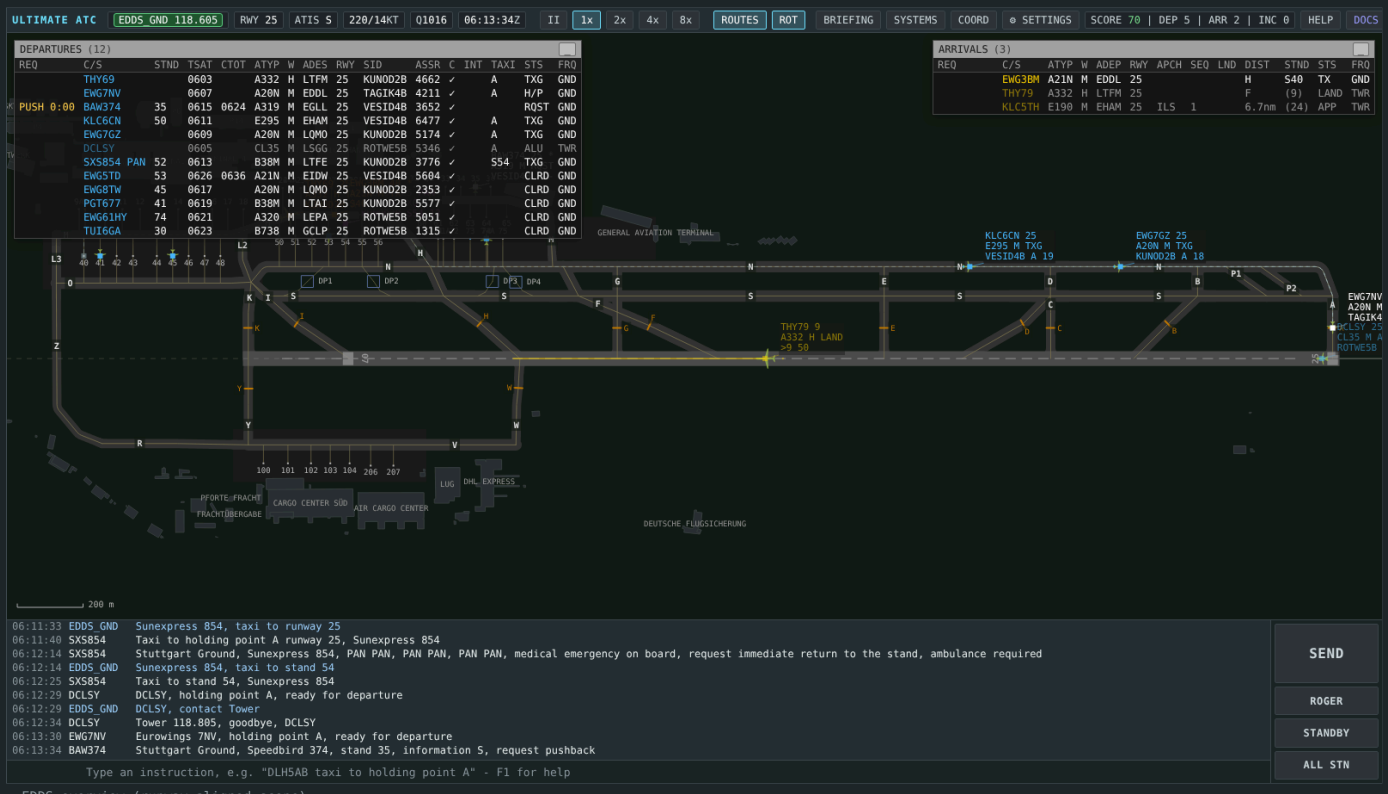
CALLSIGN	RADIO NAME	FREQUENCY	SIMULATED BY
EDDS_DEL	**Stuttgart Delivery**	**121.915**	**You** (since v0.6), or AI: clearances are then issued before the aircraft calls Ground
EDDS_GND	**Stuttgart Ground**	**118.605**	**You**, or AI Ground when you only staff Delivery
EDDS_TWR	Stuttgart Tower	118.805 (also 119.055)	AI Tower
EDDS_APP	Langen Radar	125.050 (sector Stuttgart); 119.200 for the southern departures (sector Reutlingen)	(not simulated)
EDDS_ATIS	Stuttgart Information	126.130	You (ATIS editor)

SYSTEMS AND PROCEDURES

Stuttgart has been a **full A-CDM airport** since November 2014 (TOBT from the airlines, TSAT from the pre-departure sequencer, start-up at the TSAT) and offers **departure clearance by datalink (DCL)**; start-up is requested by voice. The simulator gives EDDS all systems: A-SMGCS surveillance, RMCA, CATC and routing (the services offered for training; which A-SMGCS levels are installed at Stuttgart is not modelled from a published source), A-CDM, DCL and the ILS (localizer and glide path) of the runway in use.

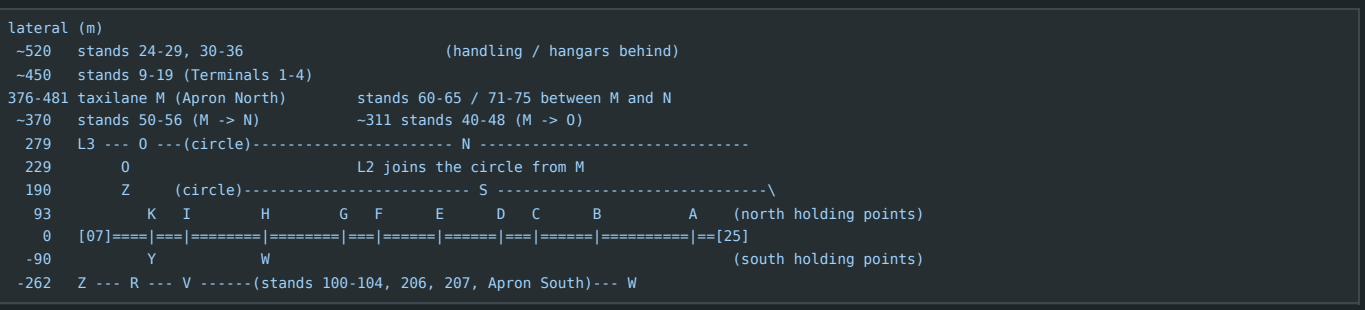
Delivery procedure in the simulator: clearance request about 10 minutes before off-block, IFR clearance `cleared to <destination> via <SID> departure, climb 5000 feet, squawk <code>`, start-up at the TSAT, then `contact ground 118.605` for pushback.

LAYOUT



▸ EDDS overview (runway-aligned scope)

The layout uses a runway-aligned frame, the same orientation as the aerodrome chart: "along" is measured from the runway 07 end towards 25, "lateral" is the perpendicular distance in metres (positive = north). In the simulator, the **ROT** button switches the scope between this orientation (the default) and north-up.



Taxiways

TAXIWAY	DESCRIPTION
N	Outer parallel taxiway north of the runway (279 m), from the circle to the 25 end, where it joins A
S	Inner parallel taxiway north of the runway (190 m), from the circle to A
O	West of the circle (229 m), along the drive-through stands 40-48, to L3
M	Apron North taxilane. It runs from L3 along the terminal stands 9-19 (376 m), up past 24-29 (437 m) and 30-36 (481 m), then down to N east of stand 75
L2	From M (near stand 18) down into the circle
L3	West end of the apron: links M and O; Z continues south
K / Y	At the 07 end: K runs from the circle to the runway (holding point K), Y from the runway south to R/V (holding point Y)
I	Angled from the circle to the runway near the 07 threshold: rapid exit for 25, intersection entry for 07 (holding point I)
H / W	H runs from Apron North (M) across N and S, angled to the runway: rapid exit for 25, entry for 07 (holding point H). W continues south from the same runway point to V (holding point W)
G	Perpendicular from N/S to the runway (holding point G); crosses F just north of the holding points
F	Angled rapid exit for 25 from the runway to S/N (holding point F)
E	Perpendicular (holding point E)
D	Angled from S/N to the runway: rapid exit for 07, intersection entry for 25 (holding point D)
C	Perpendicular from S; shares the junction with D (holding point C)
B	Angled: rapid exit for 07, intersection entry for 25 (holding point B)
A	At the 25 end, where N and S end (holding point A, full length for 25)
Z / R / V	South: Z runs from L3/O around the 07 end to R (reserved for military aircraft - automatic routes avoid it); R to Y; V along Apron South to W
P1 / P2	Holding bays at the 25 end: angled links from N down to S just before A (code C, wingspan up to 36 m)
DP1-DP4	De-icing pads between N and S: DP1 / DP2 between I and H, DP3 / DP4 between H and F, see below

Holding points carry the name of their taxiway (`holding point A`, `holding point K`, ...). The 07 end has K (north) and Y (south); H and W share one runway entry.

Closing taxiways (scenario option, e.g. for construction works): every taxiway can be closed except **N**, **O**, **V** and **A** - they are the only way to or from some stands (N: stands 50-56 and 71-75; O: 40-48; V: Apron South) or the only full-length entry of runway 25 (A). Closing several at once is checked in the scenario builder. The preset *Construction works* closes E and S: departures and arrivals then share N.

Holding bays and de-icing pads

- **P1 / P2** (bypass): two angled links from N down to S at the 25 end - P1 from N 2998 m to S 3123 m, P2 from N 3104 m to S 3230 m from the 07 end (OpenStreetMap) - limited to code C (36 m wingspan; the SOP says max. B739 / A321). A departure that is not ready can wait on P1 or P2 while the next one passes it via S to A (re-sequencing), or use them as `taxi to holding point A via N, P1, S`.
- **DP1-DP4** (de-icing pads, positions from the VATSIM Germany ground radar map): drive-through bays between N and S, **all entered from N and left onto S** along their painted lead lines - DP1 (N 246 m -> S 124 m from the 07 end) and DP3 (N 767 m -> S 717 m) lead west, DP2 (N 364 m -> S 450 m) and DP4 (N 793 m -> S 885 m) east. Wingspan limits: **DP1 65 m** (code E, no B747), **DP2 68.5 m** (code E plus B747-8 / C-5, AIP AD 2.20), **DP3 and DP4 36 m** (code C) - every aircraft in the simulator fits DP1 or DP2. The lead lines allow no sharp turn: eastbound on N (from the circle) DP2 and DP4 lie ahead to the right, DP1 is entered westbound (from H or M). After the pad, runway 25 departures from DP2 / DP4 continue on S towards A; runway 07 departures from DP1 / DP3 on S towards K (from DP2 the sharp left onto H and N, from DP4 via F and N). In reality S is closed next to the pads in use (I-H for DP1 / DP2, H-F for DP3 / DP4); the simulator keeps it open. Used in the scenario option `de-icing`.

Standard taxi flows

To keep traffic on the two parallel taxiways from meeting head-on, they are used one-way: **N against the runway direction, S the other way** (as in the VATSIM Germany SOP). Automatic routes (pilots without a `via` list, the *Taxi to* menu and the quick-action bar) follow these directions strictly; only if that is impossible without a 180 degree turn they use a taxiway against the flow:

RUNWAY IN USE	N	S	IDEA
25	eastbound	westbound	departures on N to A, arrivals from F/G/E/H on S to the aprons
07	westbound	eastbound	arrivals from E/D/C/A west on N to the aprons, departures to K via M, L2 (or west on N from 50-56)

The directions follow the VATSIM Germany SOP (modelled on the real operation, not a published procedure). Your own `via` instructions are always followed as given.

Runway entries and exits

RUNWAY	DEPARTURE HOLDING POINTS	ARRIVAL EXITS (FROM THE THRESHOLD ONWARDS)
25	**A** (full length); B, C, D, E (intersections)	F (rapid), E, G, H (rapid), W (south), I (rapid), K, Y (south)
07	**K** / **Y** (full length, north / south); I, H, W, G (intersections)	G, E, D (rapid), C, B (rapid), A

Vacate points (where an arrival stops after leaving the runway and waits for your taxi instruction) lie between the runway holding position and the next taxiway. Behind F and G they are on the short connector before the F/G junction, clear of taxiway S, so a waiting arrival does not block traffic on S (approximation of the real geometry).

Take-off run available (published values): runway 25 from A **3045 m** (the 300 m behind the displaced threshold 07 do not count for take-offs 25), B 2435 m, C 2175 m, D 1990 m, E 1665 m; runway 07 from K / Y 3345 m, I 3090 m, H 2515 m, W 2520 m, G 2210 m. This decides the pilots' answer to `advise able for departure from intersection ...`. Smaller aircraft usually depart from C (25) or G (07).

About 85% of arrivals vacate to the north. On runway 25 the others vacate to the south via W or Y. They then need a **runway crossing** to reach Apron North, unless they park on Apron South - **only between K and Y** (VATSIM Germany SOP): automatic routes cross there (elsewhere only if K or Y is closed); your own `via` list may cross elsewhere.

Stands

APRON / AREA	STANDS	MAX. WINGSPAN	TYPE	LANE
Apron North	9-19	36 m; **9 60.5 m**, **11 52 m**	nose-in, pushback (jet bridges 9-16)	M
Apron North	**9A**	65 m	wide-body position overlapping 9	M
Apron North	24-29	36 m	nose-in, pushback	M
Apron North	**24A, 26A, 28A**	65 m	code D-E positions overlapping 24+25, 25+26, 28	M
Apron North	30-36	36 m	nose-in, pushback	M (code C section)
Apron North	60-65	36 m (65: 29 m)	nose-in (nose south), pushback	M (code C section)
Apron North	71-75	36 m	nose-in, pushback onto N	N
Apron North	**71A, 74A**	80 m	code E-F positions overlapping 71+72 / 74+75, pushback onto N (A380 diversions)	N
Apron North	40-48	36 m (48: 30 m)	**drive-through**: in from M, out forwards to O	M -> O
Apron North	50-56	50: 36 m, 51 / 53 / 55 / 56: 45 m, 52 / 54: 52 m	**drive-through**: in from M, out forwards to N	M -> N
Apron South	100-104	52 m	nose-in (nose south), pushback (cargo)	V
Apron South	**206, 207**	80 m	nose-in, pushback (cargo wide-bodies; the former 105 / 106)	V

Stand sizes - sources. The current ground movement chart (AMDT 10/26) marks taxilane M east of the H junction as *max. code letter C* (wingspan below 36 m) and shows the alternative positions 9A, 24A, 26A, 28A, 71A and 206/207 (overlapping their neighbours). The per-stand limits come from the stand data of the VATSIM Germany ground radar (2026), which follows the airport's stand allocation: code letters or wingspan ranges per stand, the overlapping positions and the AIRAC 2512 changes on Apron South (100-104 moved, 105 / 106 renamed 206 / 207). The older aircraft parking/docking chart (AMDT 09/13) confirms that nose-in positions 9-36 and 100-107 are left by pushback and 40-56 are taxi-out positions. Stand 44 exists again (it was missing before 1.2); stands 45-48 are currently blocked by construction works in reality (AIP SUP until 30 October 2026), which the simulator ignores.

Alternative positions block their neighbours (wingtip clearance, see simulation model): a wide-body on 71A blocks 71 and 72 (and an aircraft on 71 or 72 blocks 71A), likewise 74A with 74/75, 9A with 9, 24A with 24 and 25, 26A with 25 and 26, 28A with 28. Narrow-bodies use them only when nothing else is free.

Departures on drive-through stands don't need a pushback: they call `request taxi` and leave forwards. The stand allocation prefers Apron North; business jets go to stands 60-65. Apron South is only used when Apron North is full, but you can taxi any aircraft there.

SIDS

Every flight plan is filed via the exit fix towards its destination (great-circle bearing from Stuttgart); fixes restricted to a few destinations are only used for them. The designators are those of 2026 (B = runway 25, H = runway 07; the numbers change with the AIRAC cycle):

EXIT FIX	RUNWAY 25 / 07	USED FOR (SIMULATOR)	RADAR AFTER TAKE-OFF
VESID	VESID4B / VESID2H	destinations 255-340° (UK, Ireland, Benelux, Paris)	Langen Radar 125.050
OKIBA	OKIBA4B / OKIBA4H	340-045° (Hamburg, Bremen, Berlin, Scandinavia; min. FL200)	125.050
TAGIK	TAGIK4B / TAGIK2H	Duesseldorf, Cologne, Moenchengladbach, Paderborn, Dortmund, Muenster (via Frankfurt area, max. FL240)	125.050
ETASA	ETASA4B / ETASA2H	Frankfurt (EDDF, EDFE, EDFC, ETOU)	125.050
KRH	KRH5B / KRH2H	Saarbruecken, Karlsruhe / Baden-Baden, Ramstein, Pirmasens (EDDR, EDRZ, EDSB, ETAR, ETIP)	125.050
DKB	DKB9B / DKB4H	Nuremberg area (EDDN, EDTY)	125.050
GEBNO	GEBNO7B / GEBNO6H	short flights via Z76 below FL180 - not used by the generated traffic	125.050
ABTAL	ABTAL4B / ABTAL4H	045-110° (Munich, Vienna, Budapest, Poland, Romania)	sector Reutlingen 119.200
KUNOD	KUNOD2B / KUNOD2H	110-190° (Italy, Balkans, Greece, Turkey, Egypt)	119.200
ROTWE	ROTWE5B / ROTWE7H	190-255° (Spain, Portugal, Canaries, southern France, Geneva, Morocco)	119.200
SUL	SUL3B / SUL3H	jets to Zurich, Friedrichshafen, Lahr, Altenrhein (LSZH, EDNY, EDTL, LSZR)	119.200

The restrictions (ETASA, KRH, SUL, DKB, GEBNO, TAGIK, OKIBA) are those of the VATSIM Germany Delivery SOP; the sectors in between are the simulator's approximation of the route network. Delivery checks that the cleared SID leads to the first fix of the flight plan. Not modelled: the local IFR SIDs STG and TEDGO and the RNP K-SIDs of runway 07.

TRAFFIC

The traffic generator uses the operator mix of Stuttgart (`traffic` in `edds.ts`). Shares are the airlines' passenger shares in scheduled and charter traffic from the airport's annual report 2024 as quoted in the press (Eurowings about 40 % also in 2025), used as movement shares:

OPERATOR	SHARE	TYPES	MAIN DESTINATIONS
Eurowings (EWG)	39.7 %	A319, A320, A20N, A21N	Palma, Pristina, Barcelona, Berlin, Hamburg, Düsseldorf, Antalya, Dublin, Manchester, Stockholm, Malta, Mostar, Split, Zagreb, Belgrade ...
SunExpress (SXS)	8.3 %	B738, B38M	Antalya, Izmir, Dalaman, Kayseri, Bodrum
TUIfly (TUI)	5.8 %	B738, B38M	Palma, Canaries, Hurghada, Heraklion, Antalya
Pegasus (PGT)	5.4 %	A20N, A21N, B38M	Istanbul Sabiha Gökçen, Antalya, Izmir
Lufthansa group (DLH, CLH)	5.4 %	A319, A320, A20N, CRJ9, E195	Frankfurt, Munich
Turkish Airlines (THY)	4.7 %	A321, A21N, B738, A332	Istanbul
Condor (CFG)	3.9 %	A320, A321, A21N	Palma, Canaries, Hurghada, Antalya, Greek islands
British Airways (BAW)	3.7 %	A319, A320, A20N	London Heathrow (about 359,000 passengers in 2025)
KLM Cityhopper (KLC)	2.2 %	E190, E295	Amsterdam
Air France (AFR)	1.2 %	A319, A320	Paris CDG

OPERATOR	SHARE	TYPES	MAIN DESTINATIONS
Austrian (AUA)	1.0 %	DH8D, E195	Vienna
Business aviation	3 %	C56X, CL35	various

The busiest destinations in 2025 were Palma (639,000 passengers), Antalya (571,000), Istanbul Sabiha Gökçen (541,000), Istanbul (382,000), Pristina (372,000), London Heathrow (359,000), Barcelona (347,000), Berlin (313,000) and Hamburg (296,000); the destination lists are weighted accordingly. Shares for British Airways, KLM, Air France and Austrian are estimates from their routes; smaller operators (Corendon, AJet, Croatia Airlines and others) are not modelled. Sources: Stuttgart Airport facts and figures, Stuttgarter Zeitung: top destinations, Schwäbische Zeitung: airline shares, airliners.de: 2025 figures.

TYPICAL ROUTES

Runway 25

FROM	INSTRUCTION
Stands 9-29, pushed facing east	`taxi to holding point A via M, H, N`
Stands 30-36, 60-65, pushed facing west	`taxi to holding point A via M, H, N`
Stands 71-75 (pushed onto N facing east)	`taxi to holding point A via N`
Stands 40-48 (drive-through)	`taxi to holding point A via O, N`
Stands 50-56 (drive-through)	`taxi to holding point A via N`
Apron South, pushed facing west	`taxi to holding point A via V, Y, K, N, cross runway 25` (crossings only between K and Y)
Arrival vacated via F, G or E	`taxi to stand 14 via S, H, M` / `taxi to stand 72 via S, H, N`
Arrival vacated via H	`taxi to stand 33 via H, M`
Arrival vacated via W (south)	`taxi to stand 206 via V`

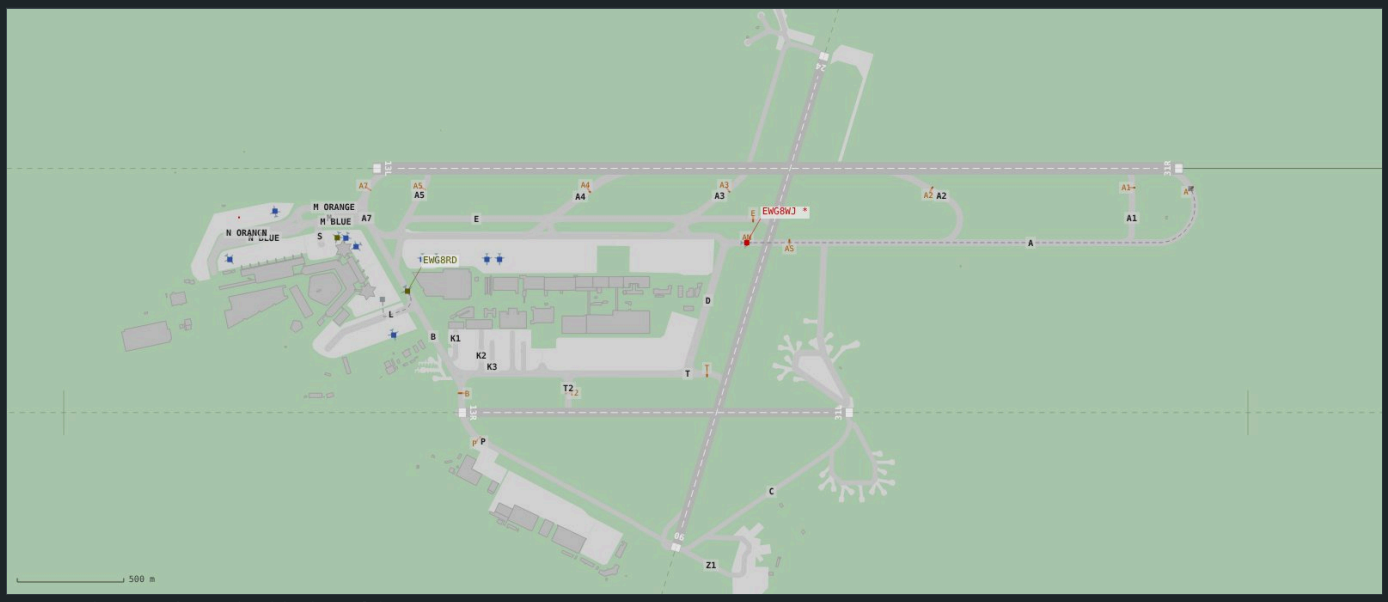
Runway 07

FROM	INSTRUCTION
Stands 9-19, pushed facing east	`taxi to holding point K via M, L2`
Stands 9-19, pushed facing west	`taxi to holding point K via M, L3, O`
Stands 30-36, pushed facing west	`taxi to holding point K via M, L2`
Stands 24-29, pushed facing west	`taxi to holding point K via M, L2`
Apron South, pushed facing west	`taxi to holding point Y via V`
Arrival vacated via D, C or E	`taxi to stand 30 via N, H, M` / `taxi to stand 14 via N, H, M`

Ground planning tips

- Taxilane **M** is a single lane used in both directions. Push departures so that they leave in the direction they will taxi. Use `give way` or conditional clearances (`behind the A320 passing left to right, push and start approved`) when arrivals are coming in on M.
- The **circle** west of N/S (a hot spot on the chart) joins N, S, O, K, I and L2. Avoid sending two aircraft through it at the same time.
- Keep the vacate points behind the exits free. The Tower won't use an exit whose vacate point is blocked, which costs runway capacity.

8 EDDK - COLOGNE/BONN



▸ EDDK at the start of a session (phx-day style)

ITEM	VALUE
ICAO / IATA	EDDK / CGN (Köln/Bonn Airport "Konrad Adenauer")
Location	Cologne, North Rhine-Westphalia, Germany
Reference point	50°51'57.30"N 007°08'33.88"E (AIP: 4 m north of the 06/24 centre line, 1469 m from threshold 06)
Elevation	300 ft (AIP)
Magnetic variation	3.7° E (AIP: 3.2° E 2024.04, +0.2° per year)
Transition altitude	5000 ft; initial climb of the SIDs 5000 ft
Runways	**13L/31R**, 3815 m x 60 m, true heading 137.5° / 317.5° (all simulated traffic); **06/24**, 2459 m x 45 m, true heading 064.4° / 244.4°, and **13R/31L**, 1863 m x 45 m (code C) - drawn and crossed, not used for take-offs and landings. Renamed from 14/32 for the magnetic variation
Data files	`src/data/airports/eddk.ts`, layout `eddk.layout.ts` (generated)

The simulator shows a condensed version of this page as the **airport briefing** (data in `briefing` in `eddk.ts`).

DATA SOURCES AND ACCURACY

DATA	SOURCE
Taxiways, holding points, aprons, buildings, stand lead-in lines	OpenStreetMap - © OpenStreetMap contributors, ODbL 1.0. Imported with `scripts/osm-import.mjs`; the derived layout is available under the ODbL
Runway thresholds and elevations, aerodrome elevation, reference point and magnetic variation, declared distances and intersection take-off distances, stand positions (AD 2.8 checkpoints of aprons A-F and W), taxiway names, widths and limits, hot spots	DFS AIP Germany AD 2 EDDK (AIRAC 2026OCT01) with its aerodrome and ground movement charts, used as a reference only: facts digitised by hand, no chart content copied
Frequencies, SIDs, runway use, taxi flows, stand sizes and stand use	VATSIM Germany Köln/Bonn SOP and stand data (facts only; a simulation resource, not an official source), checked against AIP AD 2.18 and the AIP SID charts
Traffic	Cologne/Bonn Airport route network 2026 (Wikipedia), ADV statistics (116,000 movements and 842,000 t freight in 2025); operator shares and the cargo share (about a quarter of the movements) are estimates

Simplified or not modelled:

- all take-offs and landings use runway **13L/31R** (as on a normal VATSIM day); 06/24 and 13R/31L are only crossed - departures 31R and arrivals vacating 13L at A2 / A1 cross runway 06/24 on A next to the reference point (holding points AN / AS, hot spot 1),
- the holding points carry the names of their taxiways as on VATSIM (the charts give no designators); AN / AS (A north / south of runway 06/24) are simulator names; the CAT II/III holding positions are not modelled separately,
- **apron U** (maintenance; general aviation parks there too): U10-U26 at their AIP AD 2.8 positions (U10-U14 up to code C, U16-U26 code A), reached across runway 13R/31L at B (holding points B north and P south - no designators on the charts) and along P; the lead-in lines of U10, U12 and U14 leave P where the ground movement chart shows them (digitised by hand). Business jets (registration callsigns) park there,
- the military area south-west of runway 13R/31L (C, Z1, the MIL ramps; only with prior agreement with the Bundeswehr), P beyond apron U to the 06 end, taxiway D east of runway 13L/31R to the 24 end with apron L (the dispersal pads there), and the taxilanes K1-K3 of apron V are drawn but not used: the ground movement chart shows apron V under construction with all parking positions removed. OSM still calls P "B",
- GAT 3 (light aircraft parking G01 / G02 by B) is drawn; light aircraft are not simulated,
- the blue and orange guide lines on M and TXL-N (two aircraft up to 36 m side by side) are taxiways of their own in the simulator (M BLUE, M ORANGE, N BLUE, N ORANGE): blue outbound, orange inbound (a simulator flow, not a published rule); the yellow centre lines are M and N,
- stand sizes follow the VATSIM Germany stand data (code C 36 m, D 52 m, E 65 m, F 80 m); A25-A35 (odd) are nose-out positions (the AIP checkpoint is taken as the nose wheel stop position facing L, the aircraft stands 15 m behind it); aprons E and F (cargo hub) and W (cargo feeders) take the freighters first, the B747-8 only fits E and F,
- taxi-in behind a follow-me car (AIP) and de-icing on the stand are not enforced,
- the airport is open 24 hours; the night rules of AIP AD 2.20 (noisy aircraft, runways 13R/31L, 06 and 24, intersection take-offs only from A5 / A1) are not modelled,
- no military, general aviation or light aircraft traffic,
- traffic: about the volume of Stuttgart per density step (`rateFactor` 1), a quarter of it cargo.

Never use this simulator for real-world navigation.

RUNWAYS AND APPROACHES

RUNWAY	TRUE HEADING	TORA / TODA / ASDA / LDA (M)	INTERSECTION TAKE-OFFS	APPROACHES
13L	137.5°	3815 / 3875 / 3815 / 3815	A5: 3587, A4: 2743, A3: 2121	ILS CAT II/III, LOC, RNP
31R	317.5°	3815 / 3875 / 3815 / 3815	A1: 3586, A2: 2562, A3: 1781	ILS CAT II/III, LOC, RNP
06 / 24	64.4° / 244.4°	2459 each	(06 from T 1625, 24 from E 1643 / A 1526)	06: RNP; 24: ILS, LOC, RNP
13R / 31L	137.5° / 317.5°	1863 each	(13R from T2 1354)	RNP (code C)

Elevations: aerodrome 300 ft, thresholds 13L: 229 ft, 31R: 300 ft, 06: 229 ft, 24: 271 ft, 13R: 220 ft, 31L: 258 ft. Runway 31R is preferred (up to 5 kt tailwind, VATSIM SOP).

STATIONS

STATION	CALLSIGN	FREQUENCY
Delivery	Koeln/Bonn Delivery	121.855
Ground	Koeln/Bonn Ground	121.730
Tower	Koeln/Bonn Tower	124.980
Approach	Langen Radar	135.350
ATIS	Koeln/Bonn Information	132.130

LAYOUT

All civil aprons lie between the two parallel runways, north of runway 06/24:

TAXIWAY	DESCRIPTION
A	The main parallel taxiway south-west of 13L/31R: from the B / M junction at the C pier along aprons E and F, across runway 06/24 next to the reference point (AN / AS, hot spot 1), to the 31R end (holding point A)
E	Second parallel between A and the runway, from A5 / A7 to runway 06/24
A1-A5, A7	Links to runway 13L/31R: A7 and A5 at the 13L end, A1 at the 31R end; A2 (rapid exit landing 13L), A3 (rapid exit landing 31R, hot spot 2) and A4 in between
M	From A7 / A to Terminal 2, with a yellow centre line (all sizes) and the blue and orange guide lines **M BLUE** / **M ORANGE** (up to 36 m)
N (TXL-N)	Taxilane of apron D (Terminal 2): centre line N and the guide lines **N BLUE** / **N ORANGE** (up to 36 m)
L (TXL-L), **S** (TXL-S)	Taxilanes of apron A (Terminal 1) and apron C (C pier, code C)
B	From aprons A / B over the taxiway bridge (up to 52 m) to T and the 13R end; code E
T, **T2**, **D**	T along the north side of 13R/31L (code E), T2 to 13R/31L (code C), D along runway 06/24 past apron W to A
P	From the B crossing of 13R/31L south along apron U (used), on to the 06 end (drawn)
C, Z1, K1-K3	The military area south-west of 13R/31L and apron V - drawn only

STANDS	USE IN THE SIMULATOR
A01-A35 (27)	Apron A, Terminal 1; up to 36 m, A01-A07, A11, A17, A21 and A23 up to 65 m; A25-A35 (odd) nose-out
B10-B30 (17)	Apron B, Terminal 1; 36-65 m (B11, B17, B21, B23 up to 65 m)
C01-C14, C50 (10)	Apron C, the C pier; up to 36 m, C50 up to 65 m
D10-D64 (32)	Apron D, Terminal 2, along N, N BLUE and N ORANGE; mostly up to 36 m, D15, D19, D23 up to 65 m
E09-E46 (27), F17-F34 (12)	Cargo aprons along A (UPS, FedEx, DHL); 36-80 m - the B747-8 (68.4 m) only here
W10-W34 (13)	Apron W along D (cargo feeders); up to 52 m

PROCEDURES

- **Runway 31R** (preferred): departures taxi south on **A**, **cross runway 24** at AN / AS (`cross runway 24`) and line up at **A** (3815 m) or **A1** (3586 m); intersections A2 and A3. Arrivals vacate at **A3** (rapid exit), A4, A5 or A7 and taxi north on **E**: the flow is A outbound (south-east), E inbound (north-west).
- **Runway 13L**: departures line up at **A7** (3815 m) or A5, A4, A3. Arrivals vacate at **A2** (rapid exit), A1 or A at the end and taxi north-west on A, across runway 06/24.
- Terminal 2: outbound on **N BLUE / M BLUE**, inbound on **N ORANGE / M ORANGE** (`taxi to runway 13L via november blue, mike blue`); aircraft of 36 m and more use the centre lines N and M (automatic routes of smaller aircraft keep to the guide lines and use the centre lines only for short links).
- SIDs climb to 5000 ft: WIPUR (north), PODIP (north-east), KUMIK (east, also the south-bound traffic), NVO (west); COL only for Frankfurt. Runway 13L: the F SIDs; runway 31R: R for aircraft up to 98 t, B for heavier ones (filed for the long-haul freighters). The numbers change with the AIRAC cycle.
- A-CDM (TSAT / TOBT) and datalink clearances (DCL) as on VATSIM.

TRAFFIC

Eurowings (the home carrier), Ryanair, Corendon, Pegasus, SunExpress, Turkish Airlines, Lufthansa, Austrian, Wizz Air, British Airways and Condor; cargo: UPS (European hub: B747-8, B747-400, B767, A300, B757), FedEx (B777F, B767) and DHL / European Air Transport (A300, B757). Shares are estimates.

9 EDDN - NUREMBERG



▸ EDDN at the start of a session (phx-day style)

ITEM	VALUE
ICAO / IATA	EDDN / NUE (Albrecht Dürer Airport Nuremberg)
Location	Nuremberg, Bavaria, Germany
Reference point	49°29'55.32"N 011°04'40.83"E (AIP)
Elevation	1044 ft (AIP)
Magnetic variation	4.3° E (World Magnetic Model, 2026)
Transition altitude	5000 ft; initial climb of the RNAV SIDs FL70
Runway	**10/28**, 2700 m x 45 m, true heading 098.6° / 278.6°; **28 preferred**
Data files	`src/data/airports/eddn.ts`, layout `eddn.layout.ts` (generated)

The simulator shows a condensed version of this page as the **airport briefing** (data in `briefing` in `eddn.ts`).

DATA SOURCES AND ACCURACY

DATA	SOURCE
Runway, taxiways, holding points (names from the OSM tags "Rollhalt Alpha" ...), stands, aprons, paved taxiway areas, helipads, buildings (all buildings on the aerodrome, the terminal included)	OpenStreetMap - © OpenStreetMap contributors, ODbL 1.0. Imported with `scripts/osm-import.mjs`; the derived layout is available under the ODbL
Paved areas OSM does not have: the southern half of Apron G2 / G2A, the J corridor between the aprons ("Entry GAT"), the apron strips along S3, S4 and at Hangar 8, the west edge of the GA apron, the terminal frontage, the Hangar A forecourt, the FAI and Aero-Dienst aprons, and the lane from S3 into the Aero-Dienst apron	Digitised by hand from the ground movement chart (AD 2 EDDN 2-7, AMDT 01/26), georeferenced on its printed graticule (approximate, a few metres). The FAI and Aero-Dienst aprons are the outlines of their areas of responsibility (the chart does not shade them; their paving is inferred). G3 is grass parking: not drawn
Reference point, elevations, thresholds, declared distances, approaches, taxiway widths and code limits (J, S3, S4, T2), the name M2 of the link at the east end of the apron, hot spot, helicopter stands	DFS AIP Germany AD 2 EDDN (AIRAC 2026OCT01): text pages, aerodrome chart AD 2 EDDN 2-5 (AMDT 10/26) and ground movement chart 2-7 (AMDT 01/26), used as a reference only - facts read by hand, no chart content copied. The stand positions of AD 2.8 match the OSM stands within 4 m
Frequencies, SIDs, stand use, taxiway restrictions, preferred runway, exits, intersections	VATSIM Germany Nürnberg SOP (facts only; a simulation resource, not an official source); frequencies checked against the VATSIM Germany station data and vATIS profile (2026-10-21), SID designators against a public chart index
Exit fix positions (for the SID choice), magnetic variation	public waypoint lists; NOAA World Magnetic Model (October 2026)

Simplified or not modelled:

- the general aviation areas: the lanes S3 (to Apron G2 / G2A), S4 (west to the FAI apron) and T2 (to aprons U1 / U2) are drawn and can be taxied by small aircraft (code B), but only the business jet stands 81-86 on M4 / S4 are parking positions; the helipads are drawn, helicopters are not simulated,
- the CAT II/III holding points with stop bars (A, B, C/F and J; runway 28 only) are plain junctions - no low visibility operations,
- code D / E aircraft taxi without follow-me (real: PPR and the "Higher Code Letter Aircraft" procedure with a follow-me from the east end of F or from A abeam stand 27),
- the code limits of B, C, D, E and N2 come from the VATSIM Germany SOP (the AIP publishes those of J, S3, S4 and T2),
- intersection C (AIP: 10 1461 m, 28 1289 m) is published but not used for departures (SOP),
- the helicopter stands H1-H4 are drawn (helipads), helicopters are not simulated; the current aerodrome chart (AMDT 10/26) numbers H1-H3 from east to west (H3 west), the older ground movement chart the other way round,
- SIDs: the RNAV SIDs BOLSI, ERETO, RODIS, SUKAD and SULUS (G = runway 28, C = runway 10; the numbers change with the AIRAC cycle), see SIDs; the local training SID ERL and the conventional NUB SIDs for non-RNAV aircraft are not modelled,
- traffic: estimated operator shares, about 60 % of Stuttgart's movements per density step (`rateFactor` 0.6).

Never use this simulator for real-world navigation.

STATIONS

STATION	CALLSIGN	FREQUENCY
Ground (also clearances; datalink clearances common)	Nuernberg Ground	121.760

STATION	CALLSIGN	FREQUENCY
Tower	Nuernberg Tower	118.305
Approach	Muenchen Radar (sector Franken)	129.525
ATIS	Nuernberg Information	123.080

Nuremberg is not an A-CDM airport (no TSAT); there is no Delivery position.

LAYOUT

TAXIWAY	DESCRIPTION
F	Parallel taxiway between the runway and the aprons, west part: from the 10 end to the junction with C, M2 and N2
A	Parallel taxiway, east part: from the M2 / B junction to the 28 end
F, E, D, C, B, A	Connectors to the runway with their holding points; F and A at the runway ends (full length 10 / 28), E, D (10) and B (28) intersections, D (code D) and C (code C) exits
J	From the west apron (M3 / M4 / N4) up to F: code C from F to M4 (code D / E only cross it between M3 and M4 / N4), code B from M4 south to the GA entry
M1-M4	Apron taxilanes along the terminal stands (M2, M3), the cargo apron (M1) and the west apron (M4); M2 also links the east end of the apron north to A, B and N2
N2, N4	Lanes behind the remote stands 30-35 (N2) and 40-44 (N4), used only to enter them
S3, S4	GA apron lanes south of M4 (code B): S3 east to Apron G2 / G2A and the GA entry, S4 west to the FAI apron
T2	From S4 south to the GA aprons U1 / U2 (code B)

STANDS	USE IN THE SIMULATOR
01-06, 09-16	Terminal (jet bridges 10-13), nose south, taxi in and push back onto M1 / M2 / M3 (01 for turboprops)
30-35	Remote, in via N2, nose south, out via M2 (code C)
40-44 (40D, 41D, 43D code D; 40E code E)	West apron, in via N4, out via M4
26-28 (26D, 28D code D; 27E code E)	Cargo, in via M1, nose north, out via A (used for airliners only when nothing else is free)
81-86	Business jets: 81, 83, 85 nose north (in from S4, out via M4), 82, 84, 86 nose south with pushback onto M4

Taxiways B, C, E and N2 are limited to code C (36 m), D to code D (52 m), S3 / S4 / T2 to code B (24 m), J as above. Widths (AIP AD 2.8): A, B, C, E, F, J 22.5 m, D and M1-M4 23 m, N4 21 m, N2 18 m, S3 / S4 / T2 10.5 m.

RUNWAYS AND APPROACHES

RUNWAY	THR ELEVATION	TORA (M)	INTERSECTION TAKE-OFF (TORA, M)	APPROACHES
10	1020 ft	2700	E 2300, D 1900 (C 1461 not used)	**ILS CAT I** (GP 3.0°), LOC, VOR, RNP
28	1044 ft	2700	B 1962 (C 1289 not used)	**ILS CAT II / III** (GP 3.0°), LOC Z, LOC Y, RNP

Aerodrome elevation 1044 ft; the thresholds are not displaced (the paved areas beyond them are tagged stopway in OSM, the AIP lists no stopway).

PROCEDURES

- **Runway 28 is preferred**; the direction changes only with more than 5 kt tailwind.
- **Departures**: 28 from A (full length) or intersection B; 10 from F, or intersections E and D. Aircraft up to 2 t are normally cleared from B / E (AIP). C is not used for departures.
- **Hot spot**: the junction where M3 joins M2 (next to C, F and helicopter stand H4) - "complex taxiway layout, check taxi clearance".
- **Arrivals vacate not earlier than C** (a departure may be waiting at B or E): runway 28 via C, D, E or F (end), runway 10 via C, B or A (end).
- Fill the remote stands 30-35 and 40-44 from the far end: N2 and N4 are blocked by the aircraft parked on them.

SIDS

Each flight plan is filed via the exit fix towards its destination (great-circle bearing from Nuremberg, a simulator approximation of the route network). After take-off all departures contact München Radar 129.525.

EXIT FIX	RUNWAY 28 / 10	USED FOR (SIMULATOR)
ERETO	ERETO1G / ERETO1C	000-060° (Berlin, Leipzig, Dresden, Scandinavia)
RODIS	RODIS1G / RODIS1C	060-150° (Prague, Vienna, Budapest, Balkans, Greece, Turkey, Egypt)
BOLSI	BOLSI1G / BOLSI1C	150-250° (Munich, Italy, Switzerland, Spain, Portugal, Canaries)
SUKAD	SUKAD1G / SUKAD1C	250-320° (Frankfurt, Rhine-Ruhr, Paris, Benelux, UK, Ireland)
SULUS	SULUS1G / SULUS1C	320-360° (Hamburg, Bremen, Hannover; only for flights continuing on L604)

TRAFFIC

Eurowings, Ryanair, Lufthansa (Frankfurt) and Lufthansa CityLine, TUI, Condor, SunExpress, Pegasus, Turkish, KLM Cityhopper, Air France, Wizz Air and business jets; shares are estimates from the route network.

10 EDDP - LEIPZIG/HALLE



▸ EDDP at the start of a session (phx-day style)

ITEM	VALUE
ICAO / IATA	EDDP / LEJ (Leipzig/Halle Airport)
Location	Schkeuditz, between Leipzig and Halle, Saxony, Germany
Reference point	51°25'26.37"N 012°14'10.98"E (AIP)
Elevation	469 ft (AIP)
Magnetic variation	5.0° E (AIP: 4.9° E 2026.01, +0.2° per year)
Transition altitude	5000 ft; initial climb of the SIDs FL70
Runways	**08L/26R**, 3600 m x 45 m (north), and **08R/26L**, 3600 m x 60 m (south), true heading 085.7° / 265.8°, 2.1 km apart - **both in use at the same time**, for take-offs and landings
Data files	`src/data/airports/eddp.ts`, layout `eddp.layout.ts` (generated)

The simulator shows a condensed version of this page as the **airport briefing** (data in `briefing` in `eddp.ts`). How the simulator handles two runways in use is described in parallel runways.

DATA SOURCES AND ACCURACY

DATA	SOURCE
Taxiways, holding points, aprons, buildings, stand lead-in lines	OpenStreetMap - © OpenStreetMap contributors, ODbL 1.0. Imported with `scripts/osm-import.mjs`; the derived layout is available under the ODbL
Runway thresholds and elevations, aerodrome elevation, reference point and magnetic variation, declared distances and intersection take-off distances, stand positions (AD 2.8 checkpoints), taxiway widths, rapid exits (RETIL), parallel runway operations, frequencies	DFS AIP Germany AD 2 EDDP (AIRAC 2026OCT01) with its aerodrome chart and SID charts, used as a reference only: facts read by hand and with text recognition from the published pages, no chart content copied
Runway assignment by apron and size, operating modes, stand use and the bay sizes of apron 4, airlines	VATSIM Germany airport page Leipzig/Halle (facts only; a simulation resource, not an official source)
Traffic	Leipzig/Halle Airport route network 2026 (Wikipedia), ADV statistics (about 85,000 movements and 1.4 million t of freight in 2024); operator shares and the cargo share (about two thirds of the movements) are estimates

Simplified or not modelled:

- both runways are used in **mixed mode** with **dependent parallel approaches** (1.5 NM diagonal, AIP AD 2.22); the AIP also allows segregated operations, VATSIM uses staggered approaches (3 NM) without an Arrival controller - the simulator offers the parallel runways together or one runway alone,
- one Tower position works both runways (`Leipzig Tower` 121.105; the northern runway has its own frequency 125.955, cross-coupled when one controller works both, as on VATSIM),
- the holding points carry the names of their taxiways (the charts give no designators); the CAT II/III holding positions are not modelled separately, except **H1** (08R from the north), which only exists as a CAT II/III position far from the runway (hot spot 1); A2 joins A1 before the runway,
- stands: the AIP positions of aprons 1-5 without the alternative positions (L / R / W / M and the southerly aligned ones); apron 4 sizes after the VATSIM Germany bays (A 401-412 52 m, B 421-436 36 m, C 441-452 52 m, D 461-478 52 m, E 481-487 80 m); apron 1 (101-111 36 m taxi-out positions, 112-121 and 137-143 business jets, jetway stands 126-136 52 m), apron 2 (65 m), apron 3 (80 m) and apron 5 (65 m) are estimates,
- apron 2 has no taxilane in OSM: the simulator adds one along its stands (part of V ORANGE); apron 3 gets lead-in lines from F; stands at the dead end of an apron 4 taxilane push back from the node before it (a general rule of the OSM layouts),
- the orange guide line L between stands 101-111 (follow-me only), the de-icing pads and the engine run-up stand are drawn but not used,
- taxi flows (simulator convention): direction 26 eastbound on N and T, direction 08 westbound - departures from apron 1 leave via the bridges E7 / E8 (26) or W1 (08), arrivals come the other way; the two bridges are one-way in both directions: E7 northbound (to C), E8 southbound,
- K leads west to the noise abatement facility (engine run-ups) and G2 south from C to the small pad by de-icing pad DP2, as on the aerodrome chart; the stub south-east from the H1 / N junction is P, closed (chart correction "TWY P closed"); the de-icing pads DP2-DP9 are not modelled,
- night: the airport is open 24 hours for cargo; the AIP restrictions for passenger flights (from 23:30 local) are not modelled; HIRO exits at night (S6 / S4) are not enforced,
- no general aviation, light aircraft or Antonov An-124 (no type data).

Never use this simulator for real-world navigation.

RUNWAYS AND APPROACHES

RUNWAY	TRUE HEADING	TORA (M)	INTERSECTION TAKE-OFFS (TORA)	APPROACHES
08L	085.7°	3600 (A1, A2)	A3 2450, A4 1850, A5 1745	ILS Z / Y CAT II/III, LOC, RNP
26R	265.8°	3600 (A9, A8)	A7 2450, A6 1850, A5 1745	ILS Z / Y CAT II/III, LOC, RNP
08R	085.7°	3600 (S1, H1)	S2 3270, S3 2450, S4 1850, H5 / S5 1595	ILS Z / Y CAT II/III, LOC, RNP
26L	265.8°	3600 (S10)	S9 3270, H8 2885, S8 2850, H7 2760, S7 2725, S6 2250, H5 / S5 1895	ILS Z / Y CAT II/III, LOC, RNP

TODA = TORA + 300 m everywhere (AIP AD 2.13). Thresholds: 08L 417 ft, 26R 447 ft, 08R 427 ft, 26L 469 ft. Direction **26** is preferred (up to 5 kt tailwind).

STATIONS

STATION	CALLSIGN	FREQUENCY
Delivery	Leipzig Delivery	121.680
Ground	Leipzig Ground	121.805
Tower	Leipzig Tower	121.105 (south), 125.955 (north)
Approach	Muenchen Radar	126.065 (north), 126.175 (south)
ATIS	Leipzig Information	123.955

Datalink clearances (DCL) and A-CDM as at the real airport and on VATSIM.

LAYOUT

TAXIWAY	DESCRIPTION
C	Parallel taxiway south of the northern runway; **A1-A9** link it to the runway (A4 rapid exit landing 26R, A6 landing 08L); **B**, **G1**, **G2** short links at both ends
W1, **E7 / E8**	From C south to the terminal: W1 in the west (to H1 and the 08R end), the two bridges E7 and E8 in the east (to H7 / H8 and the southern runway)
N	Along apron 1 (terminal) from W1 to E7 / E8; **L** and **M** taxilanes of apron 1; **H2**, **K** to the west; **H5** from N to the southern runway
T, **V**	Along the south side of the southern runway: T next to the runway, V along the cargo aprons 2, 4 and 5; **S1-S10** link them to the runway (S3 / S4 rapid exits landing 26L, S6 landing 08R), **Y** links T and V. S2, S3 and S4 fork into a branch to either side of T across one holding position line (chart: one holding point each), so arrivals vacating 26L turn onto T east or west
Z1-Z5	Taxilanes of apron 4 (DHL hub), each with an orange and a blue guide line (Z1.1 / Z1.2 ...)
F	Apron 3 north of the terminal

STANDS	USE IN THE SIMULATOR
101-111	Apron 1, airliners up to 36 m, taxi in from L, taxi out onto N
112-121, 137-143	Apron 1, business jets
126-136	Apron 1, jetway stands at the terminal, up to 52 m, pushback onto N
201-222	Apron 2, cargo outside the DHL network, up to 65 m
301-308	Apron 3, large freighters, up to 80 m
401-487	Apron 4, the DHL hub, bays A-E (36-80 m)
501-506	Apron 5, cargo, up to 65 m

PROCEDURES

- **Runway assignment** (VATSIM): departures from aprons 1 and 3 use the northern runway, from aprons 2, 4 and 5 the southern one; A340-600, B777 and B747 always the southern one. Arrivals land on the runway of their stand; Approach uses the other one when its final is full. The SID letter names the runway: **N** 26R, **S** 26L, **Q** 08L, **E** 08R.
- **Direction 26**: departures from apron 1 taxi east on N, over E7 / E8 to C and east to A9 / A8; from apron 4 east on V / T to S10. Arrivals 26R vacate at A4 or A3 and taxi via W1 to apron 1; arrivals 26L vacate at S4 / S3 and taxi east on V to apron 4.
- **Direction 08**: departures from apron 1 taxi west on N to W1 and A1; from apron 4 west on T to S1. Arrivals 08L vacate at A6 / A7 and come back over E7 / E8; arrivals 08R vacate at S6 - S8.
- Arrivals to the other side of the airport cross the other runway (`cross runway 26L`), only when that runway is free.
- SIDs climb to **FL70**: PENEM (north) and ORTAG (west) - Muenchen Radar 126.065; NEVKO (south-west), GOLAT (south) and DRN (Dresden, east) - Muenchen Radar South 126.175. The numbers change with the AIRAC cycle.

TRAFFIC

The DHL hub: European Air Transport (the hub airline: A300, B757, B767, A330), DHL Air UK, AeroLogic (B777F), ASL Ireland and Star Air for the network, Kalitta, Atlas, Polar and MNG for the long-haul and partner cargo; passenger flights by Eurowings, Ryanair, Condor, SunExpress, Corendon, Pegasus, Lufthansa, Austrian, Turkish Airlines, Sundair and business jets. Shares are estimates.

11 EDDW - BREMEN



▸ EDDW at the start of a session (phx-day style)

ITEM	VALUE
ICAO / IATA	EDDW / BRE (Bremen Airport "Hans Koschnick")
Location	Bremen, Germany
Reference point	53°02'50.64"N 008°47'12.29"E (AIP: 35 m north of the runway, near taxiway C)
Elevation	14 ft (AIP)
Magnetic variation	3.9° E (AIP, 2026.10)
Transition altitude	5000 ft; initial climb of the SIDs 4000 ft
Runways	**08/26**, 2040 m x 45 m, true heading 087.8° / 267.8° (renamed from 09/27); **23**, 700 m x 23 m (light aircraft, take-off only - drawn, not used)
Data files	`src/data/airports/eddw.ts`, layout `eddw.layout.ts` (generated)

The simulator shows a condensed version of this page as the **airport briefing** (data in `briefing` in `eddw.ts`).

DATA SOURCES AND ACCURACY

DATA	SOURCE
Taxiways, holding points, aprons, buildings, stand lead-in lines, the paved areas beyond the thresholds, runway 23	OpenStreetMap - © OpenStreetMap contributors, ODbL 1.0. Imported with `scripts/osm-import.mjs`; the derived layout is available under the ODbL
Runway thresholds and elevations, aerodrome elevation, reference point and magnetic variation, declared distances and intersection take-off distances, stand positions (AD 2.8 checkpoints), taxiway widths and limits, night rules	DFS AIP Germany AD 2 EDDW (AIRAC 2026OCT01) with its aerodrome and ground movement charts, used as a reference only: facts digitised by hand, no chart content copied
Frequencies, SIDs, runway use, stand use, arrival spacing	VATSIM Germany Bremen SOP and station data (facts only; a simulation resource, not an official source), checked against AIP AD 2.18 and the AIP SID charts
Traffic	Bremen Airport route network 2026 (Wikipedia), ADV statistics (28,000 movements in 2024); operator shares are estimates

Simplified or not modelled:

- the holding points carry the names of their taxiways (the charts give no designators); departures 26 are cleared via F only up to the CAT II/III holding position (noise rule, AIP AD 2.20) - that is holding point F in the simulator,
- C, D and E meet the runway in a Y: the simulator routes on one fillet of each (C: the eastern one, for departures 08; D and E: the western ones, for exits landing 08 and departures 26) and draws the other one,
- stands 12A and 18A (wide-bodies up to 52 m) are pushback stands here; stands 12-17 are taxi-in / taxi-out stands between N and S (ground movement chart); their AIP positions (and that of 12A) are taken as the nose wheel stop positions next to S, so the aircraft stand 15 m (12A: 25 m) behind them, clear of S,
- the paved sections beyond both thresholds (stopways, the special take-off runways of the Airbus Beluga), runway 05/23 (700 m, drawn as a runway: VFR take-offs only), the Airbus taxiways F1 / F2, the lanes to the LH parking and the helicopter platform and the inner GA lane of Apron 2 are drawn but not used,
- intersection take-offs 08 from D (808 m) and 26 from C (836 m) are published but not offered (VATSIM SOP),
- Beluga flights, light aircraft, helicopters and the general aviation parking along R are not simulated,
- night rules (AIP AD 2.20): jet take-offs not between 22:30 and 06:00, landings until 22:30 and delayed ones until 24:00 local; the exceptions for quieter aircraft and home carriers are simplified to these times,
- traffic: estimated operator shares, about 40 % of Stuttgart's movements per density step (`rateFactor` 0.4).

Never use this simulator for real-world navigation.

RUNWAYS AND APPROACHES

RUNWAY	TRUE HEADING	TORA / TODA / ASDA / LDA (M)	INTERSECTION TAKE-OFFS	APPROACHES
08	087.8°	2040 / 2100 / 2040 / 2040	C: 1235	ILS CAT II/III, LOC, RNP
26	267.8°	2040 / 2100 / 2040 / 2040	E: 1714, D: 1255	ILS CAT II/III, LOC, RNP

Elevations: aerodrome 14 ft, thresholds 08: 13 ft, 26: 14 ft. Westerly winds prevail: runway 26 is in use more often.

STATIONS

STATION	CALLSIGN	FREQUENCY
Ground (also IFR clearances)	Bremen Ground	121.755
Tower	Bremen Tower	120.330
Approach	Bremen Radar	124.800
ATIS	Bremen Information	132.380

Bremen has no Delivery: Ground gives the IFR clearances (VATSIM SOP).

LAYOUT

No full parallel taxiway (AIP AD 2.8: 23 m wide unless noted):

TAXIWAY	DESCRIPTION
A	From the north-west corner of Apron 1 (junction of N, S, R and L) south-west to the 08 end - the only way there; exit for jets landing 26
F	From the C / G / K junction east along the runway to the 26 end
C	From the east end of N on Apron 1 south across F to the runway (intersection departures 08); continues south of the runway as M
D, E	From F to the runway (intersection departures 26, exits landing 08); below 36 m wingspan
G	From the south end of S diagonally to the C / F / K junction; below 52 m
H	From the east end of Apron 1 to F; below 36 m
K	From R (Apron 2) to the C / F / K junction; below 24 m
N	Taxilane along stands 01-11 (Apron 1); below 52 m
S	Taxilane south of stands 12-17 to Apron 2 (stands 18, 18A, 19); below 36 m
R	Taxilane of Apron 2 (general aviation), code B
L	To Apron 3 (hangars) in the west; up to 31 m
M	South of the runway to the maintenance area (holding point M); not used by the simulated traffic
F1, F2	To the Airbus factory, Airbus traffic only - drawn

STANDS	USE IN THE SIMULATOR
01-11	Apron 1 at the terminal, pushback onto N, up to 36 m
12-17	Apron 1, taxi in from N, taxi out onto S, up to 36 m; 14-16 also for larger business jets
12A	Apron 1, wide-bodies up to 52 m, pushback onto N
18, 19	Apron 2, pushback onto S, up to 36 m
18A	Apron 2, wide-bodies up to 52 m, pushback onto the G / S junction

PROCEDURES

- **Runway 26** (preferred): departures taxi via N (or S and G, or H) to F and along F to holding point F (`taxi to runway 26`); intersections E (1714 m) and D (1255 m) on request. Jets landing 26 vacate at the end via **A** (C only after coordination) and taxi back along A to the apron.
- **Runway 08**: departures taxi via N or S to the north-west corner and along A to holding point A; intersection C (1235 m). Arrivals vacate at **D** or **E** (below 36 m) or **F** at the end.
- One SID per exit fix and runway, all climbing to 4000 ft (M = runway 08; on 26 the Z variants where they exist, otherwise L): ABDIM and OKWEL (west), WSN (north), GESTO (east), NIE (south-east), WIMCI (south). The numbers change with the AIRAC cycle.
- Approach spaces arrivals at least **6 NM** (`arrivalSpacingNm`, VATSIM SOP target spacing).

TRAFFIC

Ryanair (the largest operator), Lufthansa and Lufthansa CityLine (Munich), KLM Cityhopper (Amsterdam), Eurowings, Austrian, Swiss, Turkish Airlines, Pegasus, SunExpress, Sundair, Corendon and business jets; shares are estimates.

12 EDJA - MEMMINGEN



▸ EDJA at the start of a session (phx-day style)

ITEM	VALUE
ICAO / IATA	EDJA / FMM (Allgäu Airport)
Location	Memmingerberg, Bavaria, Germany
Reference point	47°59'19.76"N 010°14'22.14"E (AIP AD 2.2)
Elevation	2079 ft (AIP); thresholds 06: 2079 ft, 24: 2053 ft
Magnetic variation	4.1° E (World Magnetic Model, 2026; AIP: 3.2° E, 2025.01)
Transition altitude	5000 ft; initial climb of the SIDs FL70
Runway	**06/24**, 45 m wide, true heading 059° / 239°. 2630 m (AIP AD 2.12; 2629 m between the OSM runway ends); the threshold 24 is displaced by 229 m, runway 06 starts at the threshold (the area west of W is a stopway)
Data files	`src/data/airports/edja.ts`, layout `edja.layout.ts` (generated)

The simulator shows a condensed version of this page as the **airport briefing** (data in `briefing` in `edja.ts`).

DATA SOURCES AND ACCURACY

DATA	SOURCE
Taxiway R, the western part of P, P4, Apron 4, the pad south of P near E, all closed parts (crosses)	Digitised by hand from the AIP aerodrome chart (AD 2 EDJA 2-5, used as a reference only; no chart content is included). Approximate: the chart was aligned to the OSM runway ends, positions are good to a few tens of metres
Runway, taxiways, holding positions, stands, aprons, buildings	OpenStreetMap - © OpenStreetMap contributors, ODbL 1.0. Imported with `scripts/osm-import.mjs` (see airport data); the derived layout is available under the ODbL
Reference point, elevations, thresholds, declared distances (TORA from every entry), approaches (ILS CAT I or LOC both ways, RNP, NDB), taxiway widths (AD 2.8: R 23 m, N / W 18 m, C / D / E / P 12 m, P4 11 m)	DFS AIP Germany AD 2 EDJA (AIRAC 2026OCT01), used as a reference only: facts read by hand. S and the GA lanes have no published width (12 m / 11 m in the simulator)
Frequencies, SIDs, taxiway restrictions, backtracks, stand use, de-icing on stands	VATSIM Germany Memmingen SOP (facts only; a simulation resource, not an official source); frequencies checked against the VATSIM Germany station data and vATIS profile (2026-10-21), SID designators against a public chart index
Direction of the exit fixes (for the SID choice), magnetic variation	public waypoint lists and the VATSIM Germany letters of agreement (UMTEX is the transfer point to Switzerland); NOAA World Magnetic Model (October 2026)
Checked against	AIP Germany AD 2 EDJA 2-5 (aerodrome chart) and 2-7 (ground movement chart), used only as a reference: runway length, stand designators P11 / P12, the position of holding point W (before the turn towards the runway), the taxiway topology and the closed parts. The OSM intersections match the chart's take-off distances within a few metres: N 1057 m from the 06 end (chart: 2630 - 1570 = 1060 m), C 878 m (880 m), D 1055 m (1060 m), E 627 m from the 24 end (625 m), threshold 24 displaced by 227 m (229 m). No chart content is included

Accuracy: the OSM parts as good as the OSM mapping (usually a few metres), the hand-digitised parts a few tens of metres. Holding point names are the taxiway names (OSM has no references for them).

Simplified or not modelled:

- all taxiways of the chart are modelled. The south side (P, P4, C, D, E and the unnamed GA lanes) is for small aircraft: the chart limits P to an outer main gear wheel span of 6 m, the simulator uses a 24 m wingspan limit on all of them. The airline traffic therefore never uses them; there are no GA stands on Apron 3 / 4 (the chart shows none),
- the unnamed lane to Apron 2 is named N in the simulator,
- taxiway S is named after OpenStreetMap; the AIP charts leave this closed link unnamed,
- holding points of the south taxiways: C, D and E. C and D fork into two fillets onto P behind their holding point (both ways are used); E's second fillet is not used. P4's intermediate holding position is not modelled,
- the 200 m stopway beyond the runway 06 end is drawn as a paved area (not usable by the simulation); the turn loops around both runway ends are closed (crosses) and drawn as closed taxiways,
- holding point W lies before the turn towards the runway (marked on the chart with a red circle and the note "holding points located before turn towards the RWY"): mentioned in the briefing, not drawn,

- stands from OSM, named after the AIP chart: P11 and P12 at the terminal (Apron 1, OSM refs 1 and 2) with pushback, 3-6 on Apron 2 as taxi-in / taxi-out positions (the SOP has a pushback on 6; in the OSM layout its lane ends at the stand),
- no Delivery: Memmingen Ground gives the IFR clearances (the Delivery position is not offered),
- SIDs: KPT, LUPOL and UMTEX (A = runway 24, B = runway 06; the numbers change with the AIRAC cycle), each flight planned via the fix towards its destination: **KPT** (Kempten) 075-200° - Italy, the Balkans, Greece, Turkey, Romania, Bulgaria, Georgia; **UMTEX** (towards Switzerland) 200-280° - Spain, Portugal, France, Morocco, the Canaries; **LUPOL** 280-075° - UK, Ireland, Benelux, Germany north, Scandinavia, Poland. All contact München Radar 129.450 after take-off. The conventional ALG SIDs (non-RNAV aircraft, training) are not modelled,
- traffic: Ryanair, Wizz Air and a few others with estimated shares, about a third of Stuttgart's movements per density step (`rateFactor` 0.35),
- the de-icing station (Memmingen De-icing 121.930, de-icing on the stands) is not a simulator position,
- low visibility procedures (ILS CAT I only), helicopters, VFR traffic.

Never use this simulator for real-world navigation.

STATIONS

STATION	CALLSIGN	FREQUENCY
Ground (also clearances)	Memmingen Ground	121.680
Tower	Memmingen Tower	126.855
Approach	Muenchen Radar (sector Iller)	129.450
ATIS	Memmingen Information	118.855

LAYOUT AND PROCEDURES

TAXIWAY / POINT	DESCRIPTION
W	From the west end of Apron 2 to the runway 06 end (holding point W, before the turn towards the runway; the simulator follows the turn). Full-length entry for 06, last exit for landings on 24
N	Along the aprons and down to the middle of the runway (holding point N, about 1060 m from the 06 end; take-off run 06 from N 1570 m)
R	From the N / Apron 2 junction straight to the runway, about 475 m from the 06 end (ground movement chart; take-off run 06 from R 2175 m). Used on pilot request (SOP): an intersection for runway 06 departures and an exit for runway 24 landings between N and W
P	Parallel taxiway on the south side, from a cross west of P4 (next to the glide path antenna 06) to E; small aircraft only. West of the cross and east of E it is closed
P4	From P to Apron 4 (south-west, GA); small aircraft only
C, D, E	From P to the runway (holding points C, D, E; take-off run 06 from C 1750 m, from D 1570 m; 24 from E 2005 m, D 1065 m, C 885 m); small aircraft only. Apron 3 (GA parking) lies between C and D
S (closed)	From N (near Apron 1) to the runway, closed (crosses). Crews refuse it: `unable, taxiway S closed`
Closed loops	Around both runway ends, the stub at the threshold 06, P west of P4 and east of E with two links to the runway near the 24 end, and two branches from P to Apron 4 - drawn with crosses, never used
Stands P11, P12	Apron 1 at the terminal, pushback onto N
Stands 3-6	Apron 2, taxi in, turn on the stand and taxi out

Backtracking (the reason Memmingen is in the simulator):

- **Departures 24** taxi to N and need the full length: the crew calls `holding point N, ready for departure, request backtrack`. Clear them with `backtrack runway 24, line up and wait` (or `backtrack runway 24, cleared for take-off`): the aircraft enters at N, taxis along the runway to its end, turns on the runway (the turn loops at the runway ends are closed) and lines up. A take-off clearance without the backtrack gets `unable, we require a backtrack for departure, 1070 metres available from N` (AIP AD 2.13).
- **Departures 06** line up at W (full length), R (intersection, 2175 m) or N (intersection, 1570 m - enough for few airliners).
- **Landings 06** usually roll past N: they slow down, turn around on the runway and backtrack to N (`backtrack, vacate via N` - they do it on their own when N cannot be made).
- **Landings 24** vacate via N if they can make it, otherwise via R or at the end via W.
- While an aircraft backtracks, the runway is occupied for minutes. Approach spaces arrivals at least 10 NM apart when landing aircraft backtrack (runway 06) and 8 NM when departures do (runway 24).
- N is the main taxiway between the aprons and the runway: departures and arrivals meet head-on on it. The AI Ground keeps a departure on its stand while an arrival comes the other way (taxiing in, or about to land and vacate via N). As Ground, do the same.

TRAFFIC

Memmingen is a base of Ryanair and Wizz Air (point-to-point leisure and ethnic traffic). The simulator uses Ryanair (60 %, B737-800 / MAX), Wizz Air (30 %, A320 / A321neo), SunExpress (5 %) and business jets (5 %); destinations from their Memmingen route networks. Shares are estimates.

13 EDXW - SYLT



▸ EDXW at the start of a session (phx-day style)

ITEM	VALUE
ICAO / IATA	EDXW / GWT (Sylt Airport, Westerland)
Location	Westerland, Sylt, Schleswig-Holstein, Germany
Reference point	54°54'41.89"N 008°20'32.59"E (AIP: on the 14/32 centre line, 180 m SSE of the runway intersection)
Elevation	52 ft (AIP)
Magnetic variation	3.9° E (World Magnetic Model, 2026)
Transition altitude	5000 ft; initial climb of the SIDs 4000 ft
Runways	**14/32**, 2120 m x 45 m, true heading 144.5° / 324.5° (all simulated traffic); **06/24**, 1696 m x 45 m, true heading 059.7° / 239.7° (unlit, no instrument procedures; not used - crossed by departures 32)
Data files	`src/data/airports/edxw.ts`, layout `edxw.layout.ts` (generated)

The simulator shows a condensed version of this page as the **airport briefing** (data in `briefing` in `edxw.ts`).

DATA SOURCES AND ACCURACY

DATA	SOURCE
Taxiways, holding points, aprons, buildings, stands 01-03	OpenStreetMap - © OpenStreetMap contributors, ODbL 1.0. Imported with `scripts/osm-import.mjs`; the derived layout is available under the ODbL
Runway thresholds and elevations, aerodrome elevation and reference point, declared distances, intersection take-off distances, approaches per runway, taxiway widths, surfaces and weight limits, parking rules	DFS AIP Germany AD 2 EDXW (AIRAC 2026OCT01) and its aerodrome chart AD 2 EDXW 2-5 (AMDT 06/26), used as a reference only: facts digitised by hand, no chart content copied
Taxiway names L1 / L3, the closed parts (X on the chart) and the former runway	Aerodrome chart (as above); what OSM does not have - G along the east side, the G links into the 24 end and to runway 14/32 at the glide path antenna, E south of runway 06/24, the former runway (80 m wide) with the links around it, the stub west of D and the turning pad at A - is digitised by hand from the chart, georeferenced on the four runway ends (approximate, about 5 m)
Frequencies, SIDs, runway use, stands	VATSIM Germany Sylt SOP (facts only; a simulation resource, not an official source); frequencies checked against the VATSIM Germany station data and vATIS profile, the SIDs and their initial climb (4000 ft) against the VATSIM Germany vSID configuration and the AIP SID charts (2026-10-21)
Magnetic variation	NOAA World Magnetic Model (October 2026); the AIP gives 3.5° E (2023.11) + 0.2° per year

Simplified or not modelled:

- runway 06/24 is not used for take-offs and landings and not as a taxiway; departures 32 cross it near the 06 end between D2 and A2 (D meets it about 190 m, A about 130 m from threshold 06; the simulator joins both at one runway point). D2 and A2 are simulator names,
- business jets below 19 m wingspan reach Apron 2 via the grass taxiway I (AIP: 2 t; per the VATSIM SOP real jets taxi on runway 06/24 to M / P) - larger ones park on **Apron 3** as the AIP requires (general aviation from 15 t or 19 m wingspan),
- the positions G1-G4 (Apron 2) and N1-N3 (Apron 3) are simulator stands (the aprons have no published stands); stands 01-03 on Apron 1 follow the apron markings approximately,
- light aircraft are not simulated: taxiway B (5.7 t; intersection take-off 32 from B 1826 m), I (2 t, grass), Apron 4 (5.7 t), the lanes of Apron 2 west and the track to the glider site are limited accordingly,
- the paved area at the 14 end between D, L and F and the links at the 06 end between D, M, P and A lie inside the runway strips: they are drawn, but taxi routes use the holding points,
- in reality Bremen Radar gives the IFR clearances; in the simulator (as on VATSIM) Sylt Tower does,
- the ATIS is called `Sylt Information` (VATSIM); the AIP call sign is `SYLT ATIS`,
- SIDs: one per runway (OLDIP1A for 14, OLDIP1B for 32), the numbers change with the AIRAC cycle,
- traffic: estimated operator shares, about a quarter of Stuttgart's movements per density step (`rateFactor` 0.25).

Never use this simulator for real-world navigation.

RUNWAYS AND APPROACHES

RUNWAY	TRUE HEADING	TORA / TODA / ASDA / LDA (M)	INTERSECTION TAKE-OFF	APPROACHES
14	144.5°	2120 / 2180 / 2152 / 2120	C: 1748	**RNP** only (LPV, GPA 3.0°) - no ILS
32	324.5°	2120 / 2180 / 2152 / 2120	B: 1826 (5.7 t)	**ILS CAT I** (GP 3.0°), RNP
06 / 24	59.7° / 239.7°	1696 each	-	none (unlit, PCN 35)

The simulator flies runway 14 arrivals as `RNP approach runway 14` whatever the ILS switches in the systems window say; Approach spaces them like non-precision approaches. Elevations: aerodrome 52 ft, thresholds 14: 32 ft, 32: 31 ft, 06: 29 ft, 24: 52 ft.

STATIONS

STATION	CALLSIGN	FREQUENCY
Tower (also ground movements and clearances)	Sylt Tower	119.750
Approach	Bremen Radar	124.075
ATIS	Sylt Information	118.400

Sylt has no Ground or Delivery position: you play **Sylt Tower** for everything on the ground and the runway.

LAYOUT

All taxiways of the aerodrome chart (AIP AD 2.8: 15 m wide, C 12 m; I grass):

TAXIWAY	DESCRIPTION
D	From Apron 1 (terminal) north to the 14 end of runway 14/32, and south to runway 06/24 (holding point D2)
C	Short connector from D to runway 14/32 near the 14 end (intersection departures 14, exit 32)
A	From runway 06/24 near the 06 end (holding point A2) the long way round the south of the airfield to the 32 end of runway 14/32
B	From A to runway 14/32 near the 32 end; 5.7 t (light aircraft)
L	From the 14 end (holding point L, opposite D) east past Apron 3 to Apron 4; L1 leads into Apron 3, L3 along its east side
F, E	F from L round the north of the airfield to the 24 end of runway 06/24 (holding point F); E from F to runway 06/24 a little before
G	From the 32 end (holding point G) north-east to the track to the glider site; the rest of G (along the east side to the 24 end, and the link west to runway 14/32 at the glide path antenna) is closed (crosses)
Former runway	An 80 m wide paved band from D / C across runway 14/32 to the east, closed east of 14/32 (crosses) like E south of runway 06/24, the narrow links and the loop around it and the link from A along runway 06/24; C runs on its western part. Drawn only
I, M, P	To Apron 2 (general aviation) in the west: I (grass, 2 t) from D; M and P from Apron 2 to runway 06/24 at its 06 end (holding points M and P - hot spot "holding points! Runway ahead!")

STANDS	USE IN THE SIMULATOR
01-03	Apron 1 at the terminal, airliners up to code C, taxi in and out along D
G1-G4	Apron 2, business jets below 19 m wingspan, taxi in and out along P
N1-N3	Apron 3 (hangar 74), business jets from 19 m (up to 32 m), in from L1, out onto L3

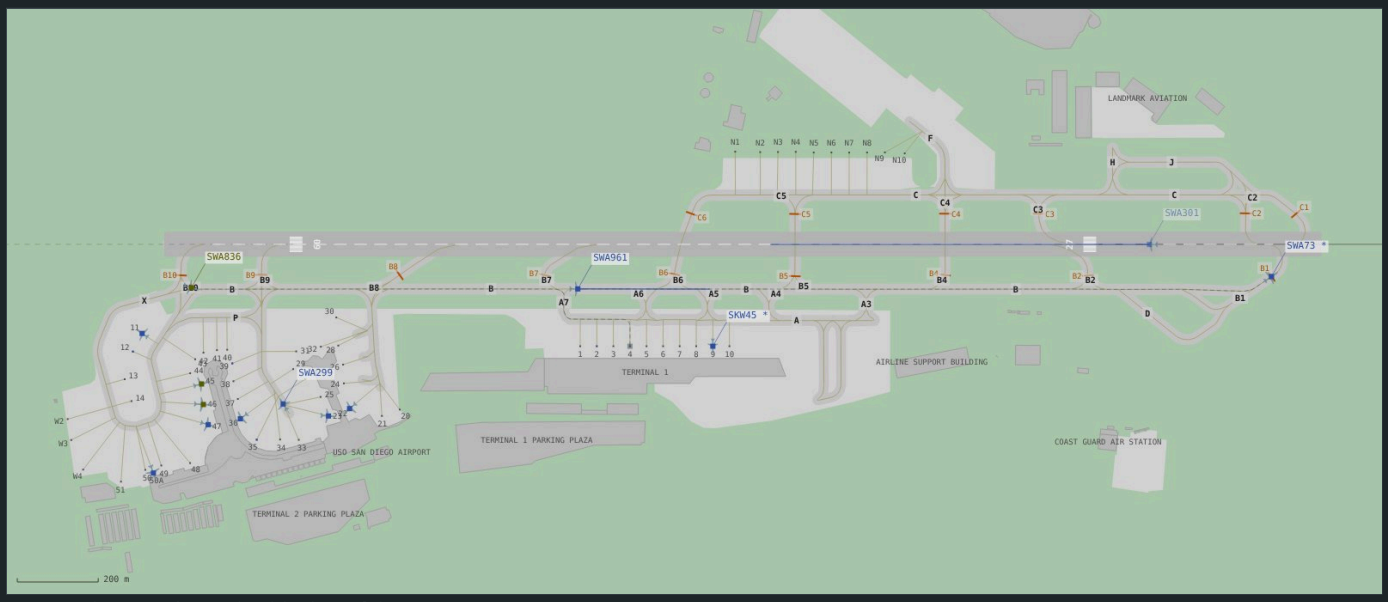
PROCEDURES

- **Runway 32** (preferred with the prevailing westerly winds, ILS): departures taxi via D, **cross runway 24** (D2 / A2) and follow A to the 32 end (`taxi to holding point A via D, A, cross runway 24`); from Apron 3 they first cross runway 14/32 at its 14 end (L to D). Arrivals vacate at the end via C or D (west: Apron 1 / 2) or L (north: Apron 3). An airliner that rolls on to L (F and the Apron 3 lanes are too narrow for it) backtracks to D instead.
- **Runway 14** (RNP approach): departures from D or L (full length) or C (intersection, 1748 m). Arrivals vacate to the right at the end via A and taxi back via A, across runway 06/24, and D.
- One SID per runway, initial climb 4000 ft: space departures at least 5 NM (2 minutes).
- Arrivals vacate only at the far end and occupy the runway for long: Approach spaces them at least **6 NM** (`arrivalSpacingNm`).
- With only three airliner stands, the last free stand is kept for arrivals (`standReserve` 1): Approach releases an arrival only when a stand is free for it.

TRAFFIC

Lufthansa (Frankfurt, Munich), Lufthansa CityLine, Eurowings, Swiss (summer) and a large share of business jets; shares are estimates.

14 KSAN - SAN DIEGO



▸ KSAN at the start of a session (phx-day style)

ITEM	VALUE
ICAO / IATA	KSAN / SAN (San Diego International Airport, Lindbergh Field)
Location	San Diego, California, United States (Los Angeles ARTCC, KZLA)
Elevation	17 ft
Magnetic variation	11° E
Transition altitude	18000 ft (US); initial altitude of the SIDs 5000 ft
Runway	**9/27**, 2865 m x 61 m (9401 ft x 200 ft), both thresholds displaced (9: 1000 ft, 27: 1810 ft)
Phraseology	**FAA** (US), see phraseology
Data files	`src/data/airports/ksan.ts`, layout `ksan.layout.ts` (generated)

The first airport outside Europe. The simulator shows a condensed version of this page as the **airport briefing** (data in `briefing` in `ksan.ts`).

DATA SOURCES AND ACCURACY

DATA	SOURCE
Taxiways, holding positions, aprons, buildings, stand lead-in lines (Terminal 2, west remote, north apron)	OpenStreetMap - © OpenStreetMap contributors, ODbL 1.0. Imported with `scripts/osm-import.mjs`; the derived layout is available under the ODbL
Runway, displaced thresholds, declared distances, elevation, frequencies, approaches	FAA Chart Supplement and the FAA airport diagram, used as a reference only: facts digitised by hand, no chart content copied
Facility name "Lindbergh", runway use, SID assignment, intersection distances, initial altitude	VATSIM Los Angeles ARTCC San Diego Tower SOP (facts only; a simulation resource, not an official source)
Traffic	San Diego International Airport route network 2026 (Wikipedia), FAA traffic counts (about 600 movements a day); operator shares are estimates

Simplified or not modelled:

- **Terminal 1** (opened 2025) has no parking positions in OpenStreetMap yet: ten gates (1-10) are placed 41 m apart along its airside front, pushback onto taxiway A - numbering and positions **approximate**,
- the cargo ramp and the general aviation (FBO) aprons on the north side are drawn but have no parking positions; cargo flights and business jets park on the north apron (N1-N10),
- Terminal 2 East gates 28 and 32 lie one behind the other on fanned OpenStreetMap lead-in lines: a parked aircraft on one blocks the way into the other,
- departures 9 from B10 and B9 are both treated as full length (TORA 2524 m, Chart Supplement 8280 ft),
- **taxi flows** (a simulator convention, not published): west traffic - departures east on B and C; east traffic west on both; A is used both ways,
- no noise curfew (in reality no take-offs between 23:30 and 06:30 local), no departure clearance times (EDCT) beyond the random slots of the simulator, no SoCal TRACON coordination,
- traffic: estimated operator shares (Southwest about 38 %).

Never use this simulator for real-world navigation.

RUNWAYS AND APPROACHES

RUNWAY	TAKE-OFF RUN (M)	LANDING DISTANCE (M)	INTERSECTION TAKE-OFFS	APPROACHES
27	2865 (B1, C1)	2313	C2 2679, C3 2167, C4 / B4 1935	LOC, RNAV - **no ILS**, over downtown
9	2524 (B10, B9)	2560	B8 2265, B7 1928, B6 1585	ILS, LOC, RNAV

West traffic (27) is used most of the time; east traffic (9) only with more than 10 kt tailwind on 27 (about 8 % of the sessions).

STATIONS

STATION	CALLSIGN	FREQUENCY
Clearance Delivery	Lindbergh Clearance Delivery	125.900

STATION	CALLSIGN	FREQUENCY
Ground	Lindbergh Ground	123.900
Tower	Lindbergh Tower	118.300
Departure	SoCal Departure	125.150 (ZZOOO), 119.600 (PADRZ), 124.350 (runway 9)
ATIS	San Diego International	134.800

LAYOUT

TAXIWAY	DESCRIPTION
B	Along the south side of the runway, from B1 (27 end) to B10 (9 end); the terminals lie south of it
C	Along the north side of the runway, from C1 to C6; the north apron and the cargo ramp lie north of it
A	South of B in front of Terminal 1, with A3-A7 to B
B1-B10, C1-C6	Connectors to the runway, named like their holding positions: `hold short of runway 27 at B4`. B4 / C4 and B6 / C6 are crossings
D, F, H, J	North side: to the cargo and general aviation ramps
P, X	Terminal 2 West and the west remote positions

STANDS	USE IN THE SIMULATOR
Terminal 1: 1-10	Narrow-bodies (approximate)
Terminal 2 East: 20-32	Narrow-bodies
Terminal 2 West: 33-51, 50A	Narrow-bodies, 40-43 regional jets, **48** and **51** wide-bodies (international)
West remote: 11-14, W2-W4	Remote; W2 / W3 up to 52 m, **W4** wide-bodies
North apron: N1-N10	N1-N5 business jets, N6-N10 cargo (N7-N10 up to 52 m)

Wide-bodies (B787 to London, A350 to Frankfurt, Hawaiian A330, cargo B767) have only three stands that take them: with a large share of heavies, the generator makes a normal flight while none is free.

PROCEDURES

- **Clearances:** "cleared to ... airport, PADRZ2 departure, then as filed, climb via SID except maintain 5000, departure frequency 125.15, squawk 1234". Pre-departure clearances (PDC) by datalink like the DCL elsewhere. There is no start-up approval: when ready, the crew asks Delivery for the Ground frequency and calls Ground "ready to push".
- **SIDs:** runway 27 **PADRZ2** (north, west; departure 119.6) or **ZZOOO4** (east, south; 125.15); runway 9 **ECHHO2** (north-west), **MMOTO2** (oceanic) or **SAYOW2** (east), all 124.35.
- **Taxi:** "runway 27, taxi via B" - the runway is the clearance limit; departures 27 to B1 (south side) or C1 (north apron), departures 9 to B10 / B9.
- **Arrivals 27** vacate to the south at B5-B10 (Terminal 2 at the west end, Terminal 1 via A) or to the north at C5 / C6 (north apron, crossing back at B6 / C6 for the terminals). Crews report "clear of runway 27 at B5".

15 AIRPORT AND ATC SYSTEMS

Real tower controllers work with a set of technical systems. Ultimate ATC models the ones a European airport of Stuttgart's size uses, and lets you check and switch them in the **systems window** (`SYSTEMS` in the toolbar, or **F3**). Switching a system off simulates maintenance or a failure, so you can train working without it.

The toolbar button reads `SYSTEMS (n OFF)` in orange while any system is off. Every switch is announced in the message window, together with what changes.

GROUP	SYSTEM	WHAT IT DOES IN THE SIMULATOR	WHEN IT IS OFF
A-SMGCS	**Surveillance** (surface movement radar + multilateration)	Every aircraft on the scope with its identity (data tag)	Targets lose their identity: no data tags on the scope. Work with the lists and the message window.
A-SMGCS	**Vehicle tracking** (ADS-B squitter beacons on vehicles)	Follow-me cars and tows are shown with their callsign (tugs are never drawn: at the aircraft's nose they merge with its radar return)	Vehicles and tows are only unlabelled targets (a towed aircraft has its transponder off)
A-SMGCS	**RMCA** (runway monitoring and conflict alerting)	Alert when an aircraft with a crossing clearance approaches the runway while it is occupied or an arrival is less than 60 s out	No runway alerts - only the incursion itself is detected
A-SMGCS	**CATC** (conflicting ATC clearances)	Warns in the command-line preview and the *Taxi to* menu before you transmit a route that meets other traffic head-on; aircraft whose cleared routes meet head-on while both are rolling towards each other flash red, with a system message (only when at least one of them is on your frequency; the simulator resolves the others). *Acknowledge CATC alert* in the aircraft menu silences an alert you have checked until that conflict ends	No route conflict warnings
A-SMGCS	**Routing service**	Proposes taxi routes (following the standard taxi flows) in the *Taxi to* menu, the quick-action bar and the *Resolve conflict* menu	Menus offer destinations without a proposed route; you plan the route yourself
A-CDM	**A-CDM / pre-departure sequencer**	TOBT from the airline, TSAT from the sequencer, CTOTs from the Network Manager; crews ask for start-up at their TSAT	No TSAT: crews ask for start-up when they are ready (first come, first served). CTOTs still apply.
Datalink	**DCL** (departure clearance by datalink)	About 4 in 10 crews request their IFR clearance by datalink; you send it from the aircraft menu, no voice and no readback	All clearances by voice
Navigation aids	**ILS localizer (LOC)**	Lateral guidance of the ILS for the runway in use; with the glide path, arrivals fly ILS approaches	No ILS: arrivals fly **RNP** (satellite) approaches, spaced at least 5 NM
Navigation aids	**ILS glide path (GP)**	Vertical guidance of the ILS	**Localizer** (non-precision) approaches, spaced at least 5 NM

Switching the ILS changes the approach procedure: the ATIS text includes it (`RNP approach runway 25, ILS runway 25 out of service`) once you broadcast a new ATIS, and the arrivals list shows the procedure in the `APCH` column (`ILS`, `LOC`, `RNP`). A runway end without a published ILS (Sylt 14: RNP only; airport data `approaches`) is always flown as an RNP approach - the ILS switches do not change it and the ATIS does not report an ILS outage there.

BACKGROUND: WHAT REAL AIRPORTS USE

- **A-SMGCS** (Advanced Surface Movement Guidance and Control System) is defined by EUROCONTROL (and ICAO Doc 9830) as a set of services: *surveillance* (surface movement radar and multilateration, which identifies aircraft by their transponder), *airport safety support* (RMCA for the runway, CATC for conflicting clearances, and alerts for restricted areas), *routing* (route proposals) and *guidance* (follow-the-greens lighting). Most larger European airports have surveillance and RMCA; CATC, routing and guidance are found at the larger hubs. The simulator offers all four services for training; guidance by airfield lighting is not modelled.
- **A-CDM** (Airport Collaborative Decision Making): airlines, ground handlers, the airport and ATC share their times. The airline gives a **TOBT** (target off-block time), the pre-departure sequencer computes a **TSAT** (target start-up approval time) that respects the runway capacity and any **CTOT**. Start-up is approved at the TSAT (EUROCONTROL tolerance -5 / +5 minutes). Stuttgart has been a full A-CDM airport since November 2014.
- **CTOT** (calculated take-off time): an ATFM slot from the EUROCONTROL Network Manager when a sector or the destination is regulated. The aircraft must take off between CTOT -5 and CTOT +10 minutes; otherwise it needs a new slot.
- **Vehicle tracking**: the surface movement radar shows every vehicle, but without identity. With a squitter beacon (an ADS-B transmitter with a unique address and callsign) a vehicle appears with a label like an aircraft. Stuttgart ordered 135 such beacons (ERA SQUID) for tugs, de-icing and fire vehicles in 2020, to work with its multilateration system.
- **ILS** (instrument landing system): the **localizer** gives lateral guidance along the runway centre line, the **glide path** vertical guidance (normally 3 degrees). Without the glide path a localizer approach (non-precision, higher minima) is possible; without the localizer, airports with satellite procedures use **RNP** approaches. In low visibility the ILS critical and sensitive areas must be kept clear, which moves departures to the CAT II/III holding points - low visibility procedures are not simulated yet. Stuttgart has ILS approaches on both runway directions.
- **DCL** (departure clearance by datalink, ED-85A / ARINC 623): the crew requests and receives the IFR clearance as a text message and confirms it with WILCO. Stuttgart offers DCL; start-up is still requested by voice.

SIMPLIFICATIONS

- Surveillance off removes all data tags at once. In reality a single radar sensor may fail while multilateration still works.
- RMCA only watches aircraft with a clearance to cross; incursions without a clearance are detected as incursions.
- CATC only looks for head-on encounters on taxiways (two aircraft meeting at more than 135 degrees, within 900 m of route ahead). Real CATC also checks, for example, a line-up clearance against a landing clearance.
- An ILS failure applies to the runway in use, whichever direction that is. Non-precision approaches only change the arrival spacing (5 NM, a simulator value); weather minima and missed approaches are not modelled.
- The pre-departure sequencer uses a fixed taxi time of 10 minutes and a TSAT spacing of 90 s. See simulation model for all numbers.

16 TOWER OPERATIONS: HOW DEPARTURES AND ARRIVALS SHARE A RUNWAY

This page summarises how tower controllers manage a **single runway in mixed mode** (departures and arrivals on the same runway, as at Stuttgart). It covers the separation minima involved and how the simulator's AI Tower implements them. It is the background for the AI Tower rules.

> These are simplified, publicly documented rules for a simulator. Real minima depend on the state's AIP, local procedures, weather and the wake category scheme in use (ICAO legacy categories or RECAT-EU). The exact current values are in ICAO Doc 4444 (PANS-ATM). Several figures below come from secondary sources and are marked as such.

1. THE THREE SEPARATIONS THAT LIMIT A SINGLE RUNWAY

1.1 Departure behind departure

Two consecutive departures need **both**:

- **Wake turbulence separation** (time-based, measured between airborne times). Behind a *Heavy*, the following departure needs **2 minutes**, or **3 minutes** if it starts from an intermediate point (intersection) of the runway. In this case ICAO adds one minute. The UK CAA applies the same idea: 2 min Heavy behind Heavy and Medium behind Heavy, 3 min behind an A380. [1][2]
- **Route separation** after take-off. If the departure routes diverge immediately after take-off (for example ≥ 45° different headings), departures can follow each other closely, typically **1 minute**. On the same route the spacing must be larger, typically **2 minutes** or a radar distance. [3][5]

  VATSIM Germany's Braunschweig SOP, for example, uses **3 NM** (or wake turbulence, whichever is greater) between departures, and **5 NM** when both use the same SID waypoint. [5]

1.2 Arrival behind arrival

On final approach, arrivals are separated by radar distance:

- **Radar minimum** of 5 NM, which the ATS authority can reduce to **3 NM**. Under specific conditions it can be reduced to **2.5 NM** on the same final within 10 NM of the threshold: runway occupancy times are monitored, exits are suitable, and braking action is good. [4]
- **Wake turbulence minima** (legacy ICAO categories): Heavy behind Heavy 4 NM, Medium behind Heavy 5 NM, Light behind Heavy 6 NM, Light behind Medium 5 NM. This table comes from a training source and should be checked against Doc 4444. [6]
- Since November 2020 ICAO also allows **seven new wake turbulence groups** (by MTOW and wingspan) instead of the legacy categories. [7]

1.3 Runway separation (who may be on the runway)

- A **landing** aircraft may only cross the threshold when the preceding landing aircraft has vacated the runway and the preceding departure is airborne and has passed the end of the runway, or has started a turn.
- **Reduced Runway Separation Minima (RRSM)**, Doc 4444 §7.11, where approved: a landing aircraft may cross the threshold when the preceding aircraft is **airborne and has passed a point at least 2400 m** from the threshold. For a preceding landing aircraft, the condition is that it has passed 2400 m, is still moving, and will vacate without backtracking. This applies to the heavier "Category 3" aircraft. States add conditions, such as daylight, visibility ≥ 5 km and tailwind ≤ 5 kt. [8][9]
- RRSM is **not** applied between a departure and a *preceding landing* aircraft. A departure waits until the landing aircraft has vacated. [8]
- **Line-up clearance is not a take-off clearance.** If a departure is lined up to use a gap, it should be told to be ready for an immediate departure. The time lined up should be short: the rate of missed approaches and simultaneous runway occupancy grows with the time between line-up and start of roll. [10]

2. MIXED-MODE FLOW: MAKING GAPS

On a single runway, every departure needs a **gap** between two arrivals large enough for:

> line-up (30-40 s) + take-off roll until airborne and clear (~45-55 s) + buffer

At a typical final approach speed of 140 kt (about 26 s per NM), that is roughly **4-5 NM** of final (arrivals fly faster further out - usually 160 kt until 4 NM - but the gap that counts is the one at the threshold). Add the runway occupancy time of the preceding landing aircraft (about 50-60 s until it has vacated), and two arrivals need to be about **6 NM apart** to fit one departure in between.

This is exactly what busy airports do:

- VATSIM Germany's Berlin SOP sets target spacings on final of **6 NM when departures are expected**, 3 NM when there are none, and 5 NM in low visibility. Tower may ask Approach for wider spacing to release departures, and must "use every gap sufficient for departing traffic". [11]
- The BEA report on a Lyon-Saint-Exupéry incident shows the risk of mixed mode: no minimum interval was defined for inserting a departure, and the controller had no aids for managing arrival-departure separation. [12]

Techniques that increase throughput:

1. **Line up behind landing traffic**: "behind the landing A320, line up and wait behind". The departure taxis onto the runway as soon as the landing aircraft has passed its entry, and takes off once that aircraft has vacated.
2. **Line up behind a departure**: the next departure lines up while the previous one is rolling.
3. **Use the departure route split**: alternate departures on diverging SIDs (1 minute) instead of the same SID (2 minutes).
4. **Intersection departures** for light or medium aircraft that don't need the full length. This gives a shorter roll and a quicker entry. Behind a heavy it costs an extra minute of wake separation.
5. **Ask Approach for gaps** when the departure queue grows (6 NM, more for longer queues), and close the gaps again when it is empty.
6. **Ground**: deliver departures to the holding point in a sensible order (mix SIDs, keep heavies apart), and keep the vacate points behind the exits free, so landing aircraft can clear the runway quickly.

3. PLAYING TOWER

Since v0.7 you can staff Tower yourself; then the rules below are yours to apply (the AI Tower only flies the aircraft). The simulator checks the departure spacing at the start of each take-off roll, the RMCA warns about clearances onto an occupied runway, and arrivals without a landing clearance go around at 0.5 NM. See the user guide.

4. WHAT THE AI TOWER DOES

RULE	IMPLEMENTATION (SEE `SRC/CORE/TOWER.TS`, `SRC/CORE/TRAFFIC.TS`)
Departure wake separation	120 s after a heavy became airborne, 180 s from an intersection
Departure route separation	60 s for different first SID fix, 120 s for the same fix

RULE	IMPLEMENTATION (SEE `SRC/CORE/TOWER.TS`, `SRC/CORE/TRAFFIC.TS`)
Line up and wait	Allowed when the separation will be met within 45 s, behind a rolling departure, or behind a landing aircraft that is > 300 m past the entry
Immediate departure	Take-off 3-8 s after lining up
Take-off vs. arrival	Next arrival more than `roll time + 8 s` (about 2 NM) from the threshold
Line-up vs. arrival	Next arrival far enough out for line-up + remaining separation + roll + 15 s (about 4-5 NM)
Landing vs. runway	Go-around at 0.5 NM if the runway is occupied; a departure already beyond 1200 m / 100 kt is accepted (RRSM-like)
Landing vs. landing	The preceding landing aircraft must have vacated
Arrival spacing	max(wake minimum, 4 NM; 6 NM with departures waiting; 8 NM with four or more); 10 NM when landing aircraft backtrack (exits marked `backtrack`), 8 NM when departures backtrack (entries marked `backtrack`)
Backtrack (departure)	From an entry marked `backtrack`: along the runway at up to 18 kt, a 180° turn on the runway (radius 11 m, 6 kt) 30 m from the runway end, lined up 80 m from the end; the line-up time counts the backtrack (distance / 14 kt + 40 s)
Backtrack (arrival)	When no exit ahead can be made (or the requested exit is behind): slows to 6 kt, turns around on the runway where it stopped, taxis back at up to 18 kt and vacates at the exit marked `backtrack`
Emergencies	No line-ups while an emergency arrival is within 8 NM
Exit choice	Skips exits whose vacate point is blocked by a waiting aircraft

Effect (automatic controller test, EDDS runway 25, two hours of heavy traffic):

MEASURE	BEFORE (V0.1)	AFTER
Mean wait at the holding point	~336 s	~110 s
Departures airborne	7	20
Go-arounds	18	0

Many v0.1 go-arounds were caused by gridlock, so the before figures are pessimistic. The model now behaves like a reasonable single-runway operation.

SOURCES

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!. BEA: Investigation report on the serious incident at Lyon-Saint-Exupéry (mixed-mode runway) - https://bea.aero/fileadmin/user_upload/PH-EXH_EN.pdf

17 SIMULATION MODEL

This page describes how the simulated world behaves: AI pilots, the AI Tower, traffic generation, ground movement, separation, radio and incident detection. All numbers are the values used in the code, so you can predict how traffic will react.

TIME

- The simulation advances in fixed **0.2 s steps**. Rendering is independent of the simulation rate.
- The **simulation rate** (1x / 2x / 4x / 8x) multiplies the real elapsed time. A frame delta is capped at 0.5 s (and a single `tick` at 5 s), so a backgrounded browser tab doesn't cause huge jumps.
- Simulation time starts at the real current UTC time.
- A seeded pseudo-random generator (mulberry32) drives every random decision. **The same seed and settings always give the same scenario**, as long as the controller's inputs are the same. A scenario code adds a fixed runway, traffic mix, more heavies and scheduled events, see scenarios.md.

AIRCRAFT LIFE CYCLE

Departures

```
parked: clearance request --(IFR clearance)--> start-up request --(start-up approved)--> frequency request --(contact ground)--> [Ground]
parked --(pushback approved)--> pushback --(push complete)--> startup --(engines running)--> taxi
   |                                                                                          |
   +--(taxi, on a taxi-out stand)-------------------------------------------------------->+
                                                                                            |
taxi --(reaches destination holding point)--> holding --(contact Tower)--> [Tower]: lineup --> takeoff --> climb --> gone
```

STEP	TIMING / BEHAVIOUR
Boarding	A departure appears on a free stand. Its **ready time** (TOBT) is 4-15 min after it appears (initial traffic 0.5-25 min).
Delivery	With Delivery staffed by you, it starts on Delivery frequency without a clearance: see Delivery, A-CDM and slots. Otherwise it already has its clearance and starts on Ground.
Pushback call	On Ground: at the ready time, or with A-CDM at **TSAT - 2 min** unless start-up was already approved by Delivery (then at the ready time, at the earliest 20 s after the hand-off).
Tug connection	6-15 s after the pushback approval, the push starts.
Pushback	1.3 m/s (~2.5 kt) backwards along the stand lead-in line, then 45 m (at most 90% of the taxilane segment) along the taxilane. Without a `facing` the crew faces the shortest way to its runway; a direction that would need a 180 degree turn afterwards (towards the dead end of a taxilane) only if there is no other. A pushback stand at the dead end of a taxilane is pushed back from the node before the dead end (OSM layouts, `buildLayout`). Drive-through stands have no pushback: the aircraft calls `request taxi` and leaves forwards.
Engine start	After the push: 15-35 s with *push and start* or *start-up approved*, otherwise 40-80 s.
Ready for taxi	The pilot calls `ready for taxi`. A taxi instruction received earlier is executed now.
Holding point	When stopped at the destination holding point, the aircraft reports `ready for departure` after 8 s if it is still on your frequency.
Tower	After the hand-off, Tower sequences the departure (see AI Tower).
Leaving	The aircraft stays on the scope until it is **10 NM** from the airport (about the extent of the Stuttgart control zone along the runway axis); then it is removed and counted as `departuresAirborne`. Passing 4000 ft above the airport (or 15 km) still on Tower frequency is a missed hand-off.

Arrivals

```
approach --(touchdown)--> landing --(passes exit holding point)--> vacating --(stops behind holding point)--> taxi (on Ground)
taxi --(reaches stand)--> arrived --(150-300 s)--> gone
```

STEP	TIMING / BEHAVIOUR
Final approach	Appears 12 NM from the threshold (initial traffic: 4-6 NM) on a 3° glide path and flies the speed profile below, on **Approach** frequency (Langen Radar); Approach hands it to Tower at 8-10 NM (typical hand-off once established on the ILS; no published value).
Touchdown	350 m past the landing threshold.
Roll-out	Brakes at 1.2-2.0 m/s² (every crew differently) to reach the chosen exit at 20-28 kt (rapid exit) or 11-16 kt (normal exit) and keeps that speed on the exit until it is behind the runway holding point (the runway is then free), then slows down at 1 m/s² to 10 kt; one in five passes the first exit it could make and rolls on to the next.
Exit choice	The first exit that can be reached at that deceleration. 85% prefer the north side, 15% the south side. Exits whose vacate point is blocked by a waiting aircraft (within 70 m) or where an aircraft stands at the holding point are skipped. Exits from which the aircraft can taxi on to its stand without turning around come first - without an assigned stand: to one of the 4 nearest free stands that fit (or, with none free, of all stands that fit); if none ahead does, it backtracks to an exit marked for it that does (Sylt D for runway 32: F and the north apron are too narrow for airliners). If no exit can be made at all, the last one is taken, the one towards a stand first. Of the branches of one exit (the same runway node, e.g. Leipzig S4 west or east onto T) it takes the one with the shorter way to the stand. A landing aircraft that waits on the runway for more than **15 s** because an aircraft stopped at the holding point of its exit (for example one waiting to cross there) takes another exit.
Vacated	Stops with its tail at least 5 m behind the runway holding point and, where it can, with its nose **40 m** from the centre line of the taxiways ahead (OSM layouts) - so traffic on the parallel taxiway passes and it can turn either way; switches to Ground and calls `vacated runway 25 via E` after a short delay.
Stand	A suggested stand is allocated when the arrival appears (if one is free). You can use any free stand that is big enough and not blocked by a neighbour.
Turn-around	150-300 s after parking, the aircraft is removed and the stand becomes free.

GROUND MOVEMENT

Aircraft follow **paths**: polylines through taxi graph nodes whose corners are replaced by smooth curves (fillet radius up to 30 m on taxi routes). The aircraft's position is a distance `s` along the path. Its heading is the path's tangent, reversed during pushback.

Speed control (taken every step as the minimum of all limits):

LIMIT	VALUE
Straight taxi speed	Aircraft type, 14-18 kt, times the crew's own **taxi pace** (0.85-1.15, drifting ±5 % along the way) (+5 kt after `expedite taxi`)

LIMIT	VALUE
Upcoming turn > 15°	14 kt
Upcoming turn > 35°	11 kt
Upcoming turn > 70°	7 kt
Next stop (holding point, hold short, destination)	Speed that allows stopping at 1.0 m/s² times the taxi pace
Traffic ahead	Speed that allows stopping at 1.2 m/s² before the traffic
Hold position / give way / stopped at a stop	0
Landing roll-out, line-up	Profile set by the Tower

Acceleration is 0.6 m/s² times the taxi pace. Braking is up to 2.5 m/s². The taxi pace applies while taxiing only (not to line-up and roll-out) and is derived from the session seed and the callsign without drawing from the simulation's random sequence (`Simulation.variation`).

Stops on a taxi route are computed when the route is assigned:

STOP	WHERE	RELEASED BY
`runway`	Every runway holding position where the route continues **towards** the runway, unless a crossing clearance was given	`cross runway ..`, or a new route with crossing
`holdShort`	At the intermediate holding position of the charts before the junction (nose at the line; OSM `holding_position:type=intermediate`, at most 250 m before it and with no other junction in between), otherwise where the nose is **40 m** from the centre line of the named taxiway - measured from where the route leaves its taxiway, so short unnamed fillets of the junction count as part of it	`continue taxi`
`destination`	End of the route (holding point, stand)	A new taxi instruction

Route finding is described in architecture.md. Routes contain no turns sharper than **150°** along a taxiway or onto an unnamed link. From one named taxiway onto another (e.g. off a rapid exit back against the landing direction, where OpenStreetMap lacks the fillet) up to **175°** is possible: a `via` route takes it at 20 m of extra cost per degree beyond 150°, an automatic route only if there is no other way (20 000 m).

Lenient via lists: a `via` list that leaves out connecting taxiways is not refused. If the exact route fails, the crew fills the gaps with other taxiways (3 times the cost, never across a runway): at most **2** added taxiways and **600 m** between the named ones, and after the last named one at most one taxiway on to the destination, however long (`taxi to runway 08L via N, E7`: on along C). The read-back then names the whole route (`taxi to holding point A via M, L2, S`), so you hear what the crew will do. For `taxi to runway`, an entry reached with the route as given comes before one that needs added taxiways. A route that needs the aircraft to turn around where it stands is only used if nothing else works: the router adds 5000 m of cost for it. A node behind the aircraft counts as "turning around" even if it is only a few metres away.

No 180 degree turns for airliners: for aircraft with a wingspan above **25 m**, routes that need a turn-around where the aircraft stands are refused (`unable, we are facing west and cannot turn around here`). Exceptions: at a runway holding point, and when the aircraft has been stuck (blocked by traffic or with a `blocked` request) for more than **30 s** at a speed below 0.2 m/s. In that case it waits **300-600 s for a tug** (ordering it, connecting, turning the aircraft) before it starts moving; the read-back says how long (`we need a tug to turn around, expect about 7 minutes`). Aircraft up to 25 m wingspan can always turn around.

Cancelled pushback: if the aircraft has already moved, the tug tows it back along the same line onto the stand. It is then parked again and calls 60-150 s later.

Automatic routes (no `via` list: pilots' own choice, menu suggestions, pushback direction) also follow the airport's **standard taxi flows** for the runway in use: they never taxi against a flow if there is any other route that doesn't need a 180 degree turn and doesn't cross a runway more often; only then a route against the flow is allowed (at 4 times the distance cost). They avoid runway crossings (3000 m extra cost per crossing). Where the airport names its crossing points (Stuttgart: only between K and Y), crossings elsewhere cost another 20 000 m, so they happen only if there is no other way (e.g. K or Y closed); taxiways reserved for others (Stuttgart Z, military) cost 20 000 m extra too. The lead lines of de-icing pads are only used to enter or leave a pad, with turns of at most 100° onto and off them.

PILOT SEE-AND-AVOID

Pilots don't collide on purpose. Every step, each aircraft that is moving along a path looks ahead along its **own path** by (this includes landing aircraft on their exit)

```
look-ahead = v² / (2 · 1.2 m/s²) + 55 m + own length
```

and checks whether another ground aircraft **ahead of it** (traffic behind or exactly beside it is ignored, because moving on only increases the distance) is (or, if it is moving, will be within 4 s) closer to the path than

```
r = 0.38 · (wingspan A + wingspan B) + 6 m        (two A320s: ~33 m)
r = 0.3 · (wingspan A + wingspan B) + 4 m         (passing an aircraft parked on its stand; two A320s: ~25 m)
```

An aircraft parked on its stand is passed closer: the apron is laid out for it, only wingtip clearance counts.

If so, it plans to stop `max(5 m, (length A + length B)/2 + 12 m - r)` before that point. This keeps in-trail traffic about 15 m nose-to-tail.

Crossing and merging traffic: in addition, each aircraft compares its path with the next 150 m of the path of every other aircraft that intends to move (within 600 m). Where the two paths come closer than `r` at an angle of at least 20 degrees, the aircraft that would reach that point **later** (distance / speed, with a minimum speed of 2 m/s) stops `own length / 2 + 10 m` before it, outside the other one's lane. An aircraft that is **already in the other one's lane** (its first conflict point is within 9 m ahead) always goes first and the other waits before entering the lane - stopping inside the other's lane would block it for good (for example nose to nose on a connector like H). Otherwise the decision is kept until the conflict is over, so the two don't take turns braking. Traffic coming from behind in the same direction whose path runs through the aircraft's current position is following it and is never given priority. Each aircraft checks the next **400 m** of the other aircraft's path, so it stops early enough where taxiways meet at a shallow angle (for example arrivals from rapid exit F crossing S).

Opposite traffic (since 0.8): every 2 s each taxiing crew looks **800 m** along its route. If an aircraft stands on that stretch heading towards it (more than 135 degrees difference) and both stand on each other's route, the two are more than **110 m** apart and there is a **junction** (a node where three or more taxiway segments meet) between them that both can still stop short of, both crews stop short of that junction - `max(length / 2, r + 10 m) + 15 m` before it, so it stays free for one of them to turn off there - and call `opposite traffic on taxiway N, EWG7TK, holding short of G, request instructions`. *Resolve conflict* (and the AI Ground) then sends one of them off via the junction; the other waits until the way is clear. A crew whose call is still waiting for a gap on the frequency drops it once the other aircraft has been given its way out.

Narrow parts without a junction: where two crews would meet on lanes too close for both wingspans (e.g. Leipzig apron 4, Z4.1 and Z4.2 31 m apart) or on a stretch without a junction between them, and their paths there run in opposite directions, the crew further from the narrow part waits short of it (15 m) without calling, until the other one has passed. The one that goes first must not have the waiting one in its way (then they swap); the waiting one stops early enough not to stand in the other's way (a wide-body next to the lane), and the crossing rule above does not make the one going first wait for it. Checked every 2 s up to 400 m apart (800 m along the routes); not if it can no longer stop in time.

Mutual conflicts: if two aircraft each see the other on their path (for example converging at an intersection), the one **closer** to the conflict point continues and the other waits. Nobody drives into an aircraft that actually stands on its path: if only one of them has the other physically in its way, that one waits and the other goes. If both have the other physically in their way, or both are stopped nose to nose, nothing moves. This is a **deadlock**: after 60 s the pilots report `we have opposite traffic ahead, request instructions`, and you have to re-route one of them (after 30 s stuck, airliners accept a route that needs a tug to turn around, see above).

Waiting aircraft blocking others: an aircraft that stands still waiting for an instruction (at a clearance limit, a hold-short, a vacate point, or with an open request) and blocks another aircraft for more than 60 s makes the blocked pilot call: `we are blocked by X, waiting on the taxiway ahead, request instructions`. The EDDS vacate points are placed so that an aircraft waiting there does not block the parallel taxiway S.

Give way and conditional clearances: an aircraft told to `give way to X`, or given a conditional clearance (`behind X, ...`), waits until X **has passed**. That means X is airborne or gone, or its distance has grown at least 40 m beyond the closest distance so far and is larger than half the combined wingspans plus 40 m. It also continues if X stopped more than 300 m away (after 20 s), or after 4 minutes at the latest.

AI PILOTS: COMMUNICATION

Pilots only talk on the frequency they are tuned to. Aircraft with Tower are silent for you.

Requests (`request` state, shown in the `REQ` column):

REQUEST	RAISED WHEN	ANSWERED BY
`pushback`	Departure ready on a pushback stand	`pushback approved` / `push and start approved`
`taxi`	Engines running after pushback, or ready on a taxi-out stand	Any taxi instruction
`taxiIn`	Arrival has vacated	Taxi instruction to a stand
`handoff`	Departure at the holding point for 8 s and still on Ground	`contact tower`
`crossing`	Stopped at a runway holding point on the route	`cross runway ..`
`blocked`	Head-on / mutual deadlock, or behind an aircraft waiting for instructions, for 60 s (cleared automatically once the aircraft moves again)	`hold position`, `give way`, a new route
`route`	Held short of a taxiway for 2 min, a taxi instruction became invalid, or Tower sent the aircraft back	`continue taxi`, a new route
`clearance`	On Delivery, 10 min before the ready time (by voice, or silently by DCL)	IFR clearance (or *Send DCL*)
`startup`	On Delivery, cleared: at TSAT - 2 min with A-CDM, otherwise at the ready time	`start-up approved`
`frequency`	On Delivery, start-up approved, 20 s after the last call	`contact ground`

Reminders: DCL requests are not repeated by voice. If a request isn't answered, the pilot calls again every 60-89 s (a fixed interval per callsign), up to 5 times. These count as an answer for the waiting-time statistics and suppress reminders:

- `standby`: for 120 s,
- `number N for ...`: for 60 s + 45 s per queue position,
- `expect ... in N minutes`: for N minutes + 20 s.

Answering a request records the waiting time (time since the first call). Every 15 s of waiting beyond 30 s costs one point.

Taxiway restrictions: taxiways and taxilanes can have a maximum wingspan (EDDS: taxilane M east of H, code C). Routes for larger aircraft avoid them.

Stand suitability: a stand can take an aircraft only if its wingspan is within the stand's maximum **and** there is enough wingtip clearance to the aircraft on (or taxiing to, or reserved for) the neighbouring stands: the stand centres must be at least `(span A + span B) / 2 + clearance` apart (and the lead-in line must pass a parked neighbour at least `max(span A, span B) / 2 + clearance` away - fanned lead-in lines, one stand behind another), with a clearance of **4.5 m** between code C aircraft (wingspan below 36 m) and **7.5 m** when one of them is code D or larger (ICAO Annex 14). So a wide-body on a large stand can block the smaller stand next to it. This applies to the automatic stand allocation and the stand menu. Crews accept your instruction to any stand their aircraft fits; **150 m before the stand** they see whether it is usable. If an aircraft stands on it (parked) or a parked neighbour is too close, they stop and call `stand 14 is occupied, request another stand` (request `route`). If the aircraft there is moving (pushing back, starting up, or taxiing onto the neighbouring stand), they hold without a call and continue once it is clear.

Validation: pilots check instructions against their state and the airport data. They reply `unable ...` or `confirm ...` instead of doing something impossible: an unknown taxiway or holding point, an impossible route, a stand too small for the aircraft (an occupied stand is only noticed when taxiing in, see *Stand suitability*), a crossing that isn't on the route, a wrong frequency, a hand-off before taxiing. See phraseology.md.

DELIVERY, A-CDM AND SLOTS

Source: `src/core/delivery.ts`.

ITEM	VALUE
Clearance request	10 min before the ready time (TOBT); immediately if the aircraft appears later than that. At session start staggered (see Traffic generation)
DCL equipped	40 % of the departures with an airline callsign (not German-registered `D-xxxx` aircraft), when DCL is on
Initial climb	from the airport data (`initialClimbFt`: EDDS 5000 ft, the altitude of every published SID; Cologne/Bonn 5000 ft; Nuremberg, Memmingen and Leipzig/Halle FL70, Bremen and Sylt 4000 ft); 5000 ft if not set
Squawk codes	next free code from the octal blocks 2101-2177, 2201-2277, 2301-2377, 2401-2477 (codes ending in 0 skipped; a simulator range, not the real ORCAM allocation). Special codes 7500, 7600, 7700, 7000, 2000, 1000, 0000 are refused
Readback error	4 % of voice clearances: two digits of the squawk read back swapped. Caught with `squawk <correct code>` (+5); passed with `readback correct` or a hand-off it stays: the crew sets the wrong code (-10)
Crew queries	missing or unknown clearance limit, wrong destination, missing or unknown SID, SID of the wrong runway (`information R says runway 25 in use`) or not leading to the first fix of the flight plan, wrong runway, missing initial climb, missing or invalid squawk
Start-up	needs the IFR clearance (`negative, we have no clearance yet`)
Hand-off to Ground	needs the IFR clearance; counts as *clearance delivered* (+10)
CTOT	12 % of the departures; CTOT = ready time + 10 min taxi time + 5-25 min, rounded to the minute
Night curfew	Scenario option: a jet still on the ground when the departure curfew starts is postponed to the morning (on its stand) or asks to taxi back to a stand; -10 if it was ready at least 15 min before the curfew. See scenarios.md
CTOT window	take-off from CTOT - 5 min to CTOT + 10 min. Tower lines a CTOT flight up only when it can be airborne inside the window (others go first). After CTOT + 10 min the flight gets a new CTOT 20-40 min later - 8-15 min if it is ready at the holding point or lined up (a ready message to the Network Manager) - (-10)

A-CDM pre-departure sequencer (every 5 s, when A-CDM is on): every parked departure without start-up approval gets a **TSAT** - not before its ready time, not before CTOT - 10 min (taxi time), rounded up to the minute and at least **90 s** from every other TSAT. A TSAT once issued is kept while it is still valid; flights that missed their TSAT (and new flights) only get a new one after that, in the earliest free slot - a late flight never bumps

one that is on time. Start-ups already approved (and aircraft off-block) keep their slot. Crews call for start-up (on Delivery) or pushback (on Ground) 2 minutes before their TSAT. Without A-CDM there is no TSAT: crews call when ready.

A-SMGCS

Source: `src/core/conflicts.ts`. Each service can be switched off in the systems window.

SERVICE	BEHAVIOUR
Surveillance	Data tags on the scope (off: no tags)
RMCA	Checked every step. Alert when a taxiing aircraft (above 0.5 m/s) with a clearance to cross comes within **120 m** of the runway holding position while the runway is occupied or an arrival is less than **60 s** from the threshold. One alert per aircraft and holding point
CATC	Every 5 s, the cleared routes (next **900 m**, sampled every 15 m) of the taxiing aircraft are compared; two routes that meet at more than **135 degrees** are a head-on conflict - but only when **both aircraft are rolling** (above about 2 kt, not stopped, holding or told to hold position). An aircraft standing still does not make the other one flash. Only reported if one of the aircraft is on your frequency; a pair is not announced again within 60 s (e.g. after a short stop). The AI Ground still uses every head-on route conflict (also with standing traffic) to resolve conflicts early. The command-line preview and the *Taxi to* menu still warn about a route towards standing traffic too
Routing	Route proposals in the menus; *Resolve conflict* options

Resolve conflict options: conflicts are aircraft blocking each other nose to nose, or blocking each other at any angle (for example a pushback into a taxiing aircraft's way), or routed head-on. An aircraft that is still pushing back gets `cancel pushback` first (the tug tows it back onto the stand). Otherwise, for each of the two aircraft (on your frequency), a route to its destination that avoids the other aircraft's next 150 m of route; it is checked to keep clear of the other aircraft's future route (from the second segment on). If its destination cannot be reached that way, another one: a departure gets another entry of its runway it accepts (an intersection with enough take-off run), an arrival one of the 3 nearest free stands that fit (`taxi to stand 12 via L2, M`). Routes across a runway come after the others (marked *crosses runway*). Last come the turn-arounds: `turn around, taxi to ...` - an airliner needs a tug for it (300-600 s), an aircraft up to 25 m turns on the spot. The AI Ground uses the first option that does not cross a runway; a re-route it gave once and that did not help (the conflict is still there) is not given again, and nothing is given to an aircraft whose tug is coming or that was re-routed less than 120 s ago.

GROUND VEHICLES

Source: `src/core/vehicles.ts`.

Tugs are not drawn: at the nose of the aircraft they move (pushback, tow, turnaround of a stuck airliner, 300-600 s) they merge with the aircraft's return on the surface movement radar, as on a real ground radar display.

Tows: when the turnaround of an arrival on a stand that needs a pushback (a contact stand) ends, **15 %** of these aircraft are towed to a free remote (drive-through) stand instead of leaving the simulation. A tow is its own entity with the tug's callsign (`TUG1` ... `TUG9`, "Tug 1") on Ground frequency:

STEP	TIMING / BEHAVIOUR
Tow request	10-60 s after the turnaround: `request tow (operator) (type) from stand X to stand Y` (request `tow`, list `TOWS`)
Approval	`tow approved [to stand ...] [via ...]`; the destination stand is reserved from the request on
Pushback	From a pushback stand the tug first pushes the aircraft back (same pushback as a departure), then repositions in 10-20 s
Towing	Along the route at no more than **10 kt**; the tug can turn the aircraft anywhere (no 180 degree restriction). Runway crossings, holds and give-way work as for aircraft
On the stand	Counts as a completed tow (+5); the aircraft stays parked for 20-40 min, then it leaves the simulation

Follow-me: two cars (`FOLLOW-ME 1`, `FOLLOW-ME 2`, radio callsigns `Follow-me 1` / `Follow-me 2`) wait at the taxi node next to the fire station, on Ground frequency. After `follow the follow-me` the next free car is assigned and its driver calls `request proceed to DCEEO at taxiway F`. After your `proceed [via ...]` it drives to a point on the aircraft's route ahead of it (at **10 m/s**, about 36 km/h, accelerating at 2.5 m/s², braking at 3 m/s², turning at most 120°/s); the aircraft waits until it is there. It then drives **35 m ahead of the aircraft's nose** (it waits at the meeting point until the aircraft is close and never backs up). The scope interpolates the cars' positions between simulation steps. (`hold position` to the car stops both), turns off when the aircraft is less than that distance plus 25 m from the end of its route (the stand entry) and asks `request return to base`; after `return to base` (or `proceed to base`) it drives back. If no car is free, the aircraft waits for the next one. With the AI running Ground, the cars do not call and drive on their own. Crews ask for a follow-me after vacating: 20 % of business aviation crews (registration callsigns), 3 % of airline crews.

SWITCHING FREQUENCIES OFF

A frequency switched off in the toolbar is run by the simulator from then on; open requests on it are dropped and the AI answers. The **AI Delivery** clears departures still waiting on Delivery frequency (next free squawk, the SID of the flight plan's fix) and sends them to Ground; new departures start on Ground with their clearance. The **AI Ground** takes over as described below. The **AI Tower** decides again itself (line-up, take-off, landing; no landing clearance needed), hands arrivals that have vacated and aircraft that have crossed to Ground. When both Ground and Tower are run by the simulator, Ground lets an aircraft at a runway holding point cross when the runway is free and the next arrival is more than 90 s away. Switching a frequency on again gives you the position back; aircraft already on it call you again.

AI GROUND

When you don't staff Ground (for example when you work Delivery alone), the AI Ground acts every 5 s, silently on its own frequency: it approves push and start for cleared departures at their ready time when no other aircraft taxis or pushes within 250 m (taxi-out stands: taxi), taxis them to the runway in use after start-up, hands them to Tower on the last 700-1400 m before the holding point (each flight differently), once they are on one of the last two taxiways of their route (runway 07 via M, L2: from L2 on), when nothing is left to coordinate (no runway crossing or hold-short ahead, no conflict, no other taxiing or pushing aircraft within 200 m except departures queuing for the same holding point, and no CTOT window opening later than 2 min from now), otherwise at the holding point, taxis arrivals to their allocated stand - without one, to the nearest free stand that fits and that they can reach without turning around (another one if the crew finds it occupied) - approves tows when nothing taxis within 250 m, keeps a departure on its stand (or, after start-up, for up to 4 minutes) while its way out would meet traffic coming the other way on a taxiway - an aircraft taxiing or vacating along it in the opposite direction, or an arrival less than 4 minutes from the threshold whose exits run against it (on one-way taxi flows this never happens; at Memmingen it keeps N free), with de-icing (scenario option) keeps departures that need it on their stand while every pad is busy, taxis them to the free pad with the shortest route and on to the runway after the de-icing (only while fewer than 3 departures are taxiing to a holding point, so the queue does not fill taxiway N), and resolves conflicts after 40 s with the first *Resolve conflict* option - once per conflict: not again while a tug is coming or within 120 s of the last resolution.

AI TOWER

When you staff Tower (v0.7), the AI Tower only flies the aircraft (final approach, landing roll and exit, line-up, take-off roll, climb-out) and detects incursions; every clearance is yours:

SITUATION	BEHAVIOUR
Take-off runway	A departure takes off from the runway in its clearance (its assigned runway), even if the ATIS has changed the runway in use since
Request hand-off	The simulator-run position hands the aircraft over 4-10 s after your request (coordination by landline)
Coordination (`COORD`)	*No more arrivals*: Approach releases no new arrivals onto the final (checked every 10 s), those already on final continue. *Hold all departures* (only while the simulator runs Ground): the AI Ground approves no pushbacks or taxi-out from stands; aircraft already moving continue. Each costs 2 points per full minute it is on. *Ask departures for the earliest intersection* (only while the simulator runs Ground, free): every 5 s Ground asks each cleared departure on the ground once which intersection with the least take-off run it accepts (`INT D`, `INT FULL` = full length only)
Departure on the way to the holding point	Handed over by the AI Ground on the last 700-1400 m before the holding point (on one of the last two taxiways of its route) once there is nothing left to coordinate on the ground (no runway crossing or hold-short ahead, no conflict, no other ground traffic within 200 m except departures queuing for the same holding point), at the latest at the holding point; calls `taxiing to holding point A, ready for departure` (more than 400 m away) or `approaching holding point A, ready for departure`. A departure waiting at the holding point without an open request (e.g. after a take-off refused in a thunderstorm) calls again after 150 s
Line-up	Not while another aircraft is on the runway within 400 m of the entry (except one rolling away), or is just lining up onto a point within 400 m (K and Y share one runway point at the 07 end). Departures from different entries may be lined up at the same time (e.g. full length A and intersection B)
Speed on final	Without instructions the **profile Approach usually gives**: 170-190 kt on the intercept, from 7-8.5 NM a reduction to **160 kt until 4 NM**, then the final approach speed (Vapp of the type, e.g. A320 about 138 kt) - reached at about 3 NM, 1000 ft above the runway. Reductions at 0.75 kt/s (1 kt/s with gear and landing flaps inside 4.5 NM), increases at 1 kt/s. An instructed speed (Vapp to 210 kt) replaces the profile until its distance (default 4 NM), then the aircraft slows to Vapp; the crew refuses one it could not reduce from in time (Vapp needed by 2.5 NM: `unable 180 knots until 4 miles, we could not be stabilised`). `reduce to final approach speed`: Vapp now; `resume normal speed`: back to the profile
Immediate take-off	Line-up at 1.5 times the normal speed, roll 0.5 s after stopping
Line-up / take-off clearance	Lines up when the clearance (and any condition, `behind ...`) allows; lined up with a take-off clearance it rolls after 2-5 s - not while another aircraft is landing or taking off, or is on the runway **ahead** of its start point. Aircraft standing behind it (e.g. lined up at A while it departs from intersection B) do not count as runway occupied - also not for the RMCA warning and the take-off menu
Departure spacing	At the start of the roll, the required spacing (2 min behind a heavy, 3 from an intersection, 2 min same SID fix, 1 min diverging) is checked; too early counts as a separation loss
Climb-out	Asks for a frequency change above 1500 ft above the airport; leaving (4000 ft) on Tower frequency counts as a missed hand-off. The crew reads back the radar frequency you give, or without one the frequency of its SID (Stuttgart southern departures: 119.200)
Arrival	First call when Approach hands it over (8-10 NM; or right away if it appears closer), reminder at 2 NM, `one mile final, request landing clearance` at 1 NM; without a landing clearance it goes around only at **0.15 NM** (just before the threshold, incident); with a clearance it goes around at 0.5 NM if the runway is not clear
Landing	Takes the exit given with `vacate via` if it can still reach it at normal braking, otherwise the usual one - never one where a departure is holding at the holding point (an exit that is also a departure entry); reports vacated and waits for `contact ground` (calls again every 20 s if the call got lost)
Backtrack (departure)	`backtrack runway 24, line up and wait` (or with the take-off clearance): the aircraft enters at its holding point, taxis along the runway towards the runway end at up to 18 kt, slows to 6 kt, turns around on the runway (a 180° turn of 11 m radius, 30 m from the end) and lines up on the centre line 80 m from the end. Only from a holding point at least 120 m from the runway end. At an entry marked for backtracking (Memmingen N for 24) the crew asks for it when the runway ahead is shorter than it needs, and refuses a line-up without it
Backtrack (arrival)	At exits marked for backtracking (Memmingen N for 06): when no exit ahead can be made at normal braking, or Tower asks for an exit behind the aircraft (`backtrack, vacate via N`), it brakes to 6 kt, turns around on the runway where it has stopped (at the latest 50 m before the end), taxis back at up to 18 kt and vacates (10 kt on the exit)
Runway crossing	The AI Ground hands an aircraft stopped at a runway holding point to Tower; after `cross runway 25` and crossing it reports vacated
CTOT	A take-off you clear before CTOT -5 min is flown and counts as a slot violation (-10, like a missed slot); missed slots as usual

The AI Ground also resolves circles of aircraft waiting for each other (A behind B, B behind C, C behind A): if one of them is pushing back, the pushback is cancelled and the tug pulls it back onto its stand.

When the AI runs Tower (you staff Delivery and/or Ground), it owns the runway. Its rules are a simplified model of ICAO PANS-ATM practice; the reasoning and the sources are in tower-operations.md. Each step it:

1. **Flies arrivals** down the final approach and lands them (see above).
2. **Sequences departures**: candidates are aircraft on Tower frequency waiting at a holding point, in the order they reached it. Only one aircraft lines up at a time. The first one may **line up** ("line up and wait") when
   - nobody is inside the runway area, except a departure that is already rolling, or a landing aircraft that is more than 300 m past the departure's entry point ("behind the landing traffic, line up"),
   - its departure separation will be met within 45 s (see below),
   - no arrival with an emergency is within 8 NM,
   - the next arrival is far enough out: its time to the threshold must exceed `35 s (line-up) + max(5 s, remaining departure separation, time until the landing aircraft ahead has vacated) + roll time + 15 s`. The roll time is `Vr / 2.0 m/s² + 12 s`, about 50 s for an A320.
3. **Lines up** the aircraft: it moves onto the runway at 9 kt and stops 50 m along the runway heading. If less than 1500 m of runway would remain (an aircraft sent to a holding point at the wrong end), Tower sends it back to Ground. From an entry marked for backtracking it backtracks to the runway end instead (see above); the line-up time then also counts the backtrack (distance at 14 kt plus 40 s).
4. **Take-off** ("ready for immediate departure"): 3-8 s after lining up, when
   - nobody else is in the runway area, the preceding departure is airborne, and the preceding landing aircraft has vacated,
   - **departure separation** is met, measured from the moment the previous departure became airborne:
     - **2 minutes behind a heavy** (wake turbulence; 3 minutes if this departure starts from an intersection),
     - otherwise **2 minutes** if both use the same departure route (same first SID fix) and **1 minute** if the routes diverge,
   - the next arrival is more than `roll time + 8 s` from the threshold (about 2 NM).

   The aircraft then accelerates at 2.0 m/s². At its rotation speed it becomes airborne, climbs at its climb rate, and accelerates to 200 kt.
5. **Stranded departures**: a departure on Tower frequency that has stopped short of the holding point (a hold-short or a clearance limit, for example after `hold short of taxiway A` instead of `taxi to holding point A`) for **15 s** is handled by Tower: at a hold-short on a route that continues, Tower lets it continue; at a clearance limit, Tower taxis it to the runway in use itself if the holding point is at most **600 m** away without turning around, otherwise it sends the aircraft back to you (`Tower sent us back to you, ... request taxi`). A message in the message window tells you what Tower did. (Only while you are not Tower yourself.)
6. **Crossings for its own traffic**: aircraft already handed to Tower that stop at a runway holding point on their route get a crossing as soon as the runway is free and the next arrival is more than 90 s away. An aircraft that has waited **120 s** to cross a runway (its `crossing` call, on any frequency) gets the next gap: no departure lines up on that runway meanwhile.

The **runway area** used for occupancy checks extends 70 m either side of the centre line (the holding positions are at 95 m) and 60 m beyond each runway end.

Arrival spacing (provided by the simulated Approach controller, see traffic generation): at least the wake turbulence minimum, and wider gaps when departures are waiting, so that one departure fits between two arrivals; on a runway where landing aircraft backtrack at least 10 NM, where departures backtrack at least 8 NM.

PARALLEL RUNWAYS

Airports with parallel runways (`parallelRunways` in the airport data, Leipzig/Halle) use both at the same time, each for take-offs and landings (mixed mode). The ATIS editor offers the parallel runways (`Runways 26R + 26L`, the default for the direction of the wind) and each runway on its own.

TOPIC	RULE
Runway of an aircraft	The runway of its apron (Leipzig: apron 1 and 3 the northern runway, the cargo aprons 2, 4, 5 the southern one); aircraft from a given wingspan always the runway named for them (Leipzig: 63 m and more, the southern runway); otherwise the runway nearest to its stand. Departures get it when they appear, with the SID of that runway (at Leipzig the SID letter names the runway)
Delivery	The SID of a clearance must belong to a runway in use; it sets the departure's runway (`DRN3S` = 26L)
Ground	`taxi to runway 26L` / `taxi to holding point S9` sends a departure to the other runway: it now departs from there, with the SID of that runway to the same exit fix (a message says so)
Approach	Puts each arrival on the runway of its stand. If there is no gap on that final but on the other, it uses the other (not for the aircraft bound to one runway by their size). The spacing on each final is the same as with one runway (wake, departure gaps from the departures waiting for *that* runway, approach type of that runway). Between the finals: **dependent parallel approaches** (Leipzig, AIP) keep **1.5 NM** diagonal radar separation to the arrivals on the adjacent final, **staggered** approaches **3 NM**, **independent** approaches none. With centre lines `d` NM apart the required stagger along the final is `sqrt(diagonal² - d²)` (Leipzig: 2.1 km apart, 0.97 NM). Wake turbulence spacing applies on each final only (the simulator assumes parallel runways more than 760 m apart, as at Leipzig)
Landing	Vacates on the side of its stand (not back across the runway) when it can reach such an exit
AI Tower	Works each runway on its own: one line-up at a time per runway, occupancy, arrivals and departure spacing per runway (departures from parallel runways are independent - AIP: independent parallel departures). A crossing of a runway in use is cleared when nobody is on it or lining up and its next arrival is more than 90 s away (no departure lines up while an aircraft is being cleared to cross, nor for 30 s after the crossing clearance until it has entered the runway); runways not in use can always be crossed
You on Tower	Your line-up, take-off and landing clearances name the aircraft's runway (`runway 26L, cleared for take-off`); the take-off menu shows the next arrival to that runway; RMCA warns per runway
Runway incursion	Per runway: entering a runway in use while an arrival to that runway is within 2.5 NM or an aircraft is lining up, taking off or landing on it
Runway change	Aircraft whose runway stays in use keep it (e.g. from `26R + 26L` to `26L` only the northern ones change)
ATIS	`ILS approach runways 26R and 26L, dependent parallel approaches in progress, runways 26R and 26L in use` (independent and dependent parallel approaches are announced, ICAO Doc 4444)

RUNWAY INCURSIONS AND GO-AROUNDS

- **Runway incursion**: a taxiing aircraft (phase `taxi`) enters the runway area while an arrival is within 2.5 NM, or while another aircraft is lining up, taking off or landing. Taxiing aircraft only enter the runway area with a crossing clearance, so every incursion is caused by a crossing clearance given at the wrong moment. **-50 points.**
- **Go-around**: an arrival inside **0.5 NM** goes around if
  - any aircraft is on the ground inside the runway area: lined up, crossing, or a preceding landing aircraft that hasn't vacated yet,
  - except a departure on its take-off roll that is already beyond 1200 m or faster than 100 kt. That departure will be airborne and beyond 2400 m when the arrival crosses the threshold, which is the reduced runway separation rule.

  It climbs away, is removed, and is reported together with the aircraft that blocked the runway. **-15 points.**

COLLISIONS

Two ground aircraft whose reference points come closer than `0.25 · (wingspan A + wingspan B)`, with at least one of them moving, have collided. Both stop permanently and are drawn red. **-100 points.** With pilot see-and-avoid active, collisions are rare. They mostly happen when a pushback is approved into an aircraft that is already in the way.

TRAFFIC GENERATION

DENSITY	DEPARTURES / HOUR	ARRIVALS / HOUR
light	8	7
light to medium	11	10
medium	14	12
medium to heavy	18	15
heavy	22	18

These are the Stuttgart figures. Smaller airports scale them with their `rateFactor` (see airport data): Nuremberg 0.6, Bremen 0.4, Memmingen 0.35, Sylt 0.25 (Cologne/Bonn 1, like Stuttgart; Leipzig/Halle 1.1 on two runways). At Memmingen and Sylt (`standReserve` 1) no new departure takes the last free stand and Approach releases an arrival only when a stand is free for it; an arrival that still has none gets one (checked every 20 s) as soon as one becomes free.

- **Initial situation**: about 55% of an hour's departures are already parked at stands, with ready times spread over the first 25 minutes, **denser towards the end** (`30 s + sqrt(u) x 24.5 min`), so the workload builds up gradually. Their first calls are **staggered**: the first after 15 s, each next one at least **90 s** later, in the order of their ready times. On Delivery that is the clearance request (a crew then needs at least 4 minutes until it is ready); on Ground the pushback call. One arrival is on a 4-6 NM final.
- **New departures** appear at exponentially distributed intervals (a Poisson process) with the mean given by the rate.
- **New arrivals** follow the same kind of process. A new arrival is only released onto the final (at 12 NM) when the distance to the previous arrival is at least the **required spacing**, which is the larger of:
  - the **wake turbulence minimum**: 3 NM normally; 4 NM heavy behind heavy, 5 NM medium behind heavy, 6 NM light behind heavy, 5 NM light behind medium (larger values behind an A380, category J),
  - the **runway spacing**: 4 NM when no departures are waiting, **6 NM** when at least one departure is holding or taxiing to a holding point, **8 NM** with four or more. These are the "departure gaps" Approach gives on request,
  - **5 NM** for non-precision approaches (localizer only or RNP, when the ILS glide path or localizer is switched off in the systems window, or the runway has no ILS - Sylt 14; simulator value),
  - **8 NM** while a thunderstorm is active (the arrival rate is halved too),
  - **10 NM** on a runway where landing aircraft backtrack to their exit, **8 NM** where departures backtrack (Memmingen).

  - the runway's own minimum where the airport sets one (`arrivalSpacingNm`; Sylt **6 NM**, arrivals vacate only at the far end; Bremen **6 NM**, the target spacing of its SOP).
  - The spacing applies **at the threshold**: the faster follower closes in while the leader slows down to its final approach speed (compression), so Approach releases the next arrival only when, flying its speed profile, it would reach the spacing distance no earlier than the leader lands (and not before it is that far behind the leader at 12 NM).
- **Night flying restrictions** (scenario option): no jet departure is generated that would take off after the curfew (ready time + 20 min estimate); within 75 minutes of its end, departures are planned for the morning wave (ready 10-0 minutes before it). Arrivals that would land in the landing curfew (12 NM ≈ 5 minutes) are not generated, except delayed flights (35 % of the normal rate) until delayed landings end. See scenarios.md.
- **Operators, types and destinations** come from the airport's `traffic` data (see airport data format), weighted by each operator's share. At EDDS this follows the real airline shares (Eurowings about 40 %, SunExpress 8 %, TUIfly 6 %, Pegasus and the Lufthansa group 5 % each, Turkish 5 %, Condor 4 %, British Airways 4 % ...) and the busiest destinations (Palma, Antalya, Istanbul, Pristina, London, Barcelona, Berlin, Hamburg); see EDDS. Airports without traffic data use the global list in `src/data/airlines.ts`.
- **Callsigns** are unique within a session: alphanumeric (`EWG7TK`), numeric (`THY1734`) or registrations (`DCMGB`), depending on the operator.
- **Flight plans** contain the destination, the SID of the active runway towards the destination (the exit fix whose sector contains the great-circle bearing to the destination, or a SID filed only for that destination - e.g. Stuttgart ETASA for Frankfurt; see the airport pages), cruise level and a squawk. The route starts with the exit fix (`ROTWE DCT LEPA`).
- **Stands**: aircraft up to code D (52 m wingspan) get a random free stand up to 52 m that fits, leaving out the alternative positions (stands overlapping a smaller one, like 71A or 24A at Stuttgart); larger aircraft the smallest free stand that fits. Every stand also needs enough wingtip clearance to its neighbours (see below). Business jets prefer their stands (Stuttgart 60-65), airlines the others. Cargo airlines (`cargo` in `src/data/airlines.ts`: UPS, FedEx, DHL / EAT, AeroLogic, ...) prefer the cargo stands (Cologne/Bonn aprons E, F and W, Leipzig/Halle aprons 2-5) and take other stands only when no cargo stand is free; everybody else uses cargo stands (Stuttgart Apron South) only when nothing else is free.

RADIO MODEL

- There is **one frequency** (yours). A transmission takes `0.8 s + 0.32 s per word`.
- Your transmissions go out immediately and occupy the frequency.
- Pilot transmissions are queued. They start when the frequency is free, plus a 0.6 s gap. **Read-backs** (priority 10) go before **new calls** (priority 0).
- Read-backs start 0.8-2.2 s after your transmission. Spontaneous calls are delayed by 0.3-2.5 s.
- **Radio discipline**: after your instruction the frequency is reserved for the addressed station until its read-back has been transmitted, at most 6 s after your transmission has ended. Other pilots' calls wait. A new instruction to another station moves the reservation to that station.
- Queued transmissions that wait too long are dropped (45 s for calls, 60 s for read-backs). The pilot then repeats the call later.
- When you transmit to a pilot, their queued (not yet spoken) call is cancelled: they listen first.
- A frequency change takes effect when the read-back has been transmitted.

WEATHER AND ATIS

- **Wind** is drawn at session start: 60% westerly (220-290°), 40% easterly (040-110°), 3-14 kt - Stuttgart's long-term average is about 60 % of the operations to the west (RP Stuttgart noise report).
- **Runway in use** at session start: the first runway of the airport (25 at EDDS), unless it has more than 3 kt tailwind; then the runway with the most headwind. Where it has a parallel runway, both are in use (see parallel runways).
- **QNH** 1003-1028 hPa at the start; the METAR QNH then drifts (below).
- **METAR** (`core/met.ts`): a routine observation at minute 20 and 50 of every hour (UTC), the last one before the start is current when the session begins. The weather follows from the seed (`sim.variation`, the traffic is not affected): 30 % CAVOK, 30 % few / scattered (2500-5500 ft), 25 % broken (1500-4500 ft, 40 % of them with light rain and 6-9 km visibility), 15 % overcast (600-1600 ft, 3000-6000 m in mist or light rain; snow at 1 degree C or below). The cloud base drifts by up to 300 ft per observation. Temperature: monthly mean of a Central European climate (January 1, July 19 degrees C, `climate` in the airport data overrides it) +-3 per session, plus a daily cycle with its maximum at 1500 local (half range 3-6 degrees C, 70 % under broken, 40 % under overcast cloud); dew point 1-11 degrees below. QNH: the pressure of a session rises or falls - 1 hPa in that direction in 25 % of the observations, back in 5 %. Wind: the observed wind (direction to 10 degrees) - it changes from one observation to the next (below). De-icing scenarios: -3 to +1 degrees C, overcast, light snow or mist. Thunderstorm: a **SPECI** at each phase - approaching `VCTS`, `FEW040CB`, `TEMPO TSRA`; overhead `4000 TSRA SCT015 BKN025CB`, 4 degrees colder, `BECMG NSW`; passed: light rain for 20 minutes.
- **New METAR**: it appears in the message window. If it differs from the ATIS - QNH, wind (30 degrees at 10 kt or more, 5 kt, gusts by 10 kt), visibility (across 5000 m or by 3000 m), present weather or ceiling (by 500 ft) - the message says so and the toolbar shows the ATIS letter in red.
- **Observed wind** (the toolbar): the actual surface wind. It **changes during the session**: for every routine METAR a new wind is drawn from the seed (`sim.variation`, the traffic is not affected) - the direction moves by up to 40 degrees and the speed by up to 5 kt from the last observation, plus a quarter of the way back to the prevailing wind of the session (so it wanders but rarely turns round), at most 30 kt; from 15 kt on, half of the observations have gusts 10-15 kt above the mean. The actual wind turns to the new value during the **15 minutes before the observation** (a degree at a time), so the METAR reports what the toolbar shows. A scenario **wind shift** turns it to the opposite direction (9-15 kt), a **thunderstorm** changes it, and broadcasting an ATIS with another wind sets it as well; the next observation goes on from there. Tutorials keep their wind. The ATIS editor is pre-filled with it (including gusts, `070/18G32`).
- **Runway change hint**: when a METAR shows more than **5 kt tailwind** on the runway in use and another runway is into wind, the message says so (`Tailwind 7 kt on runway 25: consider a runway change to runway 07 (ATIS editor) and re-route the traffic`). Changing the runway is your decision.
- **Thunderstorm** (scenario event): warning 5 min (gusts, 8 NM arrival spacing), overhead 15 min (tailwind gust front 14-20 kt gusting 12-18 kt more, no pushbacks, no take-offs, go-arounds at 1 NM with more than 10 kt tailwind or in windshear with a 40 % chance per arrival; weather go-arounds cost no points), then the wind settles at 6-11 kt from the new direction. Details: scenarios.md.
- **ATIS** in two forms, like a German D-ATIS: text (`ATIS EDDS K 0950 / EXPECT ILS APCH / RWY 25 / A-CDM IN PROGRESS / TRL 70 / 25008KT CAVOK T12 DP06 QNH1018 / TREND NOSIG`) and spoken (`Stuttgart information K, time 0950. Expect ILS approach runway 25, runway in use 25, A-CDM in progress. Transition level 70. Wind 250 degrees 8 knots, CAVOK, temperature 12, dew point 6, QNH 1018. Trend nosig. Information K out.`). Time, visibility, weather, clouds, temperature and trend come from the METAR the ATIS was made from; wind and QNH are the ones you broadcast. Notices after the runway: navigation aids out of service (`ILS RWY 25 U/S`, `GP RWY 25 U/S`), with parallel runways the runway of each apron and of the large aircraft (`A/C ON APRON 1 OR 3 EXPECT DEP AND ARR RWY 26R`), closed taxiways (`TWY E AND S CLSD`), de-icing (`DE-ICING IN PROGRESS ON DE-ICING PADS`), A-CDM (`A-CDM IN PROGRESS` while it is on), datalink clearances switched off (`DCL U/S`), and `CAUTION WINDSHEAR REPORTED ON FINAL` while a thunderstorm is overhead.
- **ATIS letter** starts at random. It only changes when **you** broadcast a new ATIS in the ATIS editor (without a QNH it takes the METAR's).
- **Old ATIS letters**: a crew listens to the ATIS 2-8 minutes (per callsign) before its first call and reports the letter that was current then - after a new ATIS some still report the old one. Tell them the current one: `information K is current, QNH 1018` (read back `Information K, QNH 1018`; a letter that is not broadcast is queried), or broadcast `all stations, information K is now current` (corrects everybody on your frequencies). A crew that leaves your frequencies with an old letter costs 3 points.
- **Runway change** (via the ATIS):
  - Departures that are parked, pushing back or starting up and have no taxi instruction yet get the new runway, and a SID with the same first fix for the new runway - unless their runway stays in use.
  - Arrivals further out than 3.5 NM are moved onto the new final (Approach re-sequences them, at least 6 NM out) - unless their runway stays in use.
  - Arrivals closer in land on the old runway; Tower doesn't line anyone up until they have landed.
  - Taxiing departures keep their clearance.

SPECIAL EVENTS

Set with the **Special events** slider in the Connect dialog or the in-game settings: *never*, *very rare*, *rare* (default), *occasional*, *often* multiply the chances below by 0, 0.5, 1, 2 and 4. At the default they are rare:

EVENT	CHANCE (AT "RARE")	WHAT HAPPENS
Medical emergency, arrival	2% of arrivals	Declared on final (system message). Lands with priority: no line-ups while it is within 8 NM. After vacating, the pilot calls `PAN PAN, medical emergency ..., request expedited taxi to the stand`.
Medical emergency, departure	1% of departures	40 s after starting to taxi, the pilot calls `PAN PAN ... request immediate return to the stand`. A stand is suggested; give it a route back (`taxi to stand ...`).
Rejected take-off	1.5% of take-offs	The take-off is aborted at 60-110 kt (or below 80 kt on your `stop immediately`). The aircraft brakes at once (3 m/s² planned, 2.5 m/s² hardest braking) down to 25 kt, rolls on to the next usable exit, vacates and calls you. 60% want to return to a stand (technical problem), 40% request taxi for another departure (with new line-up and take-off clearances: the old ones are used up). The runway is blocked meanwhile, so arrivals may have to go around.

Scheduled events from a scenario code happen at fixed minutes, whether random events are on or not: the next arrival declares a medical emergency, the earliest-ready departure on your frequency gets one 40 s after it starts taxiing (retry every 30 s if there is none), the next take-off is rejected, the wind shifts, or a thunderstorm crosses the airport. See scenarios.md.

Traffic mix of a scenario: a departure rush multiplies the departure rate by 1.6 and the arrival rate by 0.5, *more departures* by 1.3 and 0.75, and the arrival steps the other way round. With a **share of heavies** of 10, 20, 35 or 50 %, each new flight is a wide-body with that chance (on top of the operators' own fleets) - while a stand that fits it is free; otherwise the flight is a normal one. See scenarios.

A medical emergency counts as handled when the aircraft reaches a stand. Within **6 minutes** of the emergency call, that earns a **+15 point bonus**. Emergency aircraft are shown with `PAN` and a flashing magenta symbol and tag.

18 GLOSSARY

Terms used in the simulator and in this documentation. Real-world terms are explained the way the simulator uses them; where the simulator simplifies, the linked page says how.

TERM	MEANING
A-CDM	Airport Collaborative Decision Making: airlines, handlers, airport and ATC share their times (TOBT, TSAT, CTOT). See systems.
A-SMGCS	Advanced Surface Movement Guidance and Control System: surface surveillance plus safety nets (RMCA, CATC) and routing. See systems.
Altimeter	US name of the QNH, in inches of mercury: "altimeter 2992" (= 1013 hPa). Used at US airports (San Diego).
ATIS	Automatic Terminal Information Service: recorded broadcast with runway, wind, QNH and approach procedure, identified by a letter (information R).
CATC	Conflicting ATC clearances: the A-SMGCS alert for clearances that conflict; in the simulator, taxi routes that meet head-on.
Clearance limit	The point an aircraft is cleared to and must not pass: the destination of a taxi instruction or a hold-short point; in an IFR clearance, the destination airport.
Combined position	One controller staffing several positions at once (for example Delivery + Ground).
CTOT	Calculated take-off time: an ATFM slot from the Network Manager; take-off from CTOT -5 to +10 minutes.
DCL	Departure clearance by datalink instead of voice.
Deadlock	Two (or more) aircraft that block each other so that none of them can move on by itself - typically two aircraft nose to nose on a taxiway (**head-on**) or each waiting for the other at an intersection. Since 0.8 crews who see opposite traffic in time stop short of the junction between them and ask, so that one can turn off there. Otherwise the pilots report it after 60 s (we have opposite traffic ahead, request instructions); only the controller can resolve it (re-route one, *Resolve conflict*, a tug). See simulation model.
EDCT	Expect departure clearance time: the US counterpart of the CTOT (the pilots at San Diego say EDCT).
Gridlock	A deadlock that has spread: several aircraft stuck in a chain behind a deadlock or blocking each other's way out, so that a whole area (a taxiway, an apron exit) stops. Used informally in the changelog for traffic jams of early versions; the simulator itself detects and reports deadlocks.
FAA phraseology	The radio phraseology of the US (FAA JO 7110.65): "runway 27, taxi via B", "then as filed", runways without a leading zero. See phraseology.
Follow-me	A car (yellow, "FOLLOW ME" sign) that leads an aircraft to its stand or along a route, for crews unfamiliar with the airport or in low visibility. "Follow the follow-me".
Head-on	Two aircraft on the same taxiway in opposite directions (routes meeting at more than 135 degrees). Airliners cannot turn around on a taxiway, so it ends in a deadlock unless one turns off in time.
Holding point	Runway holding position where departures wait before entering the runway (EDDS: A, B, C, D for runway 25; K, Y, I, H, W for runway 07).
ILS / LOC / GP	Instrument landing system with localizer (lateral guidance) and glide path (vertical guidance). Without them: localizer-only or RNP approaches.
Incursion	An aircraft entering the runway without a clearance, or while it is occupied - a serious incident (-50 points).
PDC	Pre-departure clearance: the US datalink clearance (DCL elsewhere).
Proceed	The instruction word for vehicles ("Follow-me 1, proceed to DCEEO via N, F"); aircraft get "taxi".
Radio discipline	Only one station talks at a time; after an instruction the addressed station reads back before anyone else calls.
RMCA	Runway monitoring and conflict alerting: A-SMGCS alert for aircraft approaching an occupied runway.
RNP approach	Approach with satellite navigation (no ILS needed).
SID	Standard instrument departure: the published departure route from the runway to the first fix (exit fix) of the flight plan. The simulator uses the real designators; each flight plan goes via the exit fix towards its destination.
Squawk	The four-digit transponder code (octal digits 0-7) assigned in the IFR clearance.
Stand	Aircraft parking position.
Tow	Moving an aircraft without its own engines by a tug, e.g. from a contact stand to a remote stand. The tug driver talks to Ground: "request tow ... from stand 14 to stand 45" - "tow approved via ...".
TOBT	Target off-block time: when the airline expects the aircraft to be ready (the simulator's ready time).
TSAT	Target start-up approval time from the A-CDM pre-departure sequencer; start-up is approved at the TSAT.
Tug	Towing vehicle: pushes aircraft off the stand, and turns an airliner around when it is stuck (5-10 minutes in the simulator).